Air Regulations and Human Performance

Questions, Answers and Memory Aids

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Chapter 1 — International Organisations and Conventions

Practice Questions

1 Which of the following is obligating for members of ICAO?

- **A** ICAO shall approve the pricing of tickets on international airline connections
- **B** ICAO must be informed about changes in the national regulations
- **C** ICAO must be informed about differences from the standards in any of the Annexes to the convention

Answer: C. Article 38 obligates states to notify ICAO about differences from international standards in the Annexes (within 60 days for amendments).

2 For aircraft flying over the high seas, which rules shall be in force?

- **A** The rules established by the state of the operator of the aircraft
- **B** The rules established by the state(s) adjacent to the high seas over flown
- **C** The rules established under the Convention of international civil aviation

Answer: C. Article 12 of the Chicago Convention: Over the high seas, the rules in force are those established under this Convention.

3 Annex 14 to the convention on international civil aviation contains SARPs for:

- **A** Air Traffic Services
- **B** Aerodromes
- **C** Facilitation

Answer: B. Annex 14 covers Aerodromes (Vol I: Aerodrome Design and Operations; Vol II: Heliports).

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4 The Warsaw convention and later amendments deals with:

- **A** limitation of the operator's responsibility vis-a-vis passengers and goods transported
- **B** operator's license for international scheduled aviation
- **C** the security system at airports

Answer: A. The Warsaw Convention deals with the financial liability of airlines (operators) regarding passengers, baggage, and cargo.

5 Annex 17 to the Convention of Chicago covers:

- **A** Aerodromes
- **B** Security
- **C** Facilitations

Answer: B. Annex 17: Safeguarding International Civil Aviation against Acts of Unlawful Interference (Security).

6 The State has to notify the Council of ICAO if a state finds that it is impracticable to comply with an International Standard. The notice period is:

- **A** It shall give 45 days notice to ICAO
- **B** It shall give 60 days notice to ICAO of the differences between its own practices and the International Standard
- **C** It shall give immediate notice to ICAO

Answer: C. Article 38, first sentence: a State that finds it impracticable to comply with a Standard must give immediate notification to ICAO of the differences. The 60-day figure applies only to the separate case of an amendment to a Standard that the State does not adopt (notice to the Council within 60 days of adoption of the amendment).

7 The international convention defining rules relative to the responsibilities of international air carriers for the carriage of passengers, baggage and freight is the:

- **A** Hague Convention
- **B** Warsaw Convention
- **C** Tokyo Convention

Answer: B. Warsaw Convention 1929 defines the financial liability of air carriers for carriage of persons, baggage, and cargo.

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8 The 'Standards' contained in the Annexes to the Chicago convention are to be considered:

- **A** binding for the member states that have not notified ICAO about a national difference
- **B** advice and guidance for the aviation legislation within the member states
- **C** binding for all air line companies with international traffic

Answer: A. ICAO Standards are binding (mandatory) for member states unless they have formally notified ICAO of a difference under Article 38.

9 An aircraft flying over the territory of another state must:

- **A** Follow the rules of the Air of the State it is over flying, provided they follow the ICAO rules
- **B** Follow the rules of the Air of the State of the Operator, if registered in other State
- **C** Follow the rules of the Air of the State of Registration to the extent that they do not conflict with the rules published by the State having jurisdiction over the territory overflown

Answer: C. Article 12: The State of registration's rules apply, but they must not conflict with the rules of the state having jurisdiction over the territory overflown.

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10 An airplane is planning a flight that will require a technical landing in a neighboring state. Which freedom of the air will be exercised?

- **A** 3rd freedom
- **B** 4th freedom
- **C** 2nd freedom

Answer: C. 2nd Freedom: The privilege to land for non-traffic purposes (e.g., a technical/fuel stop).

11 The International Civil Organisation (ICAO) was established by the international convention of:

- **A** Warsaw
- **B** Chicago
- **C** Montreal

Answer: B. ICAO was established by the Chicago Convention, 1944.

12 The Rome Convention and later amendments deals with:

- **A** Damage caused by foreign aircraft to third parties on the surface
- **B** offences and certain other acts committed on board aircraft
- **C** Regulation of transportation of dangerous goods

Answer: A. Rome Convention 1952 covers damage caused by foreign aircraft to third parties on the surface of the earth.

13 The convention on offences and certain acts committed on board aircraft, is:

- **A** the convention of Rome
- **B** the convention of Tokyo
- **C** the convention of Chicago

Answer: B. Tokyo Convention 1963: Convention on Offences and Certain Other Acts Committed on Board Aircraft.

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14 The aircraft commander, when he has reasonable grounds to believe that a person has committed or is about to commit, on board the aircraft, an offense against penal law:

- **A** may not require or authorise the assistance of other crew members
- **B** may require the assistance of passengers to restrain such persons
- **C** may deliver such person to the competent authorities

Answer: C.

15 If a person is injured by a part falling from an aircraft flying over the state but registered in another state, which convention covers this?

- **A** Paris
- **B** Chicago
- **C** Rome

Answer: C. Rome Convention 1952 covers damage caused by foreign aircraft to third parties on the surface.

16 Has India ratified the International Air Transport Agreement, 1944 (the agreement that carried all five freedoms)?

- **A** Yes
- **B** Yes, but it does not subscribe to Technical Freedoms
- **C** No

Answer: C. India SIGNED the 1944 International Air Transport Agreement on 7 Dec 1944 but never ratified it (ICAO status of India). Stem reworded from "signatory" to "ratified".

17 The convention signed by the states and moved by a desire to endure adequate compensation for persons who suffer damage caused on the surface by foreign aircraft is:

- **A** the Paris Convention
- **B** the Rome Convention
- **C** the Warsaw Convention

Answer: B. Rome Convention 1952: Liability for damage caused on the ground by foreign aircraft to third parties.

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18 Which Annex to the Chicago Convention covers dangerous goods carried in aircraft?

- **A** Annex 16
- **B** Annex 18
- **C** Annex 17

Answer: B. Annex 18: Safe Transport of Dangerous Goods by Air. (Annex 16 = Environmental Protection; Annex 17 = Security)

19 Proposal to give effect to an International Convention in India is initiated by:

- **A** Ministry of Civil Aviation
- **B** Parliament
- **C** DGCA

Answer: C. DGCA carries out amendments to the Aircraft Act, Aircraft Rules, and CARs for complying with amendments to ICAO Annexes, and initiating proposals for amendment to any other Act to give effect to an international Convention.

20 The second freedom of the air is the:

- **A** right to operate a commercial passenger flight with passengers on board between two states
- **B** right to land for a technical stop
- **C** right to "cabotage" traffic (trans-border traffic)

Answer: B. 2nd Freedom: The privilege to land for non-traffic (technical) purposes.

21 Which convention makes acts of violence on board, destruction of aircraft in flight and destroying or damaging any air navigation facility punishable?

- **A** The Tokyo convention
- **B** The Montreal convention
- **C** The Warsaw convention

Answer: B. Montreal Convention 1971: Suppression of Unlawful Acts Against the Safety of Civil Aviation (sabotage, destroying aircraft, damaging navigation facilities).

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22 According to which Convention may an aircraft commander impose measures upon a person committing a crime or an offence on board the aircraft?

- **A** The Convention of Rome
- **B** The Convention of Warsaw
- **C** The Convention of Tokyo

Answer: C. Tokyo Convention 1963 grants the Aircraft Commander powers to impose reasonable measures including restraint upon a person committing or about to commit an offence on board.

23 'Cabotage' refers to:

- **A** domestic air services
- **B** a national air carrier
- **C** crop spraying

Answer: A. Cabotage (Article 7): The right of a State to refuse foreign airlines permission to carry passengers/cargo for hire between two points within the same territory (domestic air services).

24 In case of a conflict between ICAO regulations and national regulations, which one takes precedence?

- **A** ICAO regulations are always binding for all member states in the interest of uniformity.
- **B** ICAO regulations for the member states that have not notified ICAO about a national difference within a specified period.
- **C** National regulations always take precedence over ICAO regulations.

Answer: B. ICAO Standards bind every member State that has not told ICAO about a difference (Article 38). Once a State has notified a difference, its own national rule applies for that State.

25 Which convention is considered to be a forerunner of Chicago Convention, 1944?

- **A** Warsaw Convention, 1929
- **B** Paris Convention, 1919
- **C** Ibero-American Convention, 1926

Answer: B. Paris Convention 1919 is considered the forerunner (predecessor) of the Chicago Convention 1944. It established ICAN, while Chicago established ICAO.

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26 ICAO is:

- **A** An independent and autonomous agency, with a constituency status in the UNO
- **B** A part and parcel of UNO as a non-specialised agency
- **C** Not associated with UNO at all

Answer: A. ICAO is an independent and autonomous agency, with a constituency (specialized agency) status in the United Nations Organization, linked to the Economic and Social Council since 1947.

Mnemonic (R-A-C-J-R-P-C): "**R**eal **A**ircraft **C**arries **J**ournals, **R**adio **P**ermit, **C**argo-list" → Registration, Airworthiness, Crew License, Journey Log, Radio License, Passenger List, Cargo Manifest.

Chapter 2 — Aircraft Nationality and Registration Marks

Practice Questions

1 The registration mark shall be letters, numbers or a combination of letters and numbers, and shall be that assigned by:

- **A** The International Telecommunication Union
- **B** The state of registry or common mark registering authority
- **C** The state of registry only

Answer: B. ITU allots the underlying symbols, but the individual registration mark itself is assigned by the State of Registry (or the common-mark registering authority for joint operations).

2 The common mark shall be selected from the series of symbols included in the radio call signs allocated:

- **A** To state of the operator
- **B** To the International Civil Aviation Organisation by the International Telecommunication Union
- **C** To the state of registry by the International Civil Aviation Organisation

Answer: B. The common-mark series (4YA–4YZ) is allocated by the ITU to ICAO for use by common-mark registering authorities.

3 The assignment of the common mark to a common mark registration authority will be made by:

- **A** The International Civil Aviation Organisation
- **B** The state of registry
- **C** The International Telecommunication Union

Answer: A. ICAO takes the pool of symbols provided by ITU and assigns a common mark to each common-mark registering authority.

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4 When letters are used for the registration mark, combinations shall not be used which might be confused with urgent or distress signals, for example:

- **A** XXX
- **B** RCC
- **C** DDD

Answer: A. XXX is a radio-telegraphy urgency signal — expressly listed alongside PAN and TTT as an example to avoid.

5 When letters are used for the registration mark, combinations shall not be used which might be confused with:

- **A** Three-letter combinations used in the international code of signals
- **B** Four-letter combinations beginning with Q
- **C** Five-letter combinations used in the international code of signals

Answer: C. Annex 7 specifies avoiding five-letter combinations from Part II of the International Code of Signals (and three-letter combinations beginning with Q — the Q Code).

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6 The height of the marks on the fuselage (or equivalent structure) and on the vertical tail surfaces of heavier-than-air aircraft shall be:

- **A** At least 20 centimetres
- **B** At least 40 centimetres
- **C** At least 30 centimetres

Answer: C. Fuselage / vertical tail marks — minimum 30 cm.

7 The height of the marks on lighter-than-air aircraft other than unmanned free balloons shall be at least:

- **A** 60 centimetres
- **B** 50 centimetres
- **C** 1 meter

Answer: B. Lighter-than-air (airships, manned balloons) — minimum 50 cm. Unmanned free balloons are the exception (set by the State of Registry).

8 When letters are used for registration mark, combinations shall not be used which might be confused with urgent signals, for example:

- **A** LLL
- **B** PAN
- **C** RCC

Answer: B. PAN is the radio-telephony urgency signal — expressly listed in the rule.

9 When letters are used for registration mark, combinations shall not be used which might be confused with urgent signals, for example:

- **A** LLL
- **B** TTT
- **C** RCC

Answer: B. TTT is the radio-telegraphy safety signal — listed alongside XXX and PAN.

10 The height of the marks under the wings of heavier-than-air aircraft shall be at least:

- **A** 50 centimetres
- **B** Between 40 and 50 centimetres
- **C** 60 centimetres

Answer: A. Wing marks — minimum 50 cm on heavier-than-air aircraft.

11 Who exercises operational control over the leased aircraft?

- **A** Lessor
- **B** Lessee
- **C** IFSC

Answer: B. The lessee must hold the economic and operating authority and provide the crew, dispatchers and ground facilities — operational control sits with the lessee.

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12 A company undertaking to lease aircraft in India must be approved by:

- **A** DGCA
- **B** IFSCA
- **C** MOCA

Answer: B. Aircraft-leasing companies at IFSC (GIFT City) are approved by the IFSCA.

Memory Hook

"Fireproof, near the main entrance, 1–5 characters." Three facts, three marks in the exam.

Memory Hook

"Fifty on wings, thirty on the fuselage." Lighter-than-air airships and balloons follow the same 50 cm rule as the wings — because on those bodies the mark is *the only* mark visible.

Amber / dashed — Memory aids & exam traps.

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Chapter 3 — Rules of the Air

Practice Questions

1 Where State has not established minimum IFR altitudes, the minimum height of an aircraft above the highest obstacle over high terrain, or in mountainous areas shall be for an IFR flight:

- **A** At least 2,000 ft within 8 km of the estimated position
- **B** At least 1,000 ft within 8 km of the estimated position
- **C** At least 2,000 ft within 5 km of the estimated position

Answer: A.

2 Aircraft wishing to conduct IFR flight within advisory airspace, but not electing to use the air traffic advisory service:

- **A** Need to file a flight plan
- **B** Shall nevertheless submit a flight plan but changes made thereto are not necessary to be notified
- **C** Shall nevertheless submit a FP and notify changes made thereto to the ATS unit providing that service

Answer: C.

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3 The white dumb-bell with black perpendicular bar indicates that:

- **A** Taxiing need not be confined to the taxiways
- **B** Glider flying is performed outside the landing area
- **C** Landing, take-off and taxiing is allowed on runway and/or taxiway only

Answer: A.

4 Your aircraft is intercepted by a military aircraft. The signals given by this aircraft conflict with ATC instructions. You should:

- **A** Select code 7500 on your transponder
- **B** Follow ATC instructions
- **C** Follow the instructions of the intercepting aircraft

Answer: C.

5 An aircraft which is being subject to unlawful interference ("hijacked") and is forced to divert from the cleared track or cruising level without being able to communicate with ATS shall try to:

- **A** Fly the emergency triangle
- **B** Continue at a level that differs from the normal cruising levels by 500 ft (where the vertical separation minimum is 1,000 ft) or by 1,000 ft (where it is 2,000 ft)
- **C** As soon as possible commence emergency descent in order to minimize the difference between cabin pressure and outside pressure

Answer: B. Annex 2 Attachment B para 2.2 c): offset of 500 ft where the vertical separation minimum is 1,000 ft, and 1,000 ft where it is 2,000 ft. The old "above / below FL290" wording is not exact in RVSM airspace.

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6 An aircraft manoeuvring in an airport's circuit receives a series of red flashes from the control tower. This signifies that the aircraft must:

- **A** Not land for the moment regardless of previous instructions
- **B** Return to land and that clearance to land will be communicated in due course
- **C** Not land because the airport is not available for landing

Answer: C.

7 Whilst flying in an aerodrome's traffic circuit, an aircraft receives a series of green flashes from the tower. The aircraft:

- **A** Is cleared to land
- **B** Must land immediately and clear the landing area
- **C** Must come back to land and the landing clearance will be sent in due time

Answer: C.

8 Except when a clearance is obtained from an ATC unit, a VFR flight cannot enter or leave a control zone when ceiling is less than:

- **A** 2,000 ft or visibility is less than 5 km
- **B** 1,000 ft or visibility is less than 5 km
- **C** 1,500 ft (450m) or visibility is less than 5 km

Answer: C.

9 Which of the following flights has the greatest priority to land?

- **A** Emergency aircraft
- **B** Military aircraft
- **C** VIP (Head of state) aircraft

Answer: A.

10 An aircraft flying above the sea between 4,500 ft MSL and 9,000 ft MSL outside controlled airspace under VFR, must remain on principle at least:

- **A** 1,500m horizontally, 1,000 ft vertically from clouds; 8 km visibility
- **B** Clear of clouds and in sight of the surface; 8 km visibility
- **C** 1,500m horizontally, 1,000 ft vertically from clouds; 5 km visibility

Answer: C.

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11 The VMC minima for an airspace classified as "B" above 10,000 ft MSL are:

- **A** 2,000m horizontally, 1,000 ft vertically from clouds; 8 km visibility
- **B** 1,500m horizontally, 1,000 ft vertically from clouds; 8 km visibility
- **C** 1 nautical mile horizontally and 1,000 ft vertically from clouds; 8 km visibility

Answer: B.

12 The VMC minima for an airspace classified as "G" above 10,000 ft MSL are:

- **A** 1,500m horizontally, 1,000 ft vertically from clouds; 8 km visibility
- **B** 1 nautical mile horizontally and 1,000 ft vertically from clouds; 8 km visibility
- **C** 1 nautical mile horizontally and 1,000 ft vertically from clouds; 5 km visibility

Answer: A.

13 Minimum vertical distance from cloud for VFR flight within controlled airspace is:

- **A** 1,000 ft
- **B** 500 ft
- **C** 1,000 m

Answer: A.

14 Green flashes from Tower, this signal means that the aircraft:

- **A** May continue to taxi towards the take-off area
- **B** Must return to its point of departure
- **C** Must stop

Answer: A.

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15 Which of the following actions shall be taken in case of a controlled flight deviates from the track?

- **A** Adjust the heading of the aircraft to regain track as soon as possible
- **B** If VMC, maintain this condition, waiting for the ATC instructions
- **C** Notify ATC of the new track immediately and comply with instructions

Answer: A.

16 An aircraft is flying under IFR in an area where the visibility is unlimited and the sky is clear (free of clouds), when it totally loses radio communications. The procedure to be followed is:

- **A** Descend to En-route Minimum Safe Altitude and join closest airfield open to IFR operations
- **B** Land on the closest appropriate aerodrome, then advise Air Traffic Services of landing
- **C** Continue flight onto destination, complying with last received clearances in the filed flight plan

Answer: B.

17 While taxiing an aircraft receives the following light signal from the control tower: series of red flashes. This signal means that the aircraft:

- **A** Must vacate the landing area in use
- **B** May continue to taxi to the take-off area
- **C** Must stop

Answer: A.

18 An aircraft intercepted by another aircraft shall immediately attempt to establish radio communication with the intercepting aircraft on the following frequencies:

- **A** 121.5 MHz – 282.8 MHz
- **B** 121.5 MHz – 125.5 MHz
- **C** 121.5 MHz – 243 MHz

Answer: C.

19 If radio contact with the intercepting aircraft is established but communication on a common language is not possible, which phrase should be pronounced by the intercepted aircraft to communicate that he is unable to comply with the instructions received?

- **A** UNABLE TO COMPLY
- **B** CAN NOT
- **C** CAN NOT COMPLY

Answer: B.

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20 If radio communication is established during an interception but communications in a common language is not possible, which phrase should be pronounced by the intercepting aircraft to request the intercepted aircraft to descend for landing?

- **A** Descend for landing
- **B** Descend
- **C** You land

Answer: B.

21 A signalman will ask the pilot to apply parking brakes by the following signals:

- **A** Arms down, palms facing inwards, moving arms from extended position inwards
- **B** Crossing arms extended above his head
- **C** Raise arm and hand, with fingers extended, horizontally in front of body, then clinch fist

Answer: C.

22 An aircraft shall display, if so equipped, an anti-collision light:

- **A** Outside the daylight-period at engine-start. During the daylight-period this is not applicable
- **B** On the ground when the engines are running
- **C** While taxiing, but not when it is being towed

Answer: B.

23 A red flare addressed to a flying aircraft means:

- **A** Dangerous airfield. Do not land
- **B** Not withstanding any previous instructions, do not land for the time being
- **C** Come back and land

Answer: B.

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24 A double white cross displayed horizontally in the signal area means:

- **A** An area unit for the movement of aircraft
- **B** Need special precautions while approaching for landing
- **C** The aerodrome is being used by gliders and that glider flights are being performed

Answer: C.

25 An aircraft is considered to overtake another if it approaches the other aircraft from the rear on a line forming an angle of less than:

- **A** 50 degrees with the plane of symmetry of the latter
- **B** 70 degrees with the plane of symmetry of the latter
- **C** 60 degrees with the plane of symmetry of the latter

Answer: B.

26 During an IFR flight in VMC in controlled airspace you experience a two-way radio communication failure. You will:

- **A** Land at the nearest suitable aerodrome maintaining VMC and inform ATC
- **B** Select A7600 and continue according current flightplan to destination
- **C** Land at the nearest suitable aerodrome and inform ATC

Answer: A.

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27 When does FLIGHT TIME end?

- **A** When the aeroplane first stops after landing to disembark passengers
- **B** At touchdown
- **C** At engine shut down when the aeroplane has stopped in the parking bay

Answer: C.

28 The aerodrome at Calicut is not declared suitable for night operations. If the sunrise is at 0550, the earliest time by which an VFR aircraft operation at the station can take place is:

- **A** 0530
- **B** 0520
- **C** 0550

Answer: A.

29 What is the meaning of AGL?

- **A** Altitude of ground level
- **B** Aerodrome ground level
- **C** Above ground level

Answer: C.

30 Which of the following defines flight visibility?

- **A** Visibility determined in flight not obscured by cloud, dust, haze or precipitation
- **B** The ability to determine in the air the distance and identity of unlighted objects by day and lighted objects by night
- **C** The forward visibility from the cockpit of an aircraft in flight

Answer: C.

31 Who shall have the final authority over the disposition of the aircraft?

- **A** The Authority
- **B** The Owner
- **C** The Commander

Answer: C.

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32 When two aircraft are approaching head on, and there is a danger of collision:

- **A** Both aircraft alter heading to port
- **B** Both aircraft alter course to the right (starboard side)
- **C** The smaller aircraft alters heading to port

Answer: B.

33 A revised estimate shall be notified if the ETA destination passed by aircraft varies by:

- **A** 3 minutes
- **B** 2 minutes
- **C** 5 minutes

Answer: B.

34 An aircraft that has the right of way shall maintain its:

- **A** Heading and speed
- **B** Right of progress
- **C** Course, speed and height

Answer: A.

35 On aerodromes aircraft taxiing on the manoeuvring area of an aerodrome shall give way to:

- **A** Other converging aircraft
- **B** Aircraft taking off or about to take off
- **C** Other vehicles and pedestriants

Answer: B.

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36 What is Special VFR?

- **A** A VFR procedure to enable an aeroplane to transit a control zone or area in IMC without compliance with IFR
- **B** A VFR flight cleared by ATC to operate within a CTR in meteorological conditions below VMC
- **C** Any flight cleared by ATC to operate in conditions less than VMC in which the pilot is required to remain clear of cloud and in sight of the surface

Answer: B.

37 Which is the correct order of priority:

- **A** Gliders do not give way to balloons
- **B** Power-driven heavier-than-air aircraft shall give way to airships, gliders and balloons
- **C** Gliders shall give way to airships and balloons

Answer: B.

38 What is the minimum flight altitude permitted over towns and settlements and populated areas?

- **A** The altitude which permits the aircraft to land safely in the event of an engine failure
- **B** 1,000 ft above the highest obstacle within 600 m of the aircraft position
- **C** 500 ft

Answer: B.

39 A controlled flight is requested to inform the appropriate ATC unit whenever the average True Air Speed at cruising level varies or is expected to vary from that given in the flight plan by plus or minus:

- **A** 5%
- **B** Mach 0.02
- **C** 10%

Answer: B.

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40 An aircraft operated on, or in the vicinity of, an aerodrome shall whether or not within an ATZ. Which statement below is incorrect? (I: Observe other aerodrome traffic; II: Conform with or avoid traffic pattern; III: Make all turns to the RIGHT unless instructed; IV: Land and take-off into wind unless otherwise)

- **A** All statements except II) are correct
- **B** All statements except I) are correct
- **C** All statements except III) are correct[Turns are to the LEFT unless otherwise instructed]

Answer: C.

41 Aircraft "A" with an ATC clearance is flying in VMC conditions within a control area. Aircraft "B" with no ATC clearance is approaching at approximately the same altitude and on a converging course. Which is the right of way?

- **A** Aircraft "A" regardless of the direction which "B" is approaching
- **B** Aircraft "B" if "A" is on its left
- **C** Aircraft "A" if "B" is on its right

Answer: B.

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42 Which action shall be taken by an aircraft in the traffic pattern of an aerodrome, experiencing radio failure to indicate difficulties which compel it to land without requiring immediate assistance?

- **A** Switching on and off four times the landing lights
- **B** Switching on and off three times the landing lights
- **C** The repeated switching on and off of the landing lights

Answer: C.

43 Aircraft taxiing in the manoeuvring area must give way to:

- **A** Follow-me vehicles
- **B** Landing traffic
- **C** Departing and landing traffic

Answer: C.

44 Two or more white crosses, displayed on runways and taxiways, indicate that:

- **A** The threshold has been displaced
- **B** The section of the runway or taxiway is unfit for aircraft movement
- **C** The runway or taxiway should be used with caution

Answer: B.

45 Which manoeuvre shall be executed by an intercepting aircraft if the pilot wants to communicate to the intercepted aircraft "YOU MAY PROCEED"?

- **A** Rocking wings and flashing the navigational lights
- **B** Executing a climbing turn of 90 degrees or more without crossing the line of flight of the intercepted aircraft
- **C** Circling the intercepted aircraft in a clock-wise pattern

Answer: B.

46 Semi-circular system sectors are:

- **A** From 000° to 179° and 180° to 359°
- **B** From 000° to 089°, 090° to 179°, 180° to 269° and 270° to 359°
- **C** From 090° to 260° and 270° to 089°

Answer: A.

47 While on IFR flight, a pilot has an emergency which causes a deviation from an ATC clearance. What action must be taken?

- **A** Squawk 7700
- **B** The appropriate ATC unit shall be notified of the action taken as soon as circumstances permit
- **C** Request an amended clearance or cancel the IFR flightplan

Answer: B.

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48 An airborne flight plan must be submitted:

- **A** 30 minutes before the intended point of entry into a CTA or advisory area
- **B** 10 minutes before the intended point of entry into a CTA or advisory area
- **C** 60 minutes before the intended point of entry into a CTA or advisory area

Answer: B.

49 IFR flight in VMC, weather changes to IMC:

- **A** Fly VMC and land at nearest convenient aerodrome
- **B** Fly according to flight plan
- **C** Change to VFR plan

Answer: B.

50 PIC of an aeroplane shall:

- **A** Be responsible for safe conduct of navigation
- **B** Be responsible for safe conduct of flight at all times
- **C** All above is correct

Answer: C.

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51 On the ground, when two aircraft are approaching head on:

- **A** Both airplanes must stop or where practicable alter course to the right to keep well clear
- **B** Both should declare an urgency
- **C** Traffic patrol has to be informed

Answer: A.

52 No aircraft shall be flown over the congested area of cities, towns or settlements unless at a height that will permit, in the event of an emergency a landing to be made safely. Exceptions to this rule are:

- **A** Take-off and landing and permission from the appropriate authority
- **B** Permission from the appropriate authority
- **C** Take-off and landing

Answer: A.

53 Which type of service is provided in Class F airspace:

- **A** Air traffic control service
- **B** Air traffic advisory service and FIS
- **C** Air traffic advisory service

Answer: B.

54 When is deviation from the rules of the air permitted?

- **A** Only in the interests of flight safety
- **B** Only when it is economically advantageous
- **C** Anytime the PIC thinks it is necessary

Answer: A.

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55 One of the following statements about aircraft ground movement is correct:

- **A** A taxiing aircraft has priority over a vehicle towing an aircraft
- **B** A vehicle towing an aircraft has priority over a taxiing aircraft
- **C** Two airplanes approaching head-on will alter course to the left

Answer: A.

56 While flying at night another aircraft reports that you are on his 100 degrees relative bearing. In that case you should see his:

- **A** Red navigation light
- **B** Green navigation light
- **C** Green and white navigation lights

Answer: B.

57 The signal from pilot to marshaller which means - brakes applied - is:

- **A** Fist clenched in front of the face then fingers extended
- **B** Hands crossed in front of the face, palms outwards, then move arms outwards
- **C** Right arm and hand with fingers extended in front of face, then clench fist

Answer: C.

58 What does a red flashing light from the control tower to an aircraft on the manoeuvring area mean?

- **A** Clear to taxi
- **B** Return to the start point
- **C** Move clear of the landing area

Answer: C.

59 A series of red flashes sent to an aircraft in flight means:

- **A** Return to the airfield and land
- **B** Airfield is unsafe, do not land
- **C** Notwithstanding any other instructions do not land at the moment

Answer: B.

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60 A white dumbbell with a black bar spaced perpendicularly indicates:

- **A** All aircraft are required to land, takeoff and taxi on taxiways and runways only
- **B** Are required to land and takeoff on runways only
- **C** A white dumbbell when displayed requires a/c to land and takeoff and taxi on taxiways and runways only

Answer: B.

61 A red square panel with a yellow strip along each diagonal, displayed in the signals area of an aerodrome indicates:

- **A** Aerodrome is unfit for aircraft movements, and landing are prohibited
- **B** Exercise special care when landing
- **C** Land on paved surface only

Answer: A.

62 Two a/c of same category converging...

- **A** Both will turn to right
- **B** Aircraft on right will turn right
- **C** Aircraft on the left will give way

Answer: C.

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63 A signalman will ask the pilot to apply parking brakes by the following signals:

- **A** Raise arm and hand, with fingers extended, horizontally in front of body, then clench fist
- **B** Arms down, palms facing inwards, moving arms from extended position inwards
- **C** Horizontally moving his hands, fingers extended, palms toward ground

Answer: A.

64 An aircraft is following a track of 179° (M) on a VFR plan. The choice of flight levels available to the pilot are:

- **A** 50, 70, 90, 150, 170, 190, 210, 230, 250, 270, 290
- **B** 55, 75, 95, 115, 135, 155, 175, 195, 215, 235, 255, 275
- **C** 50, 70, 90, 110, 130, 160, 180, 200, 220, 240, 260, 280

Answer: B.

65 After an aircraft has been intercepted in flight, the intercepted aircraft is rocking its wings. This means:

- **A** Will comply
- **B** NO
- **C** You are not to enter this airspace

Answer: A.

66 All flights above _____ flight levels are to be cleared IFR irrespective of weather conditions:

- **A** F 280
- **B** F 150
- **C** F 140

Answer: B.

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67 A flashing intermittent red light from the ground to an aircraft in the air means:

- **A** Continue for another approach
- **B** Go-around
- **C** Aerodrome unsafe, do not land

Answer: C.

68 When a pilot raises his arms extended palms facing outwards and moves his hands inwards to cross in front of the face this means:

- **A** Remove chocks
- **B** Insert chocks
- **C** Brakes released

Answer: B.

69 While taxiing, an aircraft receives from the airport controller the following light signal: a series of green flashes. This signal means that the aircraft:

- **A** Must stop
- **B** May continue to taxi towards the take-off area
- **C** Must return to its point of departure

Answer: B.

70 A black letter "C" on a yellow background indicates:

- **A** The location of the control tower
- **B** The location where visiting pilots should report
- **C** Rest-room facilities for visiting pilots

Answer: B.

71 At an aerodrome special VFR may be authorized when:

- **A** Visibility falls below 5 km or cloud ceiling is less than 1,500 feet
- **B** Visibility falls below 8 km or cloud ceiling is less than 1,500 feet
- **C** Visibility falls below 8 km or cloud ceiling is less than 2,500 feet

Answer: A.

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72 A special VFR clearance may be obtained from ATC for the following airspaces:

- **A** CTR, TMA and AWY
- **B** CTR and TMA
- **C** CTR only

Answer: C.

73 If the sustained Mach number/true airspeed at cruising level varies by plus or minus _____ true airspeed or more from the current flight plan, the appropriate air traffic services unit shall be so informed:

- **A** Mach 0.2
- **B** 5%
- **C** 19 km/h (10 kt)

Answer: C.

74 On a VFR flight, your magnetic track is 005°, the magnetic heading 355°. Which of the following flight level is correct?

- **A** FL 70
- **B** FL 55
- **C** FL 60

Answer: B.

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75 The VMC minima for an airspace classified as "G" above 10,000 feet MSL are:

- **A** 1,500m horizontally and 1,000 feet vertically from clouds; 8 km visibility
- **B** 1 nautical mile horizontally and 1,000 feet vertically from clouds; 5 km visibility
- **C** 1 nautical mile horizontally and 1,000 feet vertically from clouds; 8 km visibility

Answer: A.

76 Which provisions on a VFR-flight in Class E airspace are CORRECT?

- **A** Service provided: Traffic Information as far as practical; ATC Clearance: required
- **B** Service provided: Traffic Information as far as practical; ATC Clearance: not required
- **C** Service provided: Air Traffic Control Service; ATC Clearance: required

Answer: B.

77 If you are flying IFR in IMC conditions and you experience a total communications failure, you should:

- **A** Continue the flight according to flight plan
- **B** Land at the nearest suitable aerodrome and report to ATS
- **C** Try to contact another aircraft for relay

Answer: A.

78 When flying on an airway on a heading of 255(M) the correct flight level will be:

- **A** Usually EVEN
- **B** Usually ODD
- **C** Always EVEN

Answer: A.

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79 You are outside controlled airspace on a VFR flight above 3,000 ft. Your distance from the cloud should be:

- **A** Clear of cloud and in sight of the surface
- **B** 1,000 ft vertically and 1.5 km horizontally
- **C** 2,000 ft and 3 NM horizontally

Answer: B.

80 In areas where a separation minimum of 1,000 ft is applied up to FL 410, authorisation for VFR flight shall not be granted above which flight level?

- **A** FL 200
- **B** FL 290
- **C** FL 240

Answer: B.

81 When an aircraft is operating in class F airspace below 900 m AMSL the minimum horizontal distance from cloud for VFR flight is:

- **A** 5 km
- **B** 1,000 ft
- **C** Clear of cloud and in sight of the surface

Answer: C.

82 SVFR may be authorized when the ground visibility is not less than:

- **A** 1,500 m
- **B** 1,800 m
- **C** 1,000 m

Answer: A.

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83 Which of the following aerodrome ground signals displayed in the signals area indicates that glider flying is in progress?

- **A** A white double cross
- **B** A white dumb-bell
- **C** A black letter C on a yellow background

Answer: A.

84 In areas where a vertical separation minimum (VSM) of 300 m (1,000 ft) is applied between FL 290 and FL 410 inclusive an aircraft on a magnetic track of 350 would be expected to fly at:

- **A** FL 410
- **B** FL 405
- **C** FL 400

Answer: C.

85 Above flight level FL 290 (non-RVSM) the vertical flight separation between aircraft on the same direction is:

- **A** 3,000 ft
- **B** 4,000 ft
- **C** 1,500 ft

Answer: B.

86 Aircraft shall not be flown in formation except:

- **A** Formation flying is not permitted in India for civil aircraft
- **B** By pre-arrangement among the pilots taking part
- **C** By pre-arrangement among the PICs taking part

Answer: A.

87 If the ground visibility is reported 1,000 m, can a special VFR flight take off from an aerodrome in a control zone?

- **A** Yes, provided the cloud ceiling is higher than 500 ft
- **B** No
- **C** Yes, provided the pilot remains in visual contact with the ground

Answer: B.

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88 A VFR flight when flying inside an ATS airspace classified as C has to maintain the following minima of flight visibility and distance from clouds:

- **A** 5 km at or above 3,050m (10,000ft) AMSL 1,500m horizontal and 300m vertical from clouds
- **B** 8 km at or above 3,050m (10,000ft) AMSL, and clear of clouds
- **C** 8 km at or above 3,050m (10,000ft) AMSL 1,500m horizontal and 300m vertical from clouds

Answer: C.

89 When, in airspace where VFR are permitted, the pilot in command of an IFR flight wishes to continue his flight in accordance with visual flight rules, until the destination is reached: (1 - must inform control unit "cancel IFR"; 2 - must request and obtain clearance; 3 - may request IFR FP changed to VFR FP; 4 - FP automatically becomes VFR FP)

- **A** 2, 4
- **B** 1, 3
- **C** 1, 4[Cancel IFR notification is required; FP becomes VFR FP]

Answer: B. 1, 3 [Annex 2 5.1.3.1: notify that the IFR flight is cancelled and communicate the changes to the flight plan.]

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90 In the event of a delay of a controlled flight, the submitted flightplan should be amended or cancelled and a new flightplan submitted when the delay is:

- **A** 30 minutes in excess of the estimated time of departure
- **B** 30 minutes in excess of the estimated time off blocks
- **C** 60 minutes in excess of the estimated time of departure

Answer: B.

91 Changing of flight rules from IFR to VFR is possible:

- **A** If the commander so requests
- **B** If instructed by ATC so long as VMC is forecasted during the next 30 minutes
- **C** Only when leaving controlled airspace

Answer: A.

92 ADC Validity Period is:

- **A** (-) 15 minutes to (+) 45 minutes of EOBT in RPL/FPL
- **B** (+) 60 minutes of EOBT in RPL/FPL
- **C** (-) 15 minutes to (+) 30 minutes of EOBT in RPL/FPL

Answer: A.

93 Flight Plan is to be filed at least?

- **A** One hour before EOBT
- **B** Three hours before EOBT
- **C** One hour before TOBT

Answer: B.

94 In the event of a proposed delay of _____ or more to the departure time for a flight plan when a flight plan has been previously filed, the Pilot-in-Command or his representative will be required to notify ATC of the revised ETD:

- **A** 60 minutes
- **B** 30 minutes
- **C** 15 minutes

Answer: C.

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95 If you see red and white lights of the aircraft ahead, it means:

- **A** The aircraft is moving from left to right
- **B** The aircraft is moving from right to left
- **C** The aircraft is flying near parallel to you

Answer: B.

96 If you see Green and white lights of the aircraft ahead, it means:

- **A** The aircraft is moving from left to right
- **B** The aircraft is moving from right to left
- **C** The aircraft is flying near parallel to you

Answer: A.

97 ADC validity for non-scheduled domestic flights is:

- **A** (-) 15 minutes to (+) 45 minutes of EOBT in RPL/FPL
- **B** (+) 60 minutes of EOBT in RPL/FPL
- **C** (-) 15 minutes to (+) 3 hours of EOBT

Answer: C.

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MEMORY AID — VMC MINIMA

Above 10,000 ft AMSL: **8 km vis | 1500m H | 1000ft V**
3000–10,000 ft AMSL: **5 km vis | 1500m H | 1000ft V**
Below 3000 ft (Class A-E): **5 km | 1500m H | 1000ft V**
Below 3000 ft (Class F/G): **5 km | Clear of cloud | Surface in sight**

STUDENT TIP — RED/GREEN MEMORY AID

"If you see RED — you could be DEAD" — The red light is on the other aircraft's port (left) side. This means the other aircraft is crossing from your right to your left. Under converging rules, the one with the other on its RIGHT gives way — so that's **YOU**. Give way!

Navigation lights CANNOT be seen from the rear — white only means the aircraft is ahead of or flying away from you.

SEMI-CIRCULAR RULE — CRUISING LEVELS (MNEMONIC DIAGRAM)

Chapter 4 — Air Traffic Services

Practice Questions

1 A lower limit of a Control Area shall be established at a height above the ground level or water of not less than:

- **A** 150 metres
- **B** 200 metres
- **C** 300 metres

Answer: B.

2 The Approach Control Service is an air traffic control service:

- **A** provided for the arriving and departing controlled flights
- **B** for IFR flights arriving and departing
- **C** provided for IFR traffic within a Control Zone

Answer: A.

3 The units providing Air Traffic Services are:

- **A** Area Control Center – Advisory Center – FIC – Approach Control Office and Tower
- **B** Area Control Center – FIC – Approach Control Office – Aerodrome Control Tower and ATS reporting office
- **C** Area Control Center – Approach Control Office and Aerodrome Control Tower only

Answer: B.

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4 Air traffic control service is provided for the purpose of:

- **A** Avoiding collisions between all aircraft and maintaining an orderly flow of air traffic
- **B** Preventing collisions between aircraft, between aircraft and obstacles on the manoeuvring area and expediting and maintaining an orderly flow of air traffic
- **C** Preventing collisions between controlled air traffic and expediting and maintaining an orderly flow of air traffic

Answer: B.

5 To perform a VFR flight in airspace classification E:

- **A** a clearance and/or two-way radio communication is required
- **B** a clearance is required
- **C** two-way radio communication is not required

Answer: C.

6 Which statement is correct?

- **A** The lower limit of a UIR may coincide with an IFR cruising level
- **B** The lower limit of a TMA shall be established at a height of at least 700ft AGL
- **C** The upper limit of a CTR shall be established at a height of at least 3000ft AMSL

Answer: B.

7 What is the speed limit (IAS) in an airspace class E?

- **A** 250kt VFR and IFR, all levels
- **B** 250kt for IFR and VFR up to FL 100
- **C** 250 kt only for IFR up to FL 100

Answer: B.

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8 The speed limitation for IFR flights inside ATS airspace classified as C, when flying below 3050m (1000ft) AMSL, is:

- **A** 250kt IAS
- **B** Not applicable
- **C** 250kt TAS

Answer: B.

9 RCP 10 means:

- **A** 10 is the number of seconds it takes for an instruction to travel from the ground to aircraft and acknowledgment back to the ground
- **B** 10 is the maximum number of minutes it takes for an instruction to be executed by an aircraft
- **C** Aircraft must communicate with ATC every 10 minutes

Answer: A.

10 An ATS airspace where IFR and VFR flights are permitted, all flights are subject to air traffic control service and are separated from each other is classified as:

- **A** Airspace C
- **B** Airspace B
- **C** Airspace D

Answer: B.

11 The speed limitation for both IFR flights and VFR flights inside ATS airspace classified as B, when flying below 3050m (10000ft) AMLS, is:

- **A** 260kt IAS
- **B** Not applicable
- **C** 250kt IAS

Answer: B.

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12 Air traffic services unit clocks and other time recording devices shall be checked as necessary to ensure correct time within plus or minus:

- **A** 1 minute of UTC at all times
- **B** 15 seconds of UTC at all times
- **C** 30 seconds of UTC at all times

Answer: C.

13 The following airspace falls under Kolkata FIR?

- **A** VED
- **B** VEB
- **C** VOR

Answer: A.

14 The VMC minima for a VFR flight inside an ATS airspace classified as B, is:

- **A** 8km visibility when at or above 3050m (10000ft) AMSL and clear of clouds
- **B** 8km visibility when at or above 3050m (10000ft) AMSL, and 1500m horizontal and 300m vertical from clouds
- **C** 5NM visibility when below 3050m (10000ft) AMSL, 1500m horizontal and 300m vertical from cloud

Answer: B.

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15 A VFR flight when flying inside an ATS airspace classified as B has to maintain the following minima of flight visibility and distance from clouds:

- **A** 5km visibility, 1500m horizontal and 300m vertical from clouds
- **B** 8km below 3050m (10000ft) AMSL, 1500m horizontal and 300m vertical from clouds
- **C** 5km below 3050m (10000ft) AMSL and clear of clouds

Answer: A.

16 An ATS airspace where IFR and VFR flights are permitted, all flights are subject to ATC service and IFR flights are separated from other IFR flights and from VFR flights; VFR flights are separated from IFR flights and receive traffic information in respect of other VFR flights, is classified as:

- **A** Airspace B
- **B** Airspace E
- **C** Airspace C

Answer: C.

17 Which condition is requested so that an aerodrome may be considered controlled?

- **A** The aerodrome shall be provided with a Control Tower
- **B** The aerodrome shall be located within a CTR and provided with a Control Tower
- **C** The aerodrome shall be located within a Control Zone

Answer: A.

18 An ATS airspace where IFR and VFR flights are permitted and all flights are subject to ATC service. IFR flights are separated from other IFR flights and receive traffic information in respect of VFR flights. VFR flights receive traffic information in respect of all other flights, is classified as:

- **A** Airspace B
- **B** Airspace D
- **C** Airspace A

Answer: B.

19 An ATS airspace where IFR and VFR flights are permitted are subject to ATC Service and are separated from other IFR flights. All flights receive traffic information as far as practical, is classified as:

- **A** Airspace D
- **B** Airspace A
- **C** Airspace E

Answer: C.

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20 An ATS airspace where IFR and VFR flights are permitted, all participating IFR flights receive an air traffic advisory service and all flights receive flight information service if requested, is classified:

- **A** Airspace F
- **B** Airspace E
- **C** Airspace D

Answer: A.

21 Which statement regarding approach control service is correct?

- **A** Approach control have to advise aircraft operators about substantial delays in departure when expected to exceed 45 minutes
- **B** During a visual approach an aircraft is maintaining its own separation
- **C** If it is anticipated that an aircraft has to hold for 30 minutes or more, an Expected Approach Time will be transmitted by the most expeditious means to the aircraft

Answer: C.

22 Which statement regarding aerodrome control service is correct?

- **A** An aircraft entering the traffic circuit without permission of ATC, will be cleared to land if this is desirable
- **B** Suspension of VFR operations cannot be initiated by the aerodrome controller
- **C** ATC permission is required for entering the apron with a vehicle

Answer: A.

23 In ATS route designator UA1Y what does U mean:

- **A** Upper
- **B** Undirectional
- **C** Uniform

Answer: A.

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24 Flight information service shall be provided to all aircraft which are likely to be affected by the information and which are:

- **A** Provided with the air traffic control services and otherwise known to the relevant air traffic service units
- **B** Known to the relevant air traffic services units by a filed flight plan
- **C** Provided with air traffic control services, only

Answer: A.

25 Alerting service shall be provided:

- **A** For all controlled flight, to any aircraft known or believed to be subject of unlawful interference, and in so far as practicable to all aircraft having filed a flight plan or otherwise known to the ATS
- **B** For all aircraft provided with air traffic control services, only
- **C** To any aircraft known or believed to be subject of unlawful interference, only

Answer: A.

26 The speed limitation for IFR flights inside ATS airspace classified as E, when flying below 3050m (10000ft) AMSL, is:

- **A** 250kt TAS
- **B** 250kt IAS
- **C** Not applicable

Answer: B.

27 Air Traffic Service unit means:

- **A** Air Traffic Control units and Air Services reporting offices
- **B** Flight Information Centers and Air Services reporting offices
- **C** Air Traffic Control units, Flight Information Centers or Air Services reporting offices

Answer: C.

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28 A controlled airspace extending upwards from a specified limit above the earth is:

- **A** Advisory airspace
- **B** Control area
- **C** Flight Information Region

Answer: B.

29 Tirupati aerodrome designator is?

- **A** VATP
- **B** VOTP
- **C** VITP

Answer: B.

30 Aerodrome traffic is:

- **A** All traffic on the manoeuvring area and flying in the vicinity of an aerodrome
- **B** All traffic in the aerodrome circuit
- **C** All traffic on the manoeuvring area

Answer: A.

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31 Which of the following Annexes to the Chicago convention contains international standards and recommended practices for air traffic services (ATS)?

- **A** Annex 6
- **B** Annex 14
- **C** Annex 11

Answer: C.

32 An information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather phenomena which may affect safety of low-level aircraft operations is:

- **A** An En-Route Met Report
- **B** A NOTAM
- **C** An AIRMET information

Answer: C.

33 Regarding Aerodrome Flight Information (AFIS):

- **A** Its purpose is to supply ATC services but it is not a state organisation
- **B** It can only supply limited services to the users and under no circumstances may it supply ATC services
- **C** It has the same privileges as an ATC organisation but its activity is neither continuous nor regular

Answer: B.

34 Air Traffic Control service shall be provided to:

- **A** All IFR flights in class A, B, C, D, E and F airspaces
- **B** All VFR flights in class B, C, D and E airspaces
- **C** To all aerodrome traffic at controlled aerodromes

Answer: C.

35 Flight Information Region (FIR) is an airspace within which the following services are provided:

- **A** Flight Information Service and Alerting Service
- **B** Flight Information Service and Advisory Service
- **C** Flight Information Service, Alerting Service and Advisory Service

Answer: A.

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36 Control Area (CTA) is defined as follows:

- **A** A controlled airspace extending upwards from a specified limit above the earth
- **B** A controlled airspace extending upwards from the surface of the earth to a specified limit
- **C** A controlled airspace extending upwards from a height of 900ft above the earth

Answer: A.

37 An air traffic control unit:

- **A** May ask an aircraft to temporarily change its callsign for safety reasons when there is a risk of confusion between two or more similar callsigns
- **B** May not ask an aircraft to change its callsign after accepting the flight plan
- **C** Must not ask an aircraft to change its callsign

Answer: A.

38 Required Navigation Performance (RNP) shall be prescribed:

- **A** By regional air navigation agreements
- **B** By states but not on the basis of regional air agreements
- **C** By states on the basis of regional air navigation agreements

Answer: C.

39 A Control Zone shall extend laterally to at least:

- **A** 10 miles from the centre of the aerodrome in the direction from which approaches may be made
- **B** 5 nautical miles from the centre of the aerodrome in the direction from which approaches may be made
- **C** 20 miles from the centre of the aerodrome in the direction from which approaches may be made

Answer: B.

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40 G245ZF is a designator for:

- **A** a SID
- **B** a STAR
- **C** an ATS Route

Answer: C.

41 An ATS airspace where IFR and VFR are permitted and receive flight information service if requested, is classified as:

- **A** Airspace G
- **B** Airspace E
- **C** Airspace C

Answer: A.

42 Concerning RNP types, the indication RNP 4 represents a navigation accuracy of:

- **A** Plus or minus 4 NM on a 95% containment basis
- **B** Plus or minus 4 NM on a 98% containment basis
- **C** Plus or minus 4 NM on a 90% containment basis

Answer: A.

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43 Area Control Centers issue clearances for the purpose of:

- **A** Achieving separation between IFR flights
- **B** Achieving separation between controlled flights
- **C** Providing flight information Service

Answer: B.

44 Clearances will be issued by an ATC unit for the purpose of:

- **A** Achieving separation between controlled flights
- **B** Providing alerting services
- **C** Providing advisory services

Answer: A.

45 ATIS broadcast:

- **A** Shall only be transmitted on a discrete VHF frequency
- **B** Shall not be transmitted on the voice channel of an ILS
- **C** Shall not be transmitted on the voice of VOR

Answer: B.

46 Whenever ATIS is provided, the preparation and dissemination of the ATIS message shall be the responsibility of:

- **A** Both air traffic services and the meteorological office
- **B** The meteorological office serving the aerodromes
- **C** The air traffic services

Answer: C.

47 Whenever ATIS is provided, the broadcast information shall be updated:

- **A** As prescribed by the meteorological office
- **B** Immediately a significant change occurs
- **C** As prescribed by the state

Answer: B.

48 The ATIS broadcast message should, whenever practicable, not exceed:

- **A** 30 seconds
- **B** 1 minute
- **C** 2 minutes

Answer: A.

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49 ATIS broadcast messages containing departure and arrival information should include cloud cover, when the clouds are:

- **A** Below 900m (3000ft) or below the highest minimum sector altitude, whichever is the greater
- **B** Below 2000m (600ft) or below the highest minimum sector altitude, whichever is the greater
- **C** Below 1500m (5000ft) or below the highest minimum sector altitude, whichever is the greater

Answer: C.

50 Flight Information Service shall be provided to aircraft in order to avoid collision hazards when operating in airspace classes:

- **A** F and G only
- **B** C, D, E, F and G
- **C** A, B, C, D, E, F and G

Answer: B.

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51 The lateral limit of CTR shall extend to at least ___ NMs from the center of aerodrome in directions from where approaches may be made:

- **A** 25
- **B** 10
- **C** 5

Answer: C.

52 At radio communications, "Distress" differs from "Urgency" because in the first case:

- **A** The aeroplane has suffered damages which impair its fitness to fly
- **B** There is a serious and imminent danger requiring immediate assistance
- **C** The aeroplane will not be able to reach a suitable aerodrome

Answer: B.

53 The Alerting Service is provided by:

- **A** The Area Control Centers
- **B** The ATS unit responsible for the aircraft at that moment
- **C** Only by ATC units

Answer: B.

54 The phases related to an aircraft in emergency or believed in emergency are:

- **A** Uncertainty phase, distress phase, urgency phase
- **B** Uncertainty phase, alert phase, distress phase
- **C** Uncertainty phase, urgency phase, distress phase

Answer: B.

55 When an aircraft is experiencing difficulties, triggering of the alert phase is the responsibility of:

- **A** Search and rescue coordination centers
- **B** Control centers only
- **C** Air traffic control and flight information centers

Answer: C.

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56 What type of flight is allowed in class A airspace?

- **A** IFR only
- **B** IFR and VFR
- **C** IFR and special VFR

Answer: A.

57 Alert phase is defined as follows:

- **A** A situation where an apprehension exists as to the safety of an aircraft and its occupants
- **B** An emergency event in which an aircraft and its occupants are considered to be threatened by a danger
- **C** A situation related to an aircraft and its occupants are considered to be in a state of emergency

Answer: A.

58 A situation in which apprehension exists as to the safety of an aircraft. To which emergency phase does this situation correspond?

- **A** ALERFA
- **B** INCERFA
- **C** DETRESFA

Answer: A.

59 Which of the following signals is a distress signal?

- **A** In radiotelephony the spoken words PAN, PAN
- **B** A parachute flare showing a red light
- **C** The repeated switching on and off of the navigation lights

Answer: B.

60 An aircraft is in emergency when:

- **A** It is threatened by grave danger
- **B** It is threatened by grave danger and requires immediate assistance
- **C** It is forced to land

Answer: B.

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61 The different emergency phases (in correct order) are:

- **A** INCERFA, ALERTFA and DISTRESFA
- **B** INCERFA, ALERFA, DETRESFA
- **C** ALERTFA, DISTRESFA, RESCUEFA

Answer: B.

62 Which of the following statements describes the Alert Phase (ALERFA)?

- **A** An aircraft fails to arrive within 30 minutes of the ETA which was last notified to ATC
- **B** An aircraft declares that the fuel remaining is insufficient to reach safety
- **C** An aircraft which has been cleared to land fails to land within 5 minutes of the ETA and communication has not been re-established

Answer: C.

63 DETRESFA is where:

- **A** An aircraft fails to land within 5 minutes of being cleared to land
- **B** All answers are correct
- **C** The fuel on board is considered to be exhausted

Answer: C.

64 A strayed aircraft is an aircraft:

- **A** that has deviated significantly from its intended track or reports that it is lost
- **B** that has been observed or is reported to be operating in a given area but whose identity has not been established
- **C** only that aircraft which has deviated significantly its intended track

Answer: A.

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65 TSA801(A) is an airspace:

- **A** temporarily segregated and allocated in Delhi FIR
- **B** temporarily segregated and allocated in Chennai FIR
- **C** denoting an area navigation route in India

Answer: B.

66 Temporary Reserved Area (TRA) is an airspace temporarily reserved and allocated:

- **A** through which other traffic may be allowed to transit in VMC only
- **B** through which other traffic is not allowed to transit without proper equipment
- **C** through which other traffic may be allowed to transit under ATC clearance

Answer: C.

67 After rain Controller reports that runway is WET. That indicates to a pilot that:

- **A** the surface is soaked but there is no standing water
- **B** there is no standing water with a depth of more than 3 cms on one third of the runway
- **C** the runway is closed due water logging and pilot should divert

Answer: A.

68 In AIP, Enroute 3.1, ATS routes are described as:

- **A** Lower ATS Routes
- **B** Conventional Navigation Routes
- **C** RNP routes

Answer: B.

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69 In AIP, Enroute 3.2, ATS routes are described as:

- **A** Upper ATS Routes
- **B** Area Navigation Routes
- **C** Non RNP routes

Answer: B. AIP India lists ENR 3.2 as Area Navigation Routes. ENR 3.1 holds the conventional navigation routes.

Memory Aid
"2 Prevent + 1 Expedite + 1 Advise + 1 Notify" — objectives a, b, c are served by ATC; d by Flight Information Service; e by Alerting Service.

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Chapter 5 — Separation Methods and Minima

Practice Questions

1 An aircraft is maintaining FL 150 within airspace class D. Another aircraft below at FL 140 is receiving a clearance to descend to FL 70. There is severe turbulence in the area. When is the earliest that a clearance to descend to FL 140 or below can be expected?

- **A** When the other aircraft has reported that it has left FL 140.
- **B** When the other aircraft has reported that it has reached FL 70.
- **C** When the other aircraft has reported that it has descended through FL 130.

Answer: C. In severe turbulence, normal "vacating" rules are suspended. Clearance shall be withheld until the vacating aircraft has reported at or passing another level separated by the required minimum (1000 ft). So the other a/c must report passing FL 130 (which is 1000 ft below FL 140).

2 The longitudinal separation minima between aircraft departed from the same aerodrome and following the same track, and the preceding aircraft is maintaining a true airspeed of 20kt or more faster than the succeeding aircraft, is:

- **A** 5 minutes.
- **B** 3 minutes.
- **C** 2 minutes.

Answer: A. Same aerodrome, same track, preceding a/c 20 kt (37 kmph) faster = 5-minute longitudinal separation.

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3 The longitudinal separation minima based on time between aircraft at same cruising level where navigation aids permit frequent determination of position and speed and the preceding aircraft is maintaining a true airspeed of 40kt or more faster than the succeeding aircraft, is:

- **A** 5 minutes.
- **B** 3 minutes.
- **C** 10 minutes.

Answer: B. Nav aids permit frequent position/speed determination + preceding a/c ≥40 kt faster = 3 minutes (the additional speed condition reduces from 5 min to 3 min).

4 The longitudinal separation minima based on time between aircraft at same cruising level where navigation aids permit frequent determination of position and speed and the preceding aircraft is maintaining a true airspeed of 20kt or more faster than the succeeding aircraft, is:

- **A** 15 minutes
- **B** 10 minutes
- **C** 5 minutes

Answer: C. Nav aids OK + preceding a/c ≥20 kt (37 kmph) faster = 5 minutes.

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5 The longitudinal separation minima based on time between aircraft at same cruising level where navigation aids permit frequent determination of position and speed, is:

- **A** 10 minutes.
- **B** 3 minutes.
- **C** 15 minutes.

Answer: A. Same cruising level, nav aids permit frequent determination, no speed advantage stated = 10 minutes baseline.

6 The longitudinal separation minima between aircraft departed from the same aerodrome and following the same track, and the preceding aircraft is maintaining a true airspeed of 40kt or more faster than the succeeding aircraft, is:

- **A** 10 minutes.
- **B** 8 minutes.
- **C** 3 minutes.

Answer: C. Same aerodrome, same track, preceding a/c ≥40 kt (74 kmph) faster = 3 minutes.

7 The longitudinal separation minima based on distance using DME, and each aircraft "on track" uses DME stations, is:

- **A** 10NM
- **B** 20NM
- **C** 5NM

Answer: B. Baseline DME longitudinal separation on same track, same cruising level = 20 NM.

8 When an aircraft will pass through the level of another aircraft on the same track, the following minimum longitudinal separation shall be provided:

- **A** 5 minutes at the time the level is crossed.
- **B** 15 minutes while vertical separation does not exist.
- **C** 10 minutes at the time the level is crossed.

Answer: B. Aircraft climbing/descending through the level of another on same track = 15 minutes while vertical separation does not exist.

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9 "Essential Traffic" is that controlled flight to which the provision of separation by ATC is applicable, but which, in relation to a particular controlled flight is not separated therefore by the appropriate separation minima. The following flights are considered essential traffic one to each other:

- **A** All IFR flights in controlled airspace and controlled VFR.
- **B** Only controlled IFR flights.
- **C** Controlled VFR flights and VFR flights.

Answer: A. Essential traffic applies to all controlled traffic: IFR and controlled VFR flights in controlled airspace.

10 Above FL 290 the vertical flight separation between aircraft on the same direction is:

- **A** 2000ft.
- **B** 2000ft (but 1000ft in RVSM).
- **C** 3000ft.

Answer: B. At and above FL 290 the nominal vertical separation minimum is 2,000 ft, except in RVSM airspace (both aircraft RVSM-compliant) where it is 1,000 ft. (Do not confuse this with the 4,000 ft spacing between cruising levels in the same direction in the conventional table.)

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11 Track separation between aircraft using the same NDB shall be applied requiring the aircraft to fly:

- **A** At least 30° separated at a distance of 15NM or more from the facility.
- **B** At least 45° separated at a distance of 15NM or more from the facility.
- **C** At least 30° separated at a distance of 15 miles or more from the facility.

Answer: A. NDB lateral separation: tracks diverging ≥30° with at least one aircraft ≥15 NM (nautical miles, not statute miles) from the facility.

12 If an ATC clearance is not suitable to the pilot in command of an aircraft:

- **A** He may request another clearance and the ATC concerned has to accept the pilot request.
- **B** He may request and, if practicable, obtain an amended clearance.
- **C** The pilot has to accept the ATC clearance because it has been based on the flightplan filed with ATC.

Answer: B. The pilot may request an amended clearance; ATC will issue one if practicable. ATC is not obligated to accept the pilot's request verbatim, but must consider it.

13 The "VMC and own separation" ATC clearance is used for a controlled flight to cross the level of another controlled flight when:

- **A** Requested by the pilot in airspace classes A and B and authorised by ATC.
- **B** Requested by the pilot, during the day light and authorized by the state overflown.
- **C** Requested by the pilot and authorized by the state overflown.

Answer: B. VMC own separation applies in Class D and E airspace, requested by pilot, during daylight hours, and authorized by the appropriate ATS authority (state overflown).

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14 A/c on same level converging track has separation:

- **A** 15min
- **B** 10min
- **C** 5min

Answer: A. Crossing tracks (which includes converging tracks) at same level = 15 minutes standard separation (when nav aids do not permit frequent determination).

15 What are the controlled IFR separation methods applied by ATC?

- **A** Vertical, horizontal and composite separation.
- **B** Time separation and track separation.
- **C** Composite separation.

Answer: A. ATC applies vertical separation, horizontal separation (lateral + longitudinal + surveillance-based), and composite separation.

16 The vertical IFR separation minimum being applied by ATC within a controlled airspace below FL 290 is:

- **A** 500ft (150m).
- **B** 2000ft (600m).
- **C** 1000ft (300m).

Answer: C. Below FL 290, nominal vertical separation minimum = 1000 ft (approximately 300 m).

17 The vertical IFR separation minimum being applied by ATC within a controlled airspace above FL 290 is:

- **A** 500ft (150m).
- **B** 2000ft (600m).
- **C** 1000ft (300m).

Answer: B. Above FL 290 (non-RVSM), nominal vertical separation = 2000 ft (approximately 600 m).

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18 Longitudinal separation minima based on distance using DME for aircraft at the same cruising level and track, provided that each aircraft utilizes "on Track" DME stations and separation is checked by obtaining simultaneous DME readings, is:

- **A** 20NM
- **B** 25NM
- **C** 40NM

Answer: A. Same cruising level, same track, both using "on track" DME = 20 NM baseline separation.

19 A "RNAV" distance based separation minimum may be used at the time the level is crossed, provided that each aircraft reports its distance to or from the same "on track" way-point. This minimum is:

- **A** 20NM
- **B** 80NM
- **C** 60NM

Answer: B. RNAV distance-based separation with Mach number technique = 150 km / 80 NM on same direction tracks (in lieu of 10-minute longitudinal separation).

20 A VFR flight constitutes essential traffic to other VFR flights, when operating in controlled airspace classified as:

- **A** B
- **B** B, C, D and E
- **C** B and C

Answer: A. In Class B airspace, separation is provided between ALL flights (including VFR-VFR), so VFR flights constitute essential traffic to each other there.

21 RCP 120 is the number of:

- **A** Transmissions loaded on aircraft FMS for data link exchange.
- **B** Minutes required to divert to a suitable aerodrome under ETOP conditions.
- **C** Seconds it takes for an instruction to travel from the ground to aircraft and acknowledgment back to the ground.

Answer: C. RCP is classified based on transaction time in seconds. RCP 120 = the total round-trip communication takes 120 seconds.

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22 Vertical or horizontal separation shall be provided between:

- **A** All flights in class A, B and C airspaces.
- **B** All flights in class A and B airspaces.
- **C** All flights in class B, C and D airspaces.

Answer: B. Between ALL flights in Class A and B airspaces; in C, D, E it is only between IFR-IFR (and IFR-VFR in Class C).

23 The separation method whereby the vertical and horizontal separation may be reduced till a maximum of half the standard criteria is called:

- **A** Composite separation.
- **B** Reduced separation.
- **C** Combined separation.

Answer: A. Composite separation uses minima that may be lower than, but not less than half of, those used individually for vertical or horizontal separation.

24 Within reliable ATS surveillance coverage in India, when no authorized reduction or greater separation requirement applies, what is the standard horizontal separation minimum in nautical miles?

- **A** 10
- **B** 5
- **C** 3

Answer: B. The standard horizontal surveillance minimum is 5 NM within surveillance coverage. Reduced minima apply only where specifically authorized.

25 Cruising level IFR during cruise within controlled airspace shall be given as flight level (FL):

- **A** Above the transition altitude when applicable.
- **B** Only in airspace class A.
- **C** If the obstacle clearance is more than 2000ft.

Answer: A. Cruising levels are expressed as flight levels (FL) above the transition altitude. Below transition altitude, altitudes (QNH) are used.

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26 Aircraft flying along the same track may be separated by DME-distances from the same DME and it is confirmed that the aircraft have passed each other. Specify the shortest difference in DME-distance to make it possible for one aircraft to climb or descend:

- **A** 10NM
- **B** 12NM
- **C** 15NM

Answer: A. Aircraft on reciprocal tracks using on-track DME may climb/descend after it is positively established they have passed each other and are at least 10 NM apart.

27 Track separation between aircraft using the same VOR shall be applied requiring the aircraft to fly:

- **A** At least 15° separated at a distance of 15 miles or more from the facility.
- **B** At least 15° separated at a distance of 15NM or more from the facility.
- **C** At least 30° separated at a distance of 15NM or more from the facility.

Answer: B.

28 ATC clearances are solely issued for:

- **A** Alerting pilots about other aircraft in close vicinity.
- **B** Making pilots follow laid down procedures.
- **C** Expediting and separating air traffic.

Answer: C. ATC clearances are solely issued for the purpose of expediting and separating air traffic.

29 The longitudinal separation minima based on DME, and each aircraft "on track" uses DME stations, is:

- **A** 10NM provided that the leading aircraft maintains a true airspeed of 40kt or more faster than the succeeding aircraft.
- **B** 10NM provided that the leading aircraft maintains a true airspeed of 20kt or more faster than the succeeding aircraft.
- **C** 20NM provided that the leading aircraft maintains a true airspeed of 10kt or more faster than the succeeding aircraft.

Answer: B. DME separation can be reduced from 20 NM to 10 NM when the leading aircraft is at least 20 kt faster.

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Chapter 6 — Separation Near Aerodromes

Practice Questions

1 If an arriving aircraft is making a straight-in approach, a departing aircraft may take off in any direction:

- **A** Until five minutes before the arriving aircraft is estimated to be over the instrument runway
- **B** Until ten minutes before the arriving aircraft is estimated to be over the instrument runway
- **C** Until two minutes before the arriving aircraft is estimated to be over the instrument runway

Answer: A. . For a straight-in approach, a departing aircraft may take off in any direction until 5 minutes before the arriving aircraft is estimated to be over the instrument runway (Doc 4444 / AIP India, Chapter 7).

2 For controlled traffic that shall be separated in the vicinity of an airport, separation minima may be reduced:

- **A** At the discretion of the air traffic controller
- **B** When the commander in the following aircraft has the preceding aircraft in sight and is able to maintain own separation
- **C** If the commander of the involved aircraft so requests

Answer: B. . Reduction is permissible when the flight crew of the succeeding aircraft reports the other aircraft is in sight and separation can be maintained (Condition c, Reduction in Separation Minima). It cannot be done at sole ATC discretion without such visual confirmation.

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3 At the commencement of final approach, if the controller possesses wind information in the form of components, the mean tail-wind component significant change is:

- **A** 3 kt
- **B** 4 kt
- **C** 2 kt

Answer: C. . Mean tail-wind component significant change = 2 kt . (Head-wind: 10 kt; Cross-wind: 5 kt; Tail-wind: 2 kt.) Tail-wind is most critical for landing performance, hence the smallest threshold.

4 The EAT has to be transmitted to the pilot as soon as possible, in case the expected delay is:

- **A** 10 minutes or more
- **B** 15 minutes or more
- **C** 20 minutes

Answer: A. . An Expected Approach Time (EAT) shall be determined for an arriving aircraft that will be subjected to a delay of 10 minutes or more .

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5 A so-called "Visual Approach" can be performed:

- **A** During VFR flight, there should be a visibility of 5 km or more
- **B** During IFR flights, if the cloudbase is 1000 ft more than the appropriate DA or MDA for that procedure
- **C** During IFR flights, if there is permanent sight on the movement area and the underlying ground

Answer: C. . A visual approach is specifically defined as an approach by an IFR flight where part or all of the instrument approach procedure is not completed and the approach is executed with visual reference to terrain . It is not a VFR operation.

6 "Time Approach Procedure" is used as necessary to expedite the approach of a number of arriving aircraft. This will be obtained by requesting aircraft:

- **A** To maintain a specified speed during the approach procedure
- **B** To pass the specified point inbound at the previously notified time
- **C** To apply a step-down descent between aircraft in the approach sequence

Answer: B. . A "Time Approach Procedure" sequences aircraft by requesting them to pass the specified inbound fix at previously notified times, thereby creating orderly spacing without requiring speed adjustments in the approach itself.

7 If the crew on an arriving aircraft approaching a controlled aerodrome reports "field in sight," a clearance for "visual approach" may be given under certain conditions:

- **A** The meteorological visibility must not be less than 8 km
- **B** The approach must be passing the FAF
- **C** The air traffic controller will provide separation to other controlled traffic

Answer: C. . Separation shall be provided between an aircraft cleared to execute a visual approach and other arriving and departing aircraft — this is an ATC responsibility. The pilot becomes responsible for terrain clearance, but ATC maintains traffic separation.

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8 Which statement is correct? During a "visual approach" in controlled airspace (Class C):

- **A** ATC will apply separation with other traffic
- **B** The pilot has to apply separation with other traffic
- **C** ATC will apply separation with other arriving traffic only

Answer: A. . In Class C controlled airspace, ATC provides separation for IFR from IFR and IFR from VFR traffic. During a visual approach (which is an IFR flight), ATC applies separation with all other traffic. The pilot is only responsible for terrain clearance.

9 One-minute separation may be used between departing aircraft if they are to fly on tracks diverging by at least:

- **A** 30° immediately after take-off
- **B** 15° immediately after take-off
- **C** 45° immediately after take-off

Answer: C. . One-minute separation applies when aircraft are to fly on tracks diverging by at least 45° immediately after take-off so that lateral separation is provided. This is a fundamental aerodrome separation rule.

Memory Aid — Numbers to Remember (Chapter 7 Exam)

Value	Context
45°	Track divergence for 1-minute separation between departing AC
40 kt	Speed differential (preceding faster) for 2-minute separation, same track
5 min	Separation when departing AC flies through level of preceding departing AC, same track
3 min	Dep. AC T/O at least 3 min before arriving AC over THR (45° off reciprocal, instrument approach)
5 min	Any-direction T/O window before arriving AC over runway (straight-in approach)
10 min	Delay threshold requiring EAT to be issued
5 min	EAT revision threshold (difference from previous)
30 min	Hold duration requiring EAT by most expeditious means
2 800 m	Min RVR/visibility for Cat A/B visual approach
3 200 m	Min RVR/visibility for Cat C visual approach
3 600 m	Min RVR/visibility for Cat D visual approach
5 km	Min ground visibility for visual approach to circling-only runway
10 kt / 2 kt / 5 kt	Significant wind component changes (head/tail/cross-wind) at final approach

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Value	Context
3 050 m / 10 000 ft	Max altitude for own separation in VMC clearance
Class D & E	Airspace where own separation VMC clearance is valid

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Chapter 7 — Aerodrome Control Service

Practice Questions

1 The following aircraft will be given priority to land:

- **A** An aircraft with VVIP on board
- **B** An aircraft with critical shortage of fuel
- **C** A scheduled aircraft

Answer: B. An aircraft with a critical shortage of fuel is in a compelling emergency — this is Priority 1 in the priority rules. A VVIP aircraft is Priority 5. A scheduled aircraft has no special priority.

2 Apron is an area on aerodrome where following activities take place:

- **A** Loading/unloading of passengers, mail and cargo only
- **B** Parking of aircraft only
- **C** Loading/unloading of passengers, mail and cargo, fueling, parking or maintenance

Answer: C. The apron is a defined area where aircraft stand for purposes of loading/unloading passengers, mail, cargo, fuelling, parking, or maintenance — i.e. all ground handling activities.

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3 Movement area of an aerodrome constitutes the following:

- **A** Area including runways/taxiways excluding apron
- **B** Area excluding taxiways but including runways and apron
- **C** Area including runways / taxiways / aprons

Answer: C. The Movement Area = Manoeuvring area (runways + taxiways) + Apron. It is the total area used for aircraft surface movements.

4 The manoeuvring area of an airport is that area:

- **A** Normally referred to as the ramp or apron
- **B** Used for taxiing, taking off and landing
- **C** Used when taxiing to and from the parking area

Answer: B. The Manoeuvring Area = Runways + Taxiways (excluding apron). It is the portion of an aerodrome used for take-off, landing and taxiing — but NOT the apron/ramp.

5 An Aircraft shall report "long final" when it is at _____ from the approach end of the rwy:

- **A** 8 NMs
- **B** 4 NMs
- **C** 1 NMs

Answer: A. As per the visual circuit diagram in this chapter — Long Final is defined as the position beyond 4 NM, up to 8 NM from the approach end of the runway. An aircraft at 8 NMs would report "long final".

6 What defines a controlled aerodrome?

- **A** It must be located within a CTR
- **B** It must have a control tower giving an ATC service
- **C** It must have a control tower and be in a CTR

Answer: B. A controlled aerodrome is one where an ATC service is provided to aerodrome traffic — which requires a control tower providing ATC service . It does not necessarily have to be located within a CTR.

7 Aerodrome traffic is considered to be:

- **A** Aircraft on the movement area and flying in the vicinity
- **B** Aircraft on the manoeuvring area and flying in the vicinity
- **C** Aircraft on the movement area only

Answer: B. Aerodrome traffic is all traffic on the manoeuvring area (runways + taxiways, not the apron) AND all aircraft flying in the vicinity of the aerodrome (including traffic circuits). Do not confuse it with the movement area, which is the manoeuvring area plus the apron(s).

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8 Aircraft shall not be held closer to a runway-in-use:

- **A** Than at a runway-holding position
- **B** Than at runway edge
- **C** Than at 40 m from the runway edge

Answer: A. Aircraft shall not be held closer to a runway-in-use than at a runway-holding position . The distances (50m or 30m) are from the runway edge to the holding position, not the holding limit itself.

9 Friction Co-efficient due to snow — in how many segments is runway divided based on friction level?

- **A** 1/2
- **B** 1/3
- **C** 1/4

Answer: B. The runway is divided into thirds (1/3 segments) for surface condition reporting. Information is provided for each third — referred to as first, second, or third part of the runway.

10 Runway friction coefficient 0.20 means estimated surface friction:

- **A** Good
- **B** Medium to Good
- **C** Poor

Answer: C. A coefficient of 0.25 and below = POOR (Code 1) . Since 0.20 is below 0.25, it is in the POOR category. This is the most dangerous friction level — extreme braking degradation on contaminated runways.

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Chapter 8 — Air Traffic Surveillance Systems

Practice Questions

1 The minimum radar separation to be provided to aircraft established on the localizer course shall be:

- **A** 2.0 NM between aircraft on adjacent localizer course
- **B** 2.5 NM between aircraft on the same localizer course
- **C** 5.0 NM between aircraft on the same localizer course

Answer: B. 2.5 NM (4.6 km) is the minimum radar separation on the same final approach / localizer track within 10 NM of the runway threshold. This is the reduced minimum permitted when local factors allow.

2 Unless otherwise prescribed by the appropriate ATS authority, the horizontal radar separation minimum prescribed by ICAO shall be:

- **A** 3.5 NM
- **B** 8 NM
- **C** 5.0 NM

Answer: C. Unless otherwise prescribed, the horizontal separation minimum based on radar and/or ADS-B and/or MLAT systems shall be 9.3 km (5.0 NM) per ICAO Doc 4444.

3 Where ADS-B is used for identification, aircraft may be identified by the following procedure:

- **A** Asking him to squawk assigned Mode A code C
- **B** Direct recognition of the aircraft identification in an ADS-B label
- **C** Recognition of an assigned discrete code, the setting of which has been verified

Answer: B. Option C is an SSR/MLAT identification procedure, not ADS-B. ADS-B identification uses: (a) direct recognition in ADS-B label, (b) transfer of ADS-B identification, (c) TRANSMIT ADS-B IDENT compliance.

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4 An aircraft is considered to be maintaining its assigned level as long as the SSR mode C derived level information indicated that it is within:

- **A** +/- 500 ft of the assigned level in non-RVSM airspace
- **B** +/- 200 ft of the assigned level in RVSM airspace
- **C** +/- 300 ft of the assigned level in RVSM airspace

Answer: B. Doc 4444 8.5.5.2.1: level occupancy criterion is +/-60 m (200 ft) in RVSM airspace; the body mark said C (300 ft), the explanation and the Doc say B

5 An aircraft in climb or descent is considered to have crossed a level when the SSR mode C derived level information indicates that it has passed this level in the required direction by:

- **A** +/- 300 ft
- **B** More than 300 ft
- **C** 300 ft

Answer: B. Doc 4444 8.5.5.2.4: a level is crossed when passed by MORE THAN 90 m (300 ft); the body mark said C (300 ft), the explanation and the Doc say B

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6 Secondary Surveillance Radar (SSR) information may be used alone in the provision of separation between aircraft provided:

- **A** Instructions to pilot are acknowledged by him by making specific maneuver(s)
- **B** Aircraft identification is established and maintained by use of discrete SSR codes
- **C** It cannot be used for providing separation

Answer: B. SSR alone can be used for separation provided aircraft identification is established and maintained by use of discrete SSR codes.

7 Radar identification of a departing aircraft can be achieved if a radar blip is observed within a certain distance from the end of the runway. Identification has to be achieved within:

- **A** 1 NM
- **B** 3 NM
- **C** 5 NM

Answer: A. PSR identification using the departing aircraft method must be established within 1 NM from the end of the runway used.

8 Separation based on the use of MLAT position symbols and PSR blips shall be applied so that:

- **A** The distance between the centres of the position symbols and PSR blips, representing the positions of the aircraft concerned, is never less than a prescribed minimum
- **B** The distance between the edges of the position symbols and PSR blips... is never less than a prescribed minimum
- **C** The distance between the furthest edges... is never more than a prescribed minimum

Answer: A. Separation is measured between the centres of the position symbols and PSR blips. The distance between centres must never be less than the prescribed minimum.

9 The air traffic control unit has reported "radar contact". What does that mean to the pilot?

- **A** The aircraft is subject to positive control
- **B** Position reports may be omitted
- **C** The radar identity of the aircraft has been established

Answer: C. "Radar contact" means the radar identity of the aircraft has been established. Further services are then provided based on this identification. Position reports may be omitted only when ATC specifically advises this.

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10 When a RADAR operator says the following to an aircraft: "fly heading 030°", the pilot must fly heading:

- **A** 030° magnetic in still air conditions (thereby flying the magnetic track)
- **B** 030° compass in still air conditions (thereby flying the compass heading)
- **C** 030° true, in still air conditions (thereby flying the true track)

Answer: A. Radar vectors are related to magnetic headings only. The pilot flies 030° magnetic (not compass or true). In still air the magnetic heading equals the magnetic track.

11 Which code shall be used on Mode "A" to provide recognition of an aircraft subjected to unlawful interference?

- **A** Code 7700
- **B** Code 7600
- **C** Code 7500

Answer: C. 7500 = Unlawful Interference (Hijack) . 7700 = Emergency. 7600 = Radio/Communication Failure.

12 Upon intercepting the assigned radial, the controller advises you that you are on the airway and to "resume own navigation". This phrase means that:

- **A** Radar services are terminated and you will be responsible for position reports
- **B** You are to assume responsibility for your own navigation
- **C** You are to contact the center at the next reporting point

Answer: B. "Resume own navigation" means you assume responsibility for your own navigation (self-navigate using your instruments/FMS). Radar service itself may continue, but you navigate independently using the assigned route.

13 What does the ATC term "radar contact" signify?

- **A** Your aircraft has been identified on the radar display and radar flight instructions will be provided until radar identification is terminated
- **B** You will be given traffic advisories until advised that the service has been terminated or that radar contact has been lost
- **C** ATC is receiving your transponder and will furnish vectors and traffic advisories until you are advised that contact has been lost

Answer: A. "Radar contact" = aircraft identified on radar display. Radar flight instructions (which may include vectors and traffic advisories) will be provided until radar identification is terminated.

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14 Where a "Secondary Surveillance Radar" (SSR) is not available, radar identification may be achieved by one of the following procedures:

- **A** To instruct the pilot to execute one or more changes of 10°
- **B** To instruct the pilot to execute one or more changes of 45°
- **C** To instruct the pilot to execute one or more changes of 30° or more

Answer: C. The Turn Method (PSR only) requires 30° or more heading change(s) to correlate the radar position with the specific aircraft.

15 The Air Traffic Control Service: do not prevent collisions with terrain.

- **A** Wrong, they do prevent collisions with terrain
- **B** Except when an aircraft is flying IFR in IMC
- **C** Correct, except when an IFR flight is vectored by radar

Answer: C. ATC does not generally prevent terrain collisions. However, when an IFR flight is being vectored by radar , the controller is responsible for providing terrain clearance, as the pilot is following ATC instructions rather than the published procedures that guarantee terrain clearance.

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16 Which code shall be used on mode "A" to provide recognition of an emergency aircraft?

- **A** Code 7600
- **B** Code 7500
- **C** Code 7700

Answer: C. 7700 = Emergency (Distress / Urgency) . This is the universal emergency transponder code recognised worldwide.

17 One of the functions ensured by a radar control unit for the provision of approach control service is:

- **A** To conduct surveillance radar approaches
- **B** To apply a horizontal separation less than 5 NM
- **C** To provide instructions in order to reduce separations minima, if accepted by the pilots

Answer: A. Surveillance radar is extensively used in approach control to conduct surveillance radar approaches (SRA), sequence traffic, and vector aircraft to position for approaches.

18 What is meant when departure control instructs you to "resume own navigation" after you have been vectored to an airway?

- **A** Advisories will no longer be issued by ATC
- **B** You are still in radar contact, but must make position reports
- **C** You should maintain that airway by use of your navigation equipment

Answer: C. "Resume own navigation" after being vectored to an airway means you should now maintain the airway using your own navigation equipment (VOR, FMS, GPS etc). You are responsible for staying on track.

19 All aircraft carrying serviceable transponder shall operate the transponder:

- **A** At all times during flight, regardless of whether the aircraft is within or outside airspace where SSR is used for ATS purposes
- **B** Only when the aircraft is flying within airspace where SSR is used for ATS purposes
- **C** As and when required by the pilot

Answer: A. Per DGCA / AIP India: All aircraft carrying a serviceable transponder shall operate the transponder at all times during flight within Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR — regardless of whether SSR is used for ATS in that specific area.

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Memory Aid — Emergency Codes

"7700 — Sevens, SAVE ME (Distress)"
"7600 — SIXES, SILENT (Radio Failure)"
"7500 — FIVE, HIJACK ALIVE (Unlawful Interference)"

Or remember: *7700 Emergency | 7600 Comms fail | 7500 Hijack*

Chapter 9 — Aeronautical Information Services

Practice Questions

1 Detailed description of meteorological information provided at the aerodrome and an indication of which meteorological office is responsible, is in the following part of the AIP:

- **A** AD
- **B** RAC
- **C** MET

Answer: A.

2 A checklist of AIP supplements currently in force shall be issued at intervals of:

- **A** Not more than 2 months
- **B** Not more than 28 days
- **C** Not more than one month

Answer: C. Regulations specify: checklist of valid AIP Supplements shall be issued at intervals of not more than ONE MONTH. This is issued through the monthly printed plain language list of valid NOTAM.

3 An integrated aeronautical information package consists of the following elements:

- **A** AIP, supplements to AIP; NOTAM and PIB, AIC and checklist summaries
- **B** AIP, including amendment service; supplements to AIP, NOTAM and pre-flight information bulletin (PIB), AIC, checklists and summaries
- **C** AIP, including amendment service; supplements to AIP, NOTAM, AIC and checklist summaries

Answer: B. Option B is the COMPLETE and correct list. Key difference from C: Option B correctly includes "pre-flight information bulletin (PIB)" as a distinct element along with NOTAM. Option A lacks "amendment service" as a distinct component.

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4 Temporary changes on specifications for AIP supplements of long duration and information of short duration which contains extensive text and/or graphics shall be published as AIP supplements. It is considered a long duration:

- **A** Three months or longer
- **B** Two months or longer
- **C** One year or longer

Answer: A. Per regulations: "Temporary changes of long duration (THREE MONTHS OR LONGER)" shall be published as AIP Supplements.

5 Operationally significant changes to the AIP shall be published in accordance with:

- **A** AIC procedures and identified by the acronym AIC followed by a number
- **B** NOTAM procedures and identified by acronym NOTAM followed by a number
- **C** AIRAC procedures and identified by the acronym AIRAC

Answer: C. Operationally significant changes to the AIP shall be published in accordance with AIRAC procedures and shall be clearly identified by the acronym — AIRAC. This is directly from the AIP General Specifications.

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6 A checklist of NOTAM currently in force shall be issued at the AFTN at intervals of:

- **A** Not more than 28 days
- **B** Not more than one month
- **C** Not more than 10 days

Answer: B. A checklist of valid NOTAM shall be issued over the Aeronautical Fixed Service (AFS) at intervals of not more than ONE MONTH using the NOTAM Format specified. One NOTAM shall be issued for each series.

7 Change in frequency or location of navigational aids on short notice is notified through:

- **A** NOTAMs
- **B** AIRAC
- **C** AIP

Answer: A. Short-notice temporary or permanent operationally significant changes (like nav aid frequency/location changes on short notice) are notified by NOTAM. AIRAC is for pre-planned changes; AIP is for permanent/long-term information.

8 The contents of Aeronautical Information Publication (AIP) are:

- **A** GEN, ENR (en-route) and AD (aerodromes)
- **B** GEN, ENR, RAC, AD
- **C** GEN, AGA, COM, ENR, FAL

Answer: A. The AIP has exactly THREE parts: GEN (General), ENR (En-Route), and AD (Aerodromes). "RAC", "AGA", "COM", "FAL" are older ICAO Annex designations no longer used as AIP part codes in the current Annex 15 structure.

9 The SIGMET service, is in the AIP, in the following part:

- **A** AGA
- **B** GEN
- **C** MET

Answer: B.

10 In which chapter of the AIP can you find a list with "location indicators"?

- **A** AGA
- **B** GEN
- **C** ENR

Answer: B.

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11 The informations on holding, approach and departure procedures, are found in the following part of the AIP:

- **A** ENR
- **B** MAP
- **C** AD

Answer: A.

12 The closure of a runway for a year, because of maintenance, will be published:

- **A** Only in AIP
- **B** In NOTAM and AIP, inclusive Supplement
- **C** Only in NOTAM

Answer: B. A year-long runway closure is a long-duration temporary change (≥3 months), so it requires an AIP Supplement. It would also initially be NOTAMed. Both NOTAM and AIP (Supplement) are used. "Only AIP" is wrong as only permanent changes go directly to AIP Amendment.

13 An AIRAC is:

- **A** An acronym for a system aimed at advance notification based on common effective dates, of circumstances necessitating significant changes in operating procedures
- **B** A notice distributed by means of telecommunication containing information concerning establishment, condition or change in any aeronautical facility…
- **C** A package which consists of AIP, supplements to the AIP, NOTAM, AIC, checklists and summaries

Answer: A. A is the correct definition of AIRAC. B describes a NOTAM. C describes the Integrated Aeronautical Information Package (IAIP).

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14 Select the acronym corresponding to the following definition: a special NOTAM series notifying, by means of a specific format, an important change for the aircraft operations, due to a volcano activity, a volcano eruption or a volcanic ash cloud.

- **A** GVATAM
- **B** ASHTAM
- **C** NAVTAM

Answer: B. ASHTAM = Volcanic Ash NOTAM. It is a special series NOTAM in a specific format for volcanic activity, eruptions, and ash clouds. GVATAM and NAVTAM are distractors.

15 The system notifying in advance the circumstances requiring important changes in the methods of operation, based on common effective dates, is identified by the acronym:

- **A** AIRAC
- **B** EATCHIP
- **C** IFPS

Answer: A. AIRAC = Aeronautical Information Regulation And Control. It uses common effective dates in 28-day cycles for advance notification of significant operational changes.

16 The temporary, long-term modification (3 months or more) and the short-term extensive or graphical information are published as follows:

- **A** AIP Supplements
- **B** Trigger NOTAM
- **C** NOTAM

Answer: A. AIP Supplements are published for: (1) temporary changes of long duration (3 months or more) AND (2) short-duration information with extensive text/graphics. Trigger NOTAMs merely remind about upcoming AIP Supplement changes.

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17 The informations concerning charges for aerodromes/heliports and Air Navigation Services are on the following part of the AIP:

- **A** AD
- **B** GEN
- **C** FAL

Answer: B.

18 A notice providing information on Rules of the Air, Traffic Services and Air Navigation Procedures and distributed in advance of its effective date is:

- **A** An AIRAC
- **B** An ATS NOTAM
- **C** An Advisory NOTAM

Answer: A. AIRAC provides advance notification of changes to Rules of the Air, ATS procedures, navigation procedures, etc. distributed in advance of common effective dates. It is the advance-notification system specifically for significant procedural changes.

19 A notice containing information concerning flight safety, air navigation, technical, administration or legislative matters and originated at the AIS of a state is called:

- **A** NOTAM
- **B** AIRAC
- **C** Aeronautical Information Circular (AIC)

Answer: C. AIC is originated when it is necessary to promulgate aeronautical information of a purely explanatory or advisory nature concerning technical, legislative or purely administrative matters. This is the hallmark of an AIC — advisory, explanatory, administrative or legislative in nature.

20 In which section of AIP are contained information elements relating to prohibited, restricted and dangerous areas?

- **A** AGA
- **B** ENR
- **C** GEN

Answer: B.

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21 In which section of AIP are contained information elements to refueling facilities and limitations on refuelling services?

- **A** AD
- **B** GEN
- **C** SAR

Answer: A.

22 In which section of the AIP are contained information elements relating to areas and/or routes for which meteorological service is provided?

- **A** MET
- **B** GEN
- **C** RAC

Answer: B. The GEN part of the AIP contains meteorological services information under "Meteorological services" subsection which covers area of responsibility and routes for which met service is provided. "MET" is not an AIP part designation.

23 Each contracting state shall provide an AIS in its territory and shall include the preparation and origination of:

- **A** Only AIP and NOTAM's
- **B** AIP, NOTAM's, Circular and AIRAC
- **C** Integrated Aeronautical Information Package

Answer: C. Each contracting state's AIS obligation is to prepare and originate the complete Integrated Aeronautical Information Package (IAIP), not just individual components.

24 The identification of each prohibited, restricted and danger area shall be composed by:

- **A** The nationality letters for location indicators assigned to the state or territory, followed by the letters P, R and D and figures
- **B** The letters P (Prohibited), R (Restricted) and D (Dangerous) for the area concerned and figures
- **C** The letters P (Prohibited), R (Restricted) and D (Dangerous) followed by figures

Answer: A. Identification = the nationality letters of the location indicators assigned to the State + the letter P, R or D + a number. In India the letters differ by FIR (VA Mumbai, VE Kolkata, VI Delhi, VO Chennai), e.g. VA(P)-2 = Prohibited Area No. 2 in Mumbai FIR; VO(R)-81 = Restricted Area No. 81 in Chennai FIR. It is not just P / R / D plus a number.

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25 NOF in India are established at:

- **A** All major aerodromes
- **B** All international airports
- **C** International Airports at Delhi, Mumbai, Kolkatta and Chennai only

Answer: C. Only FOUR International NOTAM Offices (NOF) in India: Mumbai, Calcutta (Kolkata), Delhi, and Chennai. Not at all major aerodromes. Not at all international airports — specifically only these four.

26 Runway Condition Report (RCR) is prepared by _____ for use by _____.

- **A** Aerodrome Operator, Flight Crew
- **B** Flight Crew, DGCA
- **C** Flight Crew, Aircraft Operator

Answer: A. The aerodrome operator assesses runway surface conditions and prepares the RWYCC/RCR. This information is then used by the flight crew for aeroplane performance calculations. Aerodrome Operator prepares → Flight Crew uses.

27 Runway Condition Code (RWYCC) is reported for:

- **A** The runway to indicate unusable portion of the operational runway
- **B** For each one third portion of the operational runway to indicate the presence of water, snow, ice, slush or frost
- **C** For non operational runway to denote its status

Answer: B. RWYCC shall be reported for each ONE THIRD of the runway to be assessed. The runway is divided into three equal sections and each gets its own code. It is for OPERATIONAL runways, not non-operational ones.

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28 Standing water on runway is notified through a:

- **A** Notam W
- **B** Snowtam
- **C** AIRRAC

Answer: B. SNOWTAM covers: snow, ice, slush, frost, STANDING WATER, and water associated with snow/slush/ice/frost on the movement area. All these hazardous surface conditions are notified via SNOWTAM.

29 NOF stands for:

- **A** Notam Office
- **B** Notam for Flight
- **C** International Notam Office

Answer: C. NOF = International NOTAM Office (Not just "NOTAM Office"). The word "International" is key — these offices handle international distribution of NOTAMs. India has 4 NOFs at the major international airports.

Memory Aid — 9 No-NOTAM Situations

"R-M-T-L-C-M-S-P-O"

Runway marking · Maintenance (aprons) · Temporary obstructions (safe) · Lighting (partial) · Communications (partial, alt. available) · Marshalling absent · Signs unserviceable · Parachuting (VFR uncontrolled) · Other similar

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Chapter 10 — Search and Rescue

Practice Questions

1 The units responsible for promoting efficient organization of search and rescue service are:

- **A** Area control centre, flight information centre and rescue coordination centre
- **B** Rescue coordination centre and rescue sub-centres
- **C** Alerting centre and rescue coordination centre

Answer: B. The RCC and RSC are specifically responsible for promoting efficient organization of SAR services and coordinating operations. ACCs and FICs serve ATS functions, not SAR management.

2 Three aircraft (1), (2) and (3) arrive successively at ten minute intervals overhead the scene of a recent aircraft accident. Aircraft (1) is unable to establish contact with the SAR Centre. Aircraft (2) is able to contact the SAR Centre. Aircraft (3) is a SAR helicopter. The command of the situation is the responsibility of:

- **A** (1), then by mutual consent (2) and then (3)
- **B** (1), and then by mutual consent to (3)
- **C** (1), then by mutual consent (2) until the completion of operations

Answer: A. The first aircraft on scene (1) takes command. When (2) arrives, command may pass by mutual consent. When the SAR helicopter (3) arrives, it should eventually take command as the designated SAR asset — command passes by mutual consent to (3).

3 What is the meaning of the symbol "V" in the ground-air visual signal code for use by survivors?

- **A** Landing here impossible
- **B** Require assistance
- **C** Require medical assistance

Answer: B. V = Require Assistance (general help). X = Require Medical Assistance. "Landing here impossible" is not part of the survivor's ground-air code.

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4 What is the meaning of the symbol "LLL" in the ground-air visual signal code for use by rescue units?

- **A** Operation completed
- **B** We are returning to base
- **C** We have found all personnel

Answer: A. LLL = Operation Completed. LL = We have found ALL personnel. XX = Cannot continue / returning to base.

5 Which of the following is NOT an international distress frequency?

- **A** 243.0 MHz
- **B** 2182 KHz
- **C** 2430 KHz

Answer: C.

6 An aircraft is flying over a mountainous region in which a search is being carried out to find the survivors of an aircraft accident. The pilot sees a ground signal in the form of an "X". This indicates:

- **A** "All occupants alive"
- **B** "Need mechanical assistance"
- **C** "Need medical assistance"

Answer: C. X = Require Medical Assistance — this is one of the five survivor ground-air visual signal codes. V = Require Assistance (general).

7 At night an aircraft observes a luminous signal requesting help. To indicate that he has received these ground signals, the pilot must:

- **A** Make at least one complete turn over the group of people in difficulty
- **B** Transmit, by luminous Morse signal, a series of the letter "R" using his navigational lights
- **C** Switch his landing lights on and off twice or, if he is not so equipped, his navigation lights twice

Answer: C. The air-to-ground signal at night confirming ground signals understood = flashing landing lights (or nav lights) ON and OFF TWICE.

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8 The color identification of the contents of droppable containers and packages containing survival equipment should take the form of coloured streamers according to the following code:

- **A** Red for medical supplies and first aid equipment
- **B** Black for food and water
- **C** Blue for blankets and protective clothing

Answer: A. The streamer colour code is: Red - medical supplies and first-aid equipment; Blue - food and water; Yellow - blankets and protective clothing; Black - miscellaneous equipment (stoves, axes, compasses, cooking utensils). Blue cannot be both food-and-water and blankets, so options B and C are wrong.

9 COSPAS-SARSAT is:

- **A** A communication system linking all aircraft
- **B** A space system for the search of vessels in distress
- **C** A SAR satellite-aided tracking system

Answer: C. COSPAS-SARSAT is a SAR satellite-aided tracking system operating on 406 MHz. India participates through INMCC Bangalore, LUTs at Bangalore and Lucknow.

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10 You are flying in Mumbai FIR when you intercept a distress call. You record the message and tell Mumbai FIC. The radio operator at Mumbai tells you to standby. What should you consider doing whilst awaiting instructions?

- **A** Make a general call on 121.5 asking other airplanes to assist
- **B** Descend to low level and carry out a surface search for ships
- **C** Proceed to the position given in the distress message

Answer: C.

11 Search and Rescue services within the territory of ICAO contracting states are provided:

- **A** On a 24-hour basis
- **B** When the authority of the contracting state decides to do so
- **C** From sunset to sunrise

Answer: A. Contracting states are obligated to provide SAR services on a 24-hour basis — no exceptions, no authority discretion required.

12 Contracting States shall establish a rescue co-ordination centre:

- **A** At every ATS facility
- **B** In each search and rescue region
- **C** If they are adjacent to the sea

Answer: B. One RCC per Search and Rescue Region. In India, four RCCs correspond to four FIRs: Delhi, Mumbai, Kolkata, Chennai.

13 A Rescue Unit is:

- **A** A unit assisting a RCC in SAR duties
- **B** A unit composed of trained personnel and provided with equipment suitable for the expeditious conduct of SAR
- **C** A unit responsible for SAR service in co-operation with a RCC

Answer: B. A Rescue Unit is defined as a unit composed of trained personnel, provided with equipment suitable for the expeditious conduct of SAR operations.

14 Which of the following statements regarding Alerting service is correct?

- **A** Aircraft in the vicinity of an aircraft known or believed to be the subject of unlawful interference, shall be informed about this
- **B** The Alert phase is established when no communication has been received from an aircraft within a period of thirty minutes after the time a communication should have been received
- **C** Alerting Service and Flight Information Service are often provided by the same ATS unit

Answer: C. Alerting Service and FIS are frequently collocated and provided by the same ATS unit (typically a FIC). Option B is wrong: no communication within 30 minutes after a communication should have been received is the Uncertainty phase (INCERFA) , not the Alert phase. The Alert phase (ALERFA) follows when further attempts to establish contact, or inquiries, fail.

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15 Whenever a distress signal and/or message or equivalent transmission is intercepted by the PIC of an aircraft, he shall:

- **A** Record the position of the craft in distress if given and if possible take a bearing on the transmission
- **B** If possible take a bearing on the transmission
- **C** Record the position of the craft in distress if given

Answer: A. Both recording the position AND taking a bearing are required actions (if feasible). Answer A correctly combines both obligations. Neither B nor C alone is the complete requirement.

16 What is the meaning of SAR signal "N" when used by survivors on the ground:

- **A** Negative
- **B** Require medical assistance
- **C** Require assistance

Answer: A. In the survivor ground-air visual code: N = No / Negative. X = Require Medical Assistance. V = Require Assistance.

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17 Survival equipment, dropped by SAR and containing food and water will be packed in containers and indicated by streamers of following colour:

- **A** Green
- **B** Blue
- **C** Yellow

Answer: B. Directly from the text: "Survival equipment, dropped by SAR and containing food and water will be packed in containers and indicated by streamers of Blue colour."

Memory Aid — Rescue Unit Symbols

LLL = All done (3 letters = Complete) · **LL** = Found ALL (2 = complete set) · **++** = Partial (plus signs = something, but not all) · **XX** = Cannot continue (X = wrong/stopped) · **NN** = Nothing found, keep going

Remember the mnemonic: **A-R-B-I-P** — Acknowledge · Record position · Bearing · Inform RCC · Proceed (at discretion).

Chapter 11 — Visual Aids and Aerodrome Services

Practice Questions

1 An aerodrome is obligated to have at least how many wind direction indicators?

- **A** One
- **B** Two
- **C** As many as required by size

Answer: A.

2 At which aerodrome is a signaling lamp mandatory?

- **A** All aerodromes
- **B** Controlled aerodrome (in the control tower)
- **C** Only international aerodromes

Answer: B.

3 What is the minimum size of the signal area?

- **A** 6 m square
- **B** 9 m square
- **C** 12 m square

Answer: B.

4 What colour are runway markings?

- **A** White
- **B** Yellow
- **C** Orange

Answer: A.

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5 What colour are taxiway markings, runway turn pad markings, and aircraft stand markings?

- **A** White
- **B** Yellow
- **C** Blue

Answer: B.

6 At an intersection of two runways, which runway's markings take priority?

- **A** Whichever runway is longer
- **B** The more important runway (precision approach > non-precision > non-instrument)
- **C** The runway with the higher code number

Answer: B.

7 Where shall runway designation markings be located?

- **A** At the threshold of a paved runway
- **B** At the mid-point of the runway
- **C** At both the threshold and the runway end

Answer: A.

8 In the case of parallel runways, each runway designation number shall be supplemented by a letter. For two parallel runways, the letters used are:

- **A** L and R
- **B** L, C, R
- **C** 1 and 2

Answer: A.

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9 The length of a stripe plus a gap in a runway centre line marking shall be:

- **A** Not less than 30 m or more than 50 m
- **B** Not less than 50 m or more than 75 m
- **C** Not less than 60 m or more than 90 m

Answer: B.

10 The minimum width of a runway centre line stripe on a CAT II & III precision approach runway is:

- **A** 0.45 m
- **B** 0.30 m
- **C** 0.90 m

Answer: C.

11 The stripes of the threshold marking shall commence how far from the threshold?

- **A** 3 m
- **B** 6 m
- **C** 9 m

Answer: B.

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12 How many threshold marking stripes are required for a runway of width 45 m?

- **A** 8
- **B** 10
- **C** 12

Answer: C.

13 An aiming point marking is mandatory on which of the following?

- **A** Non-instrument runway with code number 1 or 2
- **B** Paved instrument runway where code number is 2, 3 or 4
- **C** All paved runways regardless of code

Answer: B.

14 For an LDA of 1200 m up to but not including 2400 m, what is the distance from threshold to the beginning of the aiming point marking?

- **A** 250 m
- **B** 300 m
- **C** 400 m

Answer: B.

15 A touchdown zone marking shall be provided on a paved precision approach runway where the code number is:

- **A** 1, 2, 3 or 4
- **B** 2, 3 or 4
- **C** 3 or 4 only

Answer: B.

16 How many pairs of TDZ markings are required when the landing distance available is 2400 m or more?

- **A** 4
- **B** 5
- **C** 6

Answer: C.

17 The minimum width of a taxiway centre line marking is:

- **A** 15 cm
- **B** 20 cm
- **C** 30 cm

Answer: A.

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18 An isolated aircraft parking position must be located at least how far from other parking positions?

- **A** 50 m
- **B** 75 m
- **C** 100 m

Answer: C.

19 The enhanced taxiway centre line marking shall extend from the runway-holding position Pattern A to a distance of up to how far in the direction of travel away from the runway?

- **A** 30 m
- **B** 47 m
- **C** 60 m

Answer: B.

20 What is the diameter of a VOR aerodrome checkpoint marking circle?

- **A** 4 m
- **B** 6 m
- **C** 8 m

Answer: B.

21 A mandatory instruction marking shall consist of an inscription in:

- **A** Black on a yellow background
- **B** Yellow on a black background
- **C** White on a red background

Answer: C.

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22 An information marking that replaces a location sign shall have an inscription in which colour?

- **A** Yellow
- **B** Black
- **C** White

Answer: A.

23 The flash frequency of an aerodrome beacon shall be:

- **A** 10 to 20 per minute
- **B** 20 to 30 per minute
- **C** 30 to 60 per minute

Answer: B.

24 What is the light shown by an aerodrome identification beacon at a land aerodrome?

- **A** Flashing white
- **B** Flashing green
- **C** Flashing yellow

Answer: B.

25 What is the light shown by an aerodrome identification beacon at a water aerodrome?

- **A** Flashing green
- **B** Flashing yellow
- **C** Flashing white

Answer: B.

26 The minimum length of a Simple Approach Lighting System from the threshold is:

- **A** 300 m
- **B** 420 m
- **C** 900 m

Answer: B.

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27 The crossbar of a Simple Approach Lighting System is located at what distance from the threshold?

- **A** 150 m
- **B** 300 m
- **C** 450 m

Answer: B.

28 The total length of a CAT I precision approach lighting system from the threshold is:

- **A** 420 m
- **B** 720 m
- **C** 900 m

Answer: C.

29 In the CAT II & III approach lighting system, the side row lights show:

- **A** Variable white
- **B** Red
- **C** Alternate red and white

Answer: B.

30 In a CAT II & III approach lighting system, how far from the threshold do the side rows of lights extend?

- **A** 150 m
- **B** 300 m
- **C** 270 m

Answer: C.

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31 The PAPI system consists of how many light units?

- **A** Two
- **B** Four
- **C** Six

Answer: B.

32 In the PAPI system, when on or close to the approach slope, the pilot will see:

- **A** All four lights white
- **B** Three red and one white
- **C** Two nearest runway as red and two farthest from runway as white

Answer: C.

33 When "Too High" in a PAPI approach, the pilot sees:

- **A** All four lights white
- **B** Three white and one red
- **C** All four lights red

Answer: A.

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34 When "Too Low" in a PAPI approach, the pilot sees:

- **A** All four lights white
- **B** Three red and one white
- **C** All four lights red

Answer: C.

35 The APAPI system consists of how many light units?

- **A** Two
- **B** Three
- **C** Four

Answer: A.

36 In an APAPI system, when above the approach slope, the pilot sees:

- **A** Both units white
- **B** Both units red
- **C** One red and one white

Answer: A.

37 What colour is an aerodrome beacon at a land aerodrome?

- **A** Yellow or yellow/white
- **B** Green or green/white
- **C** White or white/green

Answer: C.

38 Runway edge lights shall be uniformly spaced at intervals of not more than:

- **A** 30 m for instrument runways; 60 m for non-instrument runways
- **B** 60 m for instrument runways; 100 m for non-instrument runways
- **C** 100 m for instrument runways; 150 m for non-instrument runways

Answer: B.

39 The last 600 m or one-third of runway length (whichever is less) of runway edge lights from the takeoff end may show:

- **A** Red
- **B** Yellow
- **C** Alternate red and white

Answer: B.

40 Runway threshold and wing bar lights shall show which colour?

- **A** Fixed green (unidirectional, in the direction of approach)
- **B** Fixed white
- **C** Flashing white

Answer: A.

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41 Runway end lights shall show which colour?

- **A** White
- **B** Yellow
- **C** Red

Answer: C.

42 Runway centre line lights from 900 m to 300 m from the runway end show:

- **A** Variable white only
- **B** Alternate red and variable white
- **C** All red

Answer: B.

43 Taxiway markings and aircraft stand markings are:

- **A** Red
- **B** Yellow
- **C** White

Answer: B.

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44 Taxiway edge lights shall be:

- **A** Fixed showing green
- **B** Fixed showing yellow
- **C** Fixed showing blue

Answer: C.

45 Mandatory instruction signs on an aerodrome shall have the following colours:

- **A** White inscription on a red background
- **B** Yellow inscription on a black background
- **C** Black inscription on a yellow background

Answer: A.

46 Which of the following describes a location sign?

- **A** Red writing on a white background
- **B** Yellow writing on a black background
- **C** Black writing on a yellow background

Answer: B.

47 Runway threshold identification lights (RTIL), when provided, should be:

- **A** Fixed green
- **B** Fixed white
- **C** Flashing white (60–120 flashes/minute)

Answer: C.

48 A barrette in a touchdown zone lighting system shall be composed of at least:

- **A** Two lights
- **B** Three lights
- **C** Five lights

Answer: B.

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49 Stopway lights shall show which colour in the direction of the runway?

- **A** Yellow
- **B** Red
- **C** White

Answer: B.

50 When a runway threshold is permanently displaced, arrows shall be provided on the portion of the runway before the displaced threshold. All markings before the displaced threshold shall be:

- **A** Painted yellow
- **B** Obscured (except the runway centre line marking, which shall be converted to arrows)
- **C** Removed entirely

Answer: B.

51 The number of threshold stripes on each side of the centre line of a runway which has a width of 45 m is:

- **A** 4
- **B** 6 (the total for 45 m is 12, so 6 each side)
- **C** 8

Answer: B.

52 What is a "barrette"?

- **A** Three or more ground lights closely spaced together to appear as a bar of lights
- **B** A frangible structure on which approach lights are fixed
- **C** A CAT 2 or 3 holding position

Answer: A.

53 The runway centre line marking on an exit taxiway shall consist of lights showing:

- **A** All green
- **B** Alternate green and yellow (from runway CL to ILS/MLS critical area boundary), then all green
- **C** All yellow

Answer: B.

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54 The longitudinal spacing of taxiway centre line lights intended for use in RVR conditions of less than 350 m should not exceed:

- **A** 30 m
- **B** 15 m
- **C** 7.5 m

Answer: B. The stem said "runway centre line lights ... on a taxiway"; CAR 4-B 5.3.17: taxiway centre line lights, longitudinal spacing not more than 15 m for RVR below 350 m. Stem corrected to taxiway.

55 To protect against laser emitters around aerodromes, which of the following zones should be established?

- **A** LFFZ only
- **B** LFFZ and LCFZ only
- **C** LFFZ, LCFZ, and LSFZ

Answer: C.

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56 For an instrument runway with code number 4, non-aeronautical lights should be extinguished within how far from the threshold and runway end?

- **A** 3000 m
- **B** 4500 m
- **C** 6000 m

Answer: B.

57 TDZ lights shall be provided in the touchdown zone of a precision approach runway of category:

- **A** I, II or III
- **B** II or III only
- **C** I only

Answer: B.

58 The LDI (Landing Direction Indicator) should be in the form of:

- **A** An L-shape
- **B** A "T" shape
- **C** An arrowhead

Answer: B.

59 Which of the following Annexes to the Chicago convention contains minimum specifications for the design of aerodromes?

- **A** Annex 14
- **B** Annex 11
- **C** Annex 6

Answer: A.

60 "Clearway" is a defined rectangular area established to:

- **A** Reduce the risk of damage to aircraft running off a runway
- **B** Protect aircraft during take-off or landing operations
- **C** Permit aircraft to make a portion of its initial climb to a specific height

Answer: C.

61 The STOPWAY is a defined rectangular area on the ground at the end of take-off run available prepared as a suitable area where:

- **A** A landing aircraft can be stopped if overcoming the end of runway
- **B** An aircraft can be stopped in the case of an abandoned take-off
- **C** A landing aircraft can be stopped only in emergency

Answer: B.

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62 "ASDA" (Accelerate Stop Distance Available) is:

- **A** The length of the take-off run available plus the length of stopway and clearway (if provided)
- **B** The length of the take-off run available plus the length of stopway (if stopway provided)
- **C** The length of the runway plus the length of stopway available (if stopway provided)

Answer: B.

63 "Instrument runways" are the following runways intended for the operation of aircraft using instrument approach procedures:

- **A** Non precision approach runways, precision approach runways category I, II and III
- **B** Instrument approach runways, precision approach runways category I, II and III
- **C** Precision approach runways category I, II and III

Answer: A.

64 "TODA" take-off distance available is:

- **A** The length of the take-off run available plus the length of the clearway (if provided)
- **B** The length of the take-off run available plus the length of stopway and clearway (if provided)
- **C** The length of the take-off run plus the length of the stopway

Answer: A.

65 The light shown by an "Aerodrome Identification Beacon" at a land aerodrome shall be:

- **A** White and green colour identification given by Morse Code
- **B** Green colour identification given by Morse Code
- **C** Blue colour identification given by Morse Code

Answer: B.

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66 Taxiway edge lights shall be:

- **A** Fixed showing green
- **B** Fixed showing yellow
- **C** Fixed showing blue

Answer: C.

67 In the case of parallel runways, each runway designation number shall be supplemented:

- **A** By a letter — for example 2 parallel runways "L" and "R"; for 3: "L", "C" and "R"
- **B** By a letter for 2 parallel runways only
- **C** By a number like "0" and "01" for 2 parallel runways

Answer: A.

68 In the "PAPI" system the pilot during an approach will see the two units nearest the runway as red and the two units farthest from the runway as white when:

- **A** Above the approach slope
- **B** On or close to the approach slope
- **C** Only on the approach slope

Answer: B.

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69 Runway threshold identification lights, when provided, should be:

- **A** Fixed green
- **B** Fixed white
- **C** Flashing white

Answer: C.

70 What is a "barrette"?

- **A** Three or more ground lights closely spaced together to appear as a bar of lights
- **B** A frangible structure on which approach lights are fixed
- **C** A CAT 2 or 3 holding position

Answer: A.

71 What is the length of an approach lighting system of a precision-approach runway CAT II?

- **A** 600 m
- **B** 900 m
- **C** 300 m

Answer: B.

72 Runway end lights shall be:

- **A** Fixed lights showing variable red
- **B** Fixed unidirectional lights showing red in the direction of the runway
- **C** Fixed lights showing variable white

Answer: B.

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73 Aerodrome signs should be in the following configuration:

- **A** Mandatory instruction signs, black backgrounds with red inscriptions
- **B** Information signs, yellow or black background with black or yellow inscriptions
- **C** Mandatory instruction signs, red background with black inscriptions

Answer: B.

74 The "PAPI" shall consist of:

- **A** Two wing bars of 4 sharp transition multi-lamp or paired units equally spaced
- **B** A wing bar of 4 sharp transition multi-lamp or paired units equally spaced
- **C** Two wing bars of 6 sharp transition multi-lamp or paired units equally spaced

Answer: B.

75 Runway threshold lights shall be:

- **A** Fixed lights green colours
- **B** Fixed lights showing green or white colours
- **C** Fixed unidirectional lights showing green in the direction of approach to the runway

Answer: C.

76 Taxiway centreline lights other than an exit taxiway shall be:

- **A** Fixed lights showing blue
- **B** Fixed lights showing green
- **C** Fixed lights showing yellow

Answer: B.

77 How many red lights must a pilot see, whose aircraft, in final approach, is following a normal glide path defined by a PAPI?

- **A** 2
- **B** None
- **C** 3

Answer: A.

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78 Runway edge lights expected in the case of a displaced threshold shall be:

- **A** Fixed lights, white or yellow colour
- **B** Flashing white
- **C** Fixed lights showing variable white

Answer: C.

79 The abbreviation PAPI stands for:

- **A** Precision Approach Path Indicator
- **B** Precision Approach Power Index
- **C** Precision Approach Path Index

Answer: A.

80 High intensity obstacle lights should be:

- **A** Flashing red
- **B** Flashing white
- **C** Fixed orange

Answer: B.

81 Low intensity obstacle lights on mobile objects shall be:

- **A** Fixed red or preferably orange
- **B** Flashing blue / yellow
- **C** Flashing red or preferably yellow

Answer: B.

82 Low intensity obstacle lights on fixed objects shall be:

- **A** Flashing red
- **B** Fixed red / Flashing Red
- **C** Fixed red (only)

Answer: C. CAR 4-B Table 6-1: low-intensity Types A and B for fixed obstacles are fixed red. The old key (fixed red / flashing red) wrongly adds flashing red, which belongs to medium-intensity Type B; the only fully correct option is "Fixed red (only)".

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83 The runway edge lights shall be:

- **A** Red
- **B** White (variable white)
- **C** Green

Answer: B.

84 The aerodrome category for rescue and fire fighting is based on:

- **A** The over-all length of the longest aeroplane normally using the aerodrome and its maximum fuselage width
- **B** The over-all length of the longest aeroplane
- **C** The longest aeroplane maximum width only

Answer: A.

85 Regarding declared airfield distances the ASDA is:

- **A** The distance from the start of the take-off run to the last point capable of bearing the full weight of the aircraft under normal operating conditions
- **B** The distance from the start of the take-off run to the end of the stopway
- **C** The area beyond the end of the prepared surface only available to aircraft stopping in an emergency

Answer: B.

86 "A defined rectangular area on the ground at the end of take-off run available prepared as a suitable area in which an aircraft can be stopped in the case of an abandoned take-off" is the definition for:

- **A** Runway strip
- **B** Runway end safety area
- **C** Stopway

Answer: C.

87 Where a parking bay is provided at an aerodrome suitable for the parking of an aircraft which is known or believed to be the subject of unlawful interference, what is the minimum distance that this parking bay needs to be separated from any other parking areas, buildings?

- **A** 100 m
- **B** 50 m
- **C** 200 m

Answer: A.

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88 Crash/Rescue (emergency) services provision is categorised according to physical characteristics of the type of aeroplane using the aerodrome. Upon what are the categories based?

- **A** Overall length and maximum takeoff mass
- **B** Overall length and fuselage width
- **C** Maximum number of passengers on board

Answer: B.

89 Which of the following systems describes an abbreviated precision approach path indicator?

- **A** Ten light units arranged on one side of the runway in the form of a single wing bar of four light units, with a bisecting longitudinal line of six lights
- **B** Twenty light units symmetrically disposed about the runway centre line in the form of two wing bars of four light units each, with bisecting longitudinal lines of six lights
- **C** A wing bar of 2 sharp transition multi-lamp units normally located on the left side of the runway unless it is physically impracticable to do so

Answer: C.

90 What colour are emergency vehicles painted that are used on the manoeuvring area of an aerodrome?

- **A** A single conspicuous colour, preferably red or yellowish green
- **B** Green
- **C** Orange

Answer: A.

91 Alternate yellow/green centre line lights of a taxiway indicate:

- **A** The proximity of a runway
- **B** An ILS/MLS critical/sensitive area
- **C** A stopway

Answer: B.

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92 When taxiing on a surface with white markings, you are rolling on a:

- **A** Movement Area
- **B** Runway
- **C** Clearway

Answer: B.

93 What colour are apron safety line markings?

- **A** White
- **B** Red
- **C** A conspicuous colour that contrasts with the aircraft stand markings

Answer: C. CAR 4-B 5.2.1.6: apron safety lines are a conspicuous colour that contrasts with the aircraft stand markings. The "in India = red" note is not in the CAR and was removed; option C reworded to the CAR wording.

94 Runway centre line lights shall be fixed lights showing variable ___ from the threshold to the point 900 m from the runway end, alternate ___ and variable ___ from 900 m to 300 m from runway end, and ___ from 300 m to the runway end.

- **A** White, red and white, red
- **B** White, yellow and white, red
- **C** Green, yellow, red

Answer: A.

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95 What colour is an aerodrome beacon at a land aerodrome?

- **A** Yellow or yellow/white
- **B** Green or green/white
- **C** White or white/green

Answer: C. Regulation: Green flashes + White flashes

96 Which of the following describes a location sign?

- **A** Red writing on a white background
- **B** Yellow writing on a black background
- **C** Black writing on a yellow background

Answer: B.

97 When on or close to the approach slope, the wing bar of a PAPI shows:

- **A** Two red lights and two white lights
- **B** Four red lights
- **C** Three white lights and one red light

Answer: A.

98 The number of stripes on each side of the centre line of a runway which has a width of 45 m is:

- **A** 3
- **B** 6
- **C** 4

Answer: B.

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99 Mandatory instructions signs on an aerodrome shall have the following colours:

- **A** White inscription on a red background
- **B** Yellow inscription on a black background
- **C** Black inscription on a yellow background

Answer: A.

100 Taxiway markings and aircraft stand markings are:

- **A** Red
- **B** Yellow
- **C** White

Answer: B.

101 Taxiway centre line lights other than an exit taxiway shall be:

- **A** Fixed lights showing green.
- **B** Fixed lights showing yellow.
- **C** Fixed lights showing blue.

Answer: A.

102 ARIWS is a system used for:

- **A** Warning a pilot about bird activity on take off path
- **B** Cautioning a pilot about wind shear on finals in real time
- **C** Warning a pilot about runway incursion in near real time

Answer: C.

103 You see a sign at the edge of a taxiway with a white inscription on a red background. This sign is a:

- **A** Information sign
- **B** Mandatory instruction sign
- **C** Holding position sign

Answer: B. The stem described a white board with red letters. CAR 4-B 5.4.2.12: a mandatory instruction sign has a white inscription on a red background. Stem corrected.

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104 An aiming point marking shall be provided at:

- **A** Each approach end of paved instrument runway where the code number is 2, 3 or 4
- **B** Paved taxy track joining precision approach runway
- **C** Paved runways not provided with touch down zone markings

Answer: A.

105 LFFZ is established around an aerodrome to:

- **A** Protect aircraft from low flying aircraft
- **B** Protect aircraft from laser emitters
- **C** Protect aircraft from low flying fighter aircraft in military areas

Answer: B.

106 VASIS shall be provided to serve the approach to a runway:

- **A** Not served by visual approach aids
- **B** Not served by non visual approach aids
- **C** Whether or not the runway is served by visual approach aids or by non-visual aids, where turbo jet operations take place

Answer: C.

107 Which of these is a correct taxiway designator?

- **A** I2
- **B** X1
- **C** Z2

Answer: C. CAR 4-B 5.4.3.37: the letters I, O and X are not used as taxiway designators (confusion with 1, 0 and the closed marking). The original option text was broken; the question is restated.

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108 Wildlife hazard at aerodromes is caused due to the presence of:

- **A** Wild animals.
- **B** Animals.
- **C** Birds and animals.

Answer: C.

Quick Reference — Key Values & Memory Aids

420 m	minimum

	Simple Approach Lighting System Length
900 m	minimum CAT I / CAT II & III Approach Lighting Length
300 m	from threshold Crossbar location (SALS & CAT I)
9 m²	minimum Signal Area Size
6 m	from threshold Threshold Marking Stripe Commencement
100 m	minimum Isolated Parking from Other Positions
47 m	max Enhanced Taxiway CL Marking Extension
6 m ⌀	diameter VOR Aerodrome Checkpoint Circle
900 m	extent

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	TDZ Lights extend from threshold
15 m	spacing Runway CL Light longitudinal spacing
20–30	flashes/min Aerodrome Beacon Flash Frequency
60–120	flashes/min RTIL Flash Frequency

Memory aid: "Black Square, you are there."

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Chapter 12 — PANS-OPS: Instrument Flight Procedures

Practice Questions

1 PANS-OPS means:

- **A** Procedures for Air Navigation Services - Aircraft operations
- **B** Pilots Alternate Navigational Systems and Operational Procedures
- **C** Procedures for Air Navigation Systems - Airfield operations

Answer: A.

2 The document that specifies recommendations for instrument procedures is called...

- **A** PANS OPS Doc 8168
- **B** The Convention of Chicago
- **C** The Air Navigation bulletin

Answer: A.

3 The Transition Level:

- **A** Published in Section ENR of the AIP
- **B** Shall be the lowest FL available for use above the transition altitude
- **C** Is calculated and declared for an approach by the Pilot-in-command

Answer: B.

4 During flight through the transition layer the vertical position should be expressed as:

- **A** Altitude above MSL during climb
- **B** Either altitude above MSL or FL during climb
- **C** Altitude above MSL during descent

Answer: C.

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5 Transition from altitude to FL, and vice versa, is done:

- **A** At TL during climb and TA during descent
- **B** At TA during climb and TL during descent
- **C** Only at transition altitude

Answer: B.

6 The transition altitude of an aerodrome should not be below:

- **A** 4000 ft
- **B** 1000 ft
- **C** 1500 ft

Answer: A.

7 Pilot of departing aircraft under IFR shall change altimeter from QNH to 1013.25 hPa when passing:

- **A** Transition layer
- **B** Level specified by ATC
- **C** Transition altitude

Answer: C.

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8 In relation to three entry sectors, entry into holding pattern shall be according to:

- **A** Course
- **B** Track
- **C** Heading

Answer: C.

9 Zone of flexibility on either side of sector boundaries in holding entry:

- **A** 5°
- **B** 20°
- **C** 10°

Answer: A.

10 What is the outbound timing in a holding pattern above FL 140?

- **A** 2 minutes
- **B** 1 minute
- **C** 1 minute 30 seconds

Answer: C.

11 In a holding pattern all turns are to be made at:

- **A** Rate of 3°/sec or bank angle of 25°, whichever lesser bank
- **B** Rate of 3°/sec
- **C** Rate of 3°/sec or bank angle of 20°, whichever lesser bank

Answer: A.

12 In a standard holding pattern turns are made:

- **A** To the left
- **B** To the right
- **C** In a direction depending on wind direction

Answer: B.

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Chapter 13 — National Law

Practice Questions

1 A pilot can fly an aircraft which is not entered in the aircraft rating of his licence for endorsement on his licence:

- **A** Within 5 nm of an aerodrome
- **B** Within Local Flying Area of the aerodrome
- **C** Anywhere

Answer: B. B — Within Local Flying Area of the aerodrome (Rule 6B)

2 A pilot can fly in 30 consecutive days:

- **A** 100 hrs PIC & 50 hrs as Co-pilot
- **B** 100 hrs PIC & 30 hrs as Co-pilot
- **C** 120 hrs PIC & 20 hrs as Co-pilot

Answer: B.

3 When an aircraft is being refuelled, the refuelling must be stopped if a jet aircraft crosses within a distance of:

- **A** 15 meters
- **B** 30 meters
- **C** 43 meters

Answer: C. C — 43 meters (Rule 25A(2)(n))

4 Mathura Refinery is a:

- **A** Danger Area
- **B** Restricted area
- **C** Prohibited area

Answer: C. C — Prohibited area (Schedule I, Rule 12)

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5 Prisoners on board an aircraft can be carried with the permission of:

- **A** DGCA
- **B** Controller of Aerodrome
- **C** Director General of Prisons

Answer: A.

6 Smoking on a private aircraft is permitted:

- **A** Provided the owner has no objection
- **B** Only if the certificate of airworthiness of the aircraft permits
- **C** Without restriction if no passengers are carried

Answer: B. B — Only if the C of A permits (Rule 25)

7 An aircraft without registration markings can be flown only:

- **A** If the sale deed is not yet completed
- **B** For purpose of test flight only, prior to purchase
- **C** If specially permitted in writing by the Central Govt.

Answer: C. C — If specially permitted in writing by the Central Govt. (Rule 5 proviso)

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8 The member of operating crew shall not have taken alcoholic drinks:

- **A** 6 hrs before commencement of flight
- **B** 12 hrs before commencement of flight
- **C** 24 hrs before commencement of flight

Answer: B. B — 12 hrs before commencement of flight (Rule 24)

9 Photography at an aerodrome can be done with prior permission in writing from:

- **A** Aerodrome officer
- **B** DGCA
- **C** Central Govt.

Answer: B.

10 While refuelling is in progress, no naked light be brought within ___ mts:

- **A** 35
- **B** 30
- **C** 100

Answer: B. B — 30 meters (Rule 25A(2)(c))

11 An aerial work aircraft on a VFR plan to Jaipur departs at 1032 hrs UTC. Sunset at Jaipur is 1318 UTC. Night landing facilities not available. Flying time available to reach Jaipur is:

- **A** 3 hrs & 06 minutes
- **B** 2 hrs & 26 min
- **C** 2 hrs & 46 min

Answer: A. A — 3 hrs & 06 minutes. VFR flights may operate from 20 minutes before sunrise to 20 minutes after sunset (CAR 9-C-I 4.3), so the day ends at 1318 + 20 = 1338 UTC; 1338 − 1032 = 3 h 06 min. (Flying only to sunset, 2 h 46 min, is the old rule.)

12 Minimum crew required on a private aircraft is:

- **A** One pilot
- **B** Two pilots
- **C** As specified in the certificate of Airworthiness

Answer: C. C — As specified in the certificate of Airworthiness (Rule 38A(7))

13 Dropping of paper leaflets require the permission of:

- **A** Aerodrome
- **B** Local District Magistrate
- **C** Operator

Answer: B. B — Local District Magistrate (or Commissioner of Police) — Rule 26(d)

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14 If your aircraft is carrying cargo for remuneration, it will be known as a:

- **A** Private aircraft
- **B** Aerial work aircraft
- **C** Public transport aircraft

Answer: C. C — Public transport aircraft (carrying persons or cargo for remuneration)

15 One pilot will be designated as PIC of a public transport aircraft for each flight by:

- **A** The operator
- **B** DGCA
- **C** Flying contract unit

Answer: A.

16 No of cabin crew required is according to:

- **A** No. of passengers excluding crew members
- **B** No. of persons on board including crew members
- **C** No. of seats

Answer: C. C — No. of seats (Rule 38B)

17 Succession of command is given:

- **A** In Operations Manual
- **B** by Operator
- **C** by DGCA

Answer: A. A — In Operations Manual (Rule 140B)

18 An international flight from India can take off:

- **A** From any aerodrome in India
- **B** From any major aerodrome in India
- **C** From any custom aerodrome only

Answer: C. C — From any custom aerodrome only (Indian Aircraft Rules 1920, Rule 54)

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19 PIC of an aeroplane shall:

- **A** Be responsible for safe conduct of navigation
- **B** Be responsible for safe conduct of flight at all times
- **C** All above is correct

Answer: C.

20 An a/c with seating capacity of 150, what is the number of flight attendants required:

- **A** 2
- **B** 3
- **C** 4

Answer: C. C — 4. Formula: 2 + 1 per unit of 50 above 99. Seats above 99 = 51 → 2 units → 2 + 2 = 4 (Rule 38B)

21 Every member of the Flight Crew shall get his log book certified for correctness by the competent authority:

- **A** At the end of each calendar month
- **B** At the end of every quarter
- **C** At a half yearly interval ending Jun and Dec

Answer: B. B — At the end of every quarter (Rule 67A(4))

22 The Pilot-in-Command may permit the access of internet services by passengers on board an aircraft in flight, through Wi-Fi on board, provided:

- **A** The manufacturer has certified the aircraft for internet usage
- **B** The aircraft is certified by DGCA for Wi-Fi and flying in VMC and pilot is in contact with ground
- **C** The Director-General has certified the aircraft for usage of internet service in flight through Wi-Fi on board

Answer: C. C — Director-General has certified the aircraft (Rule 29B(3))

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23 If any person contravenes any of the provisions of the Indian aircraft Act or the rules made thereunder:

- **A** The licence, certificate or approval issued to such person may be suspended or cancelled
- **B** The person may be arrested and will not be eligible for bail
- **C** The person may be imprisoned for up to one year or fined ₹1 crore in lieu

Answer: A. A — Licence, certificate or approval may be suspended or cancelled (Sec 10B)

24 No unauthorised person must be present within ___ mts of a/c while refuelling is on:

- **A** 15
- **B** 30
- **C** 50

Answer: A. A — 15 meters (Rule 25A(2)(h))

25 Pilot's log book is to be preserved for:

- **A** 5 years from the date of starting the log book
- **B** 5 years from the date of last entry
- **C** 3 years from the date of last entry

Answer: B. B — 5 years from the date of last entry (Rule 67A(3))

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26 Operational Manual is:

- **A** Prepared by the operator and authorized by the DGCA
- **B** Prepared by the DGCA and authorized by operator
- **C** Prepared by the operator and authorized by manufacturer

Answer: A. A — Prepared by the operator and authorised by DGCA (Rule 140B)

Chapter 14 — Personnel Licensing

Practice Questions

1 An applicant for an Airline Transport Pilot Licence shall have completed in aeroplanes not less than:

- **A** 100 hours of instrument time, of which not more than 50 hours may be instrument ground time
- **B** 150 hours of instrument time, of which not more than 75 hours may be instrument ground time
- **C** 100 hours of instrument time, of which not more than 30 hours may be instrument ground time

Answer: A.

2 The holder of a pilot's licence should inform the Authority of any illness which they are suffering which involves incapacity to undertake those functions to which the licence relates throughout a period of a certain number of days or more. The number of days is:

- **A** 15
- **B** 20
- **C** 30

Answer: A.

3 The International Civil Aviation Convention Annex containing standards and recommended practices for Personnel Licensing is:

- **A** Annex 1
- **B** Annex 2
- **C** Annex 11

Answer: A.

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4 The minimum age of obtaining a PPL is:

- **A** 18 years
- **B** 17 years
- **C** 21 years

Answer: B.

5 The privileges of the holder of a Commercial Pilot Licence — aeroplane shall be:

- **A** to act as pilot-in-command in any aeroplane engaged in commercial air transportation
- **B** to act as pilot-in-command in any aeroplane certificate for single pilot operation other than in commercial air transportation
- **C** to act as Pilot-in-Command of any aeroplane having an all-up-weight not exceeding 5700 kg and which is entered in the aircraft rating of his licence

Answer: C.

6 The duration of the period of currency of a medical assessment shall begin on the date:

- **A** the license is issued or renewed
- **B** reckoned from the date of medical examination
- **C** the license is delivered to the pilot

Answer: B. DGCA CAR Section 7 Series C Part I: the date of the medical assessment is reckoned from the date of the medical examination.

7 Type ratings shall be established:

- **A** for any type of aircraft whenever considered necessary by the authority
- **B** all the answers are correct
- **C** only aircraft certificated for operation with a minimum crew of at least two pilots

Answer: A.

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8 An applicant holding a private or commercial pilot license aeroplane for the issue of an instrument rating shall have completed …… hours of cross-country flight time as pilot-in-command of aircraft in categories acceptable to the licensing Authority:

- **A** 20 hours
- **B** 40 hours
- **C** 50 hours

Answer: C.

9 An applicant for an Airline Transport Pilot Licence shall have completed in aeroplanes not less than:

- **A** 75 hours of night flight as pilot-in-command or as copilot
- **B** 100 hours of night flight as pilot-in-command or as copilot
- **C** 100 hours of night flight only as pilot-in-command

Answer: B.

10 For Commercial Pilot Licence aeroplane the applicant shall have completed in aeroplanes not less than ….. if the privileges of the licence are to be expected at:

- **A** 5 hours of night flight time including 3 take-offs and 3 landings as pilot-in-command
- **B** 5 hours of night flight time including 5 take-offs and 5 landings as pilot-in-command
- **C** 5 hours of night flight time including 10 take-offs and 10 landings as pilot-in-command

Answer: C.

11 An applicant for a Commercial Pilot License aeroplane shall have completed in aeroplanes not less than:

- **A** 150 hours of flight time and 100 hours as pilot-in-command
- **B** 200 hours of flight time and 70 hours as pilot-in-command
- **C** 200 hours of flight time and 100 hours as pilot-in-command

Answer: C.

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12 An applicant for a Commercial Pilot Licence shall have completed in aeroplanes not less than:

- **A** 20 hours of instrument instruction time of which not more than 10 hours may be instrument ground time
- **B** 20 hours of instrument instruction time of which not more than 5 hours may be instrument ground time
- **C** not less than ten hours of instrument time of which not more than five hours may be on an approved simulator

Answer: C.

13 When the holders of aircraft transport pilot licenses aeroplane and helicopter have passed their 40th birthday the medical examination shall be reduced to:

- **A** 6 months for single crew commercial operations
- **B** 6 months for all commercial operations
- **C** 6 months for all operations

Answer: A.

14 The applicant for an Airline Transport Pilot License shall have completed in aeroplanes not less than …. hours of cross-country flight time, including not less than …. hours of flight time by night as Pilot-in-Command or as Co-Pilot performing, under the supervision of a pilot who fulfils the flying experience requirements of a check pilot, the duties and functions of a pilot-in-command.

- **A** 200 hours and 50 hours
- **B** 250 hours and 50 hours
- **C** 200 hours and 75 hours

Answer: A.

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15 An applicant for a Commercial Pilot Licence — aeroplane shall have completed not less than …. hours of cross country flight time as pilot-in-command including a cross country flight totaling not less than …. km (…. NM), in the course of which full stop landings at two different aerodromes shall be made. The hours and distance referred are:

- **A** 15 hours and 540 km (300 NM)
- **B** 20 hours and 540 km (300 NM)
- **C** 10 hours and 270 km (150 NM)

Answer: B.

16 An applicant for a Commercial Pilot Licence shall hold a current:

- **A** class 1 medical assessment
- **B** class 3 medical assessment
- **C** class medical assessment as prescribed by the state issuing the licence

Answer: A.

17 The age of an applicant for a Commercial Pilot Licence shall not be less than:

- **A** 17 years of age
- **B** 18 years of age
- **C** 16 years of age

Answer: B.

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18 To be able to execute a public transport flight, the minimum and maximum age (with ATPL) is:

- **A** 21 and 65 years
- **B** 16 and 60 years
- **C** 18 and 60 years

Answer: A.

19 Which of the following Annexes to the Chicago convention contains minimum specifications for a crew licence to have international validity?

- **A** Annex 1
- **B** Annex 3
- **C** Annex 4

Answer: A.

20 An applicant for the ____ licences shall have the ability to speak and understand the English language used for radiotelephony communications to the level of proficiency specified by the Director General.

- **A** PPL, CPL, ATPL, Flt Navigator and Flt Engg
- **B** PPL, CPL, ATPL only
- **C** PPL only

Answer: A.

21 The privileges of the holder of a Private Pilot's Licence shall be to act as:

- **A** Pilot-in-command or as Co-pilot of any aeroplane provided no passengers are carried
- **B** Pilot-in-command or as Co-pilot of any aeroplane and carry passengers therein without remunerations
- **C** Pilot-in-command or as Co-pilot of any aeroplane employed on aerial work for remuneration

Answer: B.

22 If the age of a CPL holder flying a single pilot helicopter for charters is more than 40 years the validity of medical is for ____.

- **A** 6 months
- **B** 12 months
- **C** 12 months if he is not carrying VIPs

Answer: A.

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Chapter 15 — Airworthiness of Aircraft

Practice Questions

1 The continuing airworthiness of an aircraft, according to ICAO Annex 8, shall be determined by:

- **A** ICAO
- **B** The operator's state
- **C** The state of registry

Answer: C. Responsibility for an aircraft's airworthiness — including its continuing airworthiness — rests with the State of Registry (the State on whose register the aircraft is entered). ICAO only sets broad Standards; it does not itself determine the airworthiness of an individual aircraft.

2 According to ICAO Annex 8, a certificate of airworthiness shall be renewed or shall remain valid subject to the:

- **A** Laws of the state of registry and operation
- **B** Laws of the state of registry
- **C** Requirements laid down by ICAO

Answer: B. The C of A is tied to the State of Registry . Its renewal and continued validity are governed by the laws of the State of Registry — the same State that is responsible for the aircraft's airworthiness requirements. The State of operation does not control the C of A's validity.

3 The technical standards dealing with certification of aeroplanes are at present limited to multi-engine aircraft of:

- **A** Up to 5700 Kg maximum certificated take-off and landing mass
- **B** Over 5700 Kg maximum certificated take-off mass
- **C** Up to 5700 Kg maximum certificated take-off mass

Answer: B. The technical Standards apply to multi-engined aeroplanes of OVER 5 700 kg maximum certificated take-off mass . The trap options use "up to" instead of "over", and one adds "and landing mass" — the rule speaks only of take-off mass .

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4 When an aircraft has sustained damage, the aircraft shall be allowed to resume its flight, if:

- **A** The state of registry, the state of design and the state of manufacture consider that the aircraft is still airworthy
- **B** The state of manufacture informs the state of registry that the damage sustained is of a nature such that the aircraft is still airworthy
- **C** The state of registry considers that the damage sustained is of a nature such that the aircraft is still airworthy

Answer: C. Airworthiness decisions for an aircraft are made by its State of Registry . After damage, it is the State of Registry that judges whether the damage is of a nature that leaves the aircraft still airworthy — and only then may the flight resume. Options needing the State of Design or State of Manufacture to also decide are incorrect.

5 Jet Airways flight is to take off from London to Bosnia. C of A will be as per the rules of:

- **A** UK
- **B** India
- **C** Bosnia

Answer: B. The C of A always follows the State of Registry , not the departure or destination State. Jet Airways is an Indian operator with aircraft registered in India , so the C of A is governed by India's rules — even though the flight operates between London (UK) and Bosnia.

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Memory Hook

"Def — Cert — Design — Heli" → Part I Definitions, Part II Certification & continuing airworthiness procedures, Part III new large aeroplane Designs, Part IV Helicopters.

Chapter 16 — Operational Procedures

Practice Questions

1 Flight Duty Period and Flight Time Limit for a 2-crew international scheduled flight is _____ hrs, _____ hrs and _____ landings.

- **A** 11, 8 and 6.
- **B** 12, 8 and 5.
- **C** 13, 8 and 4.

Answer: A.

2 A pilot can fly in 28 consecutive days

- **A** 100 hrs PIC & 50 hrs as Co-pilot
- **B** 100 hrs
- **C** 120 hrs

Answer: B.

3 The maximum duty time limit per day extension for a scheduled flight is

- **A** 1.5 hrs
- **B** 2 hrs
- **C** 3 hrs

Answer: B.

4 Max flying hours for a flight crew in 90 consecutive days is:

- **A** 200 hrs.
- **B** 300 hrs
- **C** 330 hrs

Answer: B.

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5 With regard to flight time limitations, "flight time" means:

- **A** The total time from when an aircraft first moves under its own power for the purpose of taking off until it comes to rest after the flight
- **B** The time between take-off and landing
- **C** The time between an aircraft starting to move and coming to a complete stop plus one and a half hours

Answer: A.

6 The maximum permitted flight time for flight crew is:

- **A** 81 hours in the 27 days prior to the flight
- **B** 120 hours in the 27 days before the current flight
- **C** 1000 hours in 365 consecutive days

Answer: C.

7 A flight to be operated with a pressurized aeroplane will not be commenced unless a sufficient quantity of stored breathing oxygen is carried to supply:

- **A** All crew members
- **B** All crew members and passengers
- **C** All crew members and passengers, when the atmospheric pressure in any compartment occupied by them is less than 700 hPa

Answer: C.

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8 Number of first aid kits to be carried on board an aircraft carrying 250 passengers is:

- **A** 1
- **B** 2
- **C** 3

Answer: C.

9 For an unplanned event, such as unforecast weather, equipment malfunction, or air traffic delay that is beyond the control of the operator, 'Flight Time' and 'Flight Duty Period' for scheduled operations may be extended as follows:

- **A** Flight Time by max 2 hours and FDP by max 4 hours, subject to a cumulative limit of max 8 hours and max 16 hours respectively in 30 consecutive days.
- **B** Flight Time by max 4 hours and FDP by max 8 hours.
- **C** Flight Time by max 1 hour and FDP by max 2 hours

Answer: C.

10 Minimum trip fuel required shall be the amount of fuel required to enable the aeroplane to fly from take-off or the point of in-flight re-planning,

- **A** until landing at the destination aerodrome taking into account the operating conditions.
- **B** until overhead the destination aerodrome taking into account the operating conditions.
- **C** until overhead the destination aerodrome taking into account the fuel required to go to a diversion.

Answer: A.

11 If, after commencing an instrument approach, the reported RVR/Visibility falls below the applicable minimum, the approach shall not be continued:

- **A** below MDA/DA, as applicable
- **B** below AOM published for the aerodrome.
- **C** below 1 000 ft above the aerodrome.

Answer: C.

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12 All FDRs shall be capable of retaining the information recorded during at least the last ____ hours of their operation. CVRs at least ____ hrs and Data link recorders shall be capable of retaining the information recorded during at least the last ____ hours of their operation.

- **A** 24, 2, 4
- **B** 25, 2, 2
- **C** 25, 3, 2

Answer: B.

13 Filed minima is 300 m. Touch down zone RVR is 400 m. Relevant mid-point RVR is 100 m and stop-end RVR 300 m. The runway is:

- **A** Below minima
- **B** Within minima
- **C** Equal to minima.

Answer: A. DGCA CAR Section 8 Series C Part I, 4.6.5: the touchdown zone RVR always controls. If reported and relevant, the mid-point must be at least 125 m (or the touchdown value if that is less) and the stop-end at least 50 m. Here the mid-point is 100 m, so the runway is below minima.

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14 CARS is a:

- **A** Light weight voice recorder.
- **B** Heavy weight flight recorder.
- **C** Crash protected flight recorder.

Answer: A. DGCA CAR Section 2 Series I Part V, 3.2: lightweight flight recorders include the cockpit audio recording system (CARS).

Memory Aid

Think **"1-6-9 / 8-6"** — General Aviation Part came in **'69**, Helicopters in **'86**. Part I (the big airline part) is the original core. Helicopters were added *last* because the rules for them matured later.

Memory Aid

"**One hundred up, four hundred out.**" Add **100 ft** to the decision/minimum descent height and **400 m** to the required visibility/RVR. Both additives apply together.

Memory Aid

"**10% or 3 — take the bigger.**" If 10% of a rating works out to less than 3, you must still carry 3 of that rating.

Memory Aid

PBE protects the "**three openings**" — eyes, nose, mouth — for at least **15 minutes**. The 15-minute figure applies to *both* flight crew and cabin crew PBE.

Memory Aid

Better rest facility = longer FDP allowed. A **Bunk** (lie-flat) always permits the highest FDP; an **Isolated Rest Seat** is next; a **Basic Rest Seat** the least — and is **not available (NA)** for 4-crew or ULR operations.

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Chapter 17 — Special Operational Procedures and Hazards

Practice Questions

1 MMEL is drawn up by:

- **A** The operator and approved by the certification authority
- **B** The manufacturer (the organisation responsible for the type design) and approved by the State of Design
- **C** The operator from a main list drawn up by the manufacturer

Answer: B. DGCA CAR Section 8 Series O Part II (definitions): the master minimum equipment list is established by the organisation responsible for the type design, with the approval of the State of Design. The operator then writes its own minimum equipment list from it, and DGCA approves that.

2 The Minimum Equipment List (MEL) is established by:

- **A** The airline operator
- **B** The aeronautical authority the airline operator depends on
- **C** The manufacturer

Answer: A. The operator tailors the MMEL into its own MEL (DGCA-approved).

3 A piece of equipment fails while still parked. The reference document used first to decide the procedure is:

- **A** The minimum equipment list
- **B** The flight manual
- **C** The operation manual's chapter "Abnormal and Emergency Procedures"

Answer: A. The MEL gives the go/no-go despatch decision for unserviceable items.

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4 The minimum equipment list of a public transport airplane is to be found in the:

- **A** Flight manual
- **B** Operation manual
- **C** CAR OPS

Answer: B. DGCA CAR Section 8 Series O Part II, 6.1.3: the operator shall include in the Operations Manual a minimum equipment list approved by the DGCA.

5 In public transport, prior to take-off in icing conditions, the captain must check that:

- **A** External surfaces are free from any ice accretion which may impede performance and manoeuvrability, except within the limits specified by the flight manual
- **B** External surfaces are free from any ice accretion greater than 5 mm
- **C** Possible ice accretions do not cause weight and balance limits to be exceeded

Answer: A. Surfaces must be clean except for amounts the AFM specifically permits.

6 Accumulation of snow or ice on an aircraft in flight induces an increase in the:

- **A** Value of the stall angle of attack
- **B** Roll rate
- **C** Stalling speed

Answer: C. Contamination destroys lift, so the wing stalls at a higher speed.

7 Which requirement should be met when planning a flight with icing conditions:

- **A** Flight planned so a change of cruising level can be initiated rapidly
- **B** The aircraft shall be sprayed with anti-icing fluid before flight
- **C** The aircraft shall be equipped with approved ice-protection systems

Answer: C. The aircraft must be certificated and equipped for icing conditions.

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8 As captain, after take-off you notice a flock of birds which may present a bird strike hazard. You must:

- **A** Immediately inform the appropriate ground station
- **B** Draft a bird strike hazard report upon arrival within at most 48 hours
- **C** Inform the appropriate ground station within a reasonable period of time

Answer: A. Warning following traffic immediately is the only effective defence.

9 Who is responsible for establishing operating procedures for noise abatement during instrument flight in compliance with ICAO PANS-OPS?

- **A** Operator
- **B** State in which the aeroplane is operating
- **C** Commander

Answer: A. The operator specifies the noise-abatement procedure for each aeroplane type.

10 About procedures for noise attenuation during landing:

- **A** Such procedures will not involve the prohibition of using reverse thrust
- **B** Such procedures do not exist
- **C** They prohibit the use of reverse thrust

Answer: A. Only reverse thrust above reverse idle may be restricted — idle reverse is never fully prohibited.

11 Noise abatement shall not be the determining factor in runway designation when: (1) crosswind incl. gust >15 kt; (2) tailwind incl. gust >5 kt; (3) runway not clear or dry. Correct combination:

- **A** 1, 3
- **B** 2, 3
- **C** 1, 2, 3

Answer: C. Any one of the three conditions removes noise abatement's say.

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12 In establishing noise preferential routes:

- **A** No turns should be required coincident with a reduction of power associated with a noise-abatement procedure
- **B** Turns during take-off/climb not required unless bank angle limited to 28° (climbing at V2 + 10 to 20 kt)
- **C** Turns during take-off/climb not required unless the aeroplane has reached and can maintain a height of no less than 100 m above terrain and obstacles

Answer: A. Turning while reducing power compounds workload — it is avoided.

13 What precludes a runway being used for noise abatement procedures if landing in VMC?

- **A** Tailwind up to 3 kts
- **B** No ILS or visual guidance
- **C** Crosswind including gusts of 10 kts

Answer: B. Absence of ILS/visual guidance rules a runway out for VMC noise abatement.

14 During a noise-abatement approach, the aeroplane is to be in the final landing configuration after passing the ___ or at a point ___ from the threshold, whichever is earlier:

- **A** Final marker, 6 NM
- **B** Middle marker, 5 NM
- **C** Outer marker, 5 NM

Answer: C. Stabilised by the outer marker or 5 NM — whichever comes first.

15 To extinguish a fire in the cockpit you use: (1) water; (2) powder/chemical; (3) halon; (4) CO_2. Correct combination:

- **A** 2, 3, 4
- **B** 1, 2
- **C** 3, 4

Answer: C. Halon and CO_2 are non-conducting and clean — safe around electrics.

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16 A Class A fire is a fire of:

- **A** Electrical origin
- **B** Solid material, generally of organic nature
- **C** Liquid or liquefiable solid

Answer: B. Class A = wood, paper, cloth, plastic — ordinary solids.

17 After landing, in case of high temperature of the brakes you:

- **A** Release the parking brake and approach the wheels sidewards
- **B** Release the parking brake and approach the wheels either from aft or fore
- **C** Apply the parking brake and approach the wheels sidewards

Answer: B. A bursting tyre throws energy sideways — stay fore/aft.

18 H_2O extinguishers are fit to fight:

- **A** Class A fires
- **B** Electrical source fires
- **C** Class B fires

Answer: A. Water cools Class A fires; it is dangerous on B and C.

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19 Fire-extinguisher types which may be used on Class B fires: (1) H_2O; (2) CO_2; (3) Dry-chemical; (4) Halogen. Correct combination:

- **A** 3, 4
- **B** 1, 2, 3, 4
- **C** 2, 3, 4

Answer: C. Class B (flammable liquid) cannot be fought with water.

20 Following a heavy-mass landing on a short runway, you should check the:

- **A** Temperature of the brakes
- **B** Pressure of the hydraulic fluid
- **C** Pressure of the pneumatic tyres

Answer: A. A heavy stop dumps huge kinetic energy as heat into the brakes.

21 The correct statement about extinguishing agents on board aeroplanes is:

- **A** Water may only be used for minor fires
- **B** Burning cargo in cargo-aeroplanes is usually extinguished by carbon dioxide
- **C** Halon is an effective extinguishing agent for use in aeroplanes

Answer: C. Halon is highly effective — three times CO_2 for the same quantity.

22 If smoke appears in the air conditioning, the first action to take is to:

- **A** Determine which system is causing the smoke
- **B** Begin an emergency descent
- **C** Put on the mask and goggles

Answer: C. Protect the crew first — then diagnose and act.

23 Beneath fire extinguishers, the following equipment for fire fighting is on board:

- **A** A hydraulic winch and a big box of tools
- **B** A big bunch of fire-extinguishing blankets
- **C** Crash axes or crowbars

Answer: C. Crash axes/crowbars allow access to hidden fires behind panels.

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24 A Class B fire is a fire of:

- **A** Liquid or liquefiable solid
- **B** Solid material usually of organic nature
- **C** Special fire: metal, gas, chemical product

Answer: A. Class B = flammable liquids, oil, fuel, hydraulic fluid.

25 The system which must be switched off in case of a belly compartment fire is generally the:

- **A** Ventilation of the cargo compartment
- **B** Trim air
- **C** Pressurization

Answer: A. Cutting ventilation deprives the cargo fire of oxygen.

26 You will use a water fire-extinguisher (straight jet) on a fire of: (1) solid (fabric, carpet); (2) liquids (ether, gasoline); (3) gas; (4) metals (sodium). Correct combination:

- **A** 3
- **B** 1
- **C** 2

Answer: B. A straight water jet suits only Class A solids.

27 You will use a halon extinguisher for a fire of: (1) solids; (2) liquids; (3) gas; (4) metals. Correct combination:

- **A** 1, 2, 4
- **B** 1, 2, 3, 4
- **C** 1, 2, 3

Answer: C. Halon covers A, B and C — never use it on metals (Class D).

28 A 1211 halon fire-extinguisher can be used for: (1) paper; (2) fabric; (3) electric; (4) wood; (5) hydrocarbon. Correct combination:

- **A** 2, 4, 5
- **B** 2, 3, 4
- **C** 1, 2, 3, 4, 5

Answer: C. BCF/Halon 1211 is effective across Class A, B and C.

29 A water fire-extinguisher can be used without restriction for: (1) paper; (2) hydrocarbon; (3) fabric; (4) electrical; (5) wood. Correct combination:

- **A** 2, 3, 4
- **B** 1, 2, 3, 4, 5
- **C** 1, 3, 5

Answer: C. Paper, fabric and wood are Class A — water's safe domain.

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30 After a landing with overweight and over-speed conditions, the fireguards should approach the landing-gear tyres:

- **A** Only from left or right sides
- **B** From any side
- **C** Only from front or rear side

Answer: C. A tyre burst releases energy sideways — approach fore/aft.

31 For a flight deck fire, which do you use? (1) BCF; (2) Halon; (3) Dry Powder; (4) Water.

- **A** 1, 2 & 3
- **B** 1, 2, 3 & 4
- **C** 1 & 2

Answer: C. BCF/Halon are clean, non-conducting agents safe on the flight deck.

32 An aircraft is configured for seating 61 to 200 passengers. The requirement for hand-held fire extinguishers is:

- **A** 4 conveniently located in the passenger compartment
- **B** 2 conveniently located in the passenger compartment
- **C** 3 conveniently located in the passenger compartment

Answer: C. 61–200 seats requires three conveniently located extinguishers.

33 The number of hand fire extinguishers in the passenger compartment when seating configuration is between 401 and 500 is:

- **A** 3
- **B** 6
- **C** 5

Answer: B. 401–500 seats requires six conveniently located extinguishers.

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34 A water fire extinguisher with a directed spray can be used on which fires?

- **A** Gas fires
- **B** Liquid fires
- **C** Solid fires

Answer: C. Water targets Class A solids only.

35 You will use a Halon extinguisher for a fire of: (1) solids; (2) liquids; (3) gas; (4) metals. Correct combination:

- **A** 2, 3, 4
- **B** 1, 2, 4
- **C** 1, 2, 3

Answer: C. Halon = Class A, B, C — never on metals (Class D).

36 Fire extinguishers should be located in the pilots' compartment and…

- **A** At each passenger compartment that is separate from the pilots' compartment and not readily accessible to the flight crew
- **B** In the passenger cabin
- **C** At each door

Answer: A. Each separate passenger compartment needs its own extinguisher.

37 If cabin altitude increases during level flight, the differential pressure:

- **A** Decreases
- **B** Remains constant
- **C** Increases

Answer: A. Higher cabin altitude = lower cabin pressure = smaller differential.

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38 Minimum requirements for supplemental oxygen during/following an emergency descent: for pilots it shall be available for the entire flight time the cabin pressure altitude exceeds a minimum of X feet. X is:

- **A** 13000 ft
- **B** 14000 ft
- **C** 15000 ft

Answer: A. DGCA CAR Section 8 Series O Part II, 4.3.9.1: oxygen must be carried for all crew for any period the cabin pressure is below 620 hPa, which is 13,000 ft.

39 A slow decompression may be caused by:

- **A** Loss of a cabin window
- **B** A leak in a door seal during normal pressurised flight
- **C** Loss of a door

Answer: B. A small seal leak bleeds pressure slowly; a lost door is sudden.

40 At FL290 in straight & level flight, a small leak causes a slow depressurisation. The cabin rate-of-climb indicator will show:

- **A** A rate of climb
- **B** Zero
- **C** A rate of descent of approximately 300 fpm

Answer: A. Falling cabin pressure reads as a rising (climbing) cabin altitude.

41 An aeroplane suffers explosive decompression at 31000 ft. The initial action by the operating crew is:

- **A** Place the seat belt signs to ON
- **B** To put on oxygen masks
- **C** Transmit a MAYDAY message

Answer: B. Crew oxygen first — hypoxia incapacitates within seconds at altitude.

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42 A slow decompression may be caused by: (1) cracked window; (2) bad functioning of pressurization; (3) minor leak in fuselage; (4) loss of a door. Correct combination:

- **A** 1, 2
- **B** 3, 4
- **C** 1, 2, 3

Answer: C. Loss of a door is rapid/explosive, not slow.

43 A fast decompression is recognizable by: (1) mist in the cabin; (2) blast towards the exterior; (3) expansion of body gases; (4) blast of air released violently from the lungs. Correct combination:

- **A** 1, 4
- **B** 1, 2, 3, 4
- **C** 1, 2, 3

Answer: B. All four are recognised signatures of a fast decompression.

44 Supplemental oxygen is used to:

- **A** Provide people on board with oxygen during a cabin depressurisation
- **B** Protect a crew who fights a fire
- **C** Assist a passenger with breathing disorders

Answer: A. Its purpose is preventing hypoxia during loss of pressurisation.

45 Oxygen should be used after rapid decompression in an emergency descent until what altitude?

- **A** 14000 ft
- **B** 13000 ft
- **C** 10000 ft

Answer: C. 10,000 ft is the altitude at which oxygen is no longer required.

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46 In case of an unexpected encounter with windshear, you will: (1) set maximum take-off thrust; (2) increase pitch-up to the limit actuating the stick shaker; (3) pull in the drag devices (gear/flaps); (4) keep the airplane's current configuration; (5) reach maximum lift-to-drag ratio. Correct combination:

- **A** 1, 2, 4
- **B** 1, 3, 5
- **C** 3, 5

Answer: A. Full power, max safe pitch, configuration unchanged — do not retract gear/flap.

47 If you encounter a microburst just after taking off, at the beginning you will have: (1) head wind; (2) strong rear wind; (3) better climb performance; (4) diminution of climb gradient; (5) important thrust drop. Correct combination:

- **A** 1, 4
- **B** 1, 3
- **C** 4, 5

Answer: B. The initial headwind momentarily improves climb performance.

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48 During a landing approach, the aircraft is subjected to windshear with an increasing tailwind. In the absence of pilot action, the aircraft: (1) flies above glide path; (2) flies below glide path; (3) increasing TAS; (4) decreasing TAS. Correct combination:

- **A** 2, 3
- **B** 1, 4
- **C** 2, 4

Answer: C. Increasing tailwind robs airspeed — aircraft sinks below the glide path.

49 To counter the effects of windshear, the amount of control action required is:

- **A** Substantial
- **B** Medium
- **C** Null

Answer: A. Recovery may need full power and a near-stall pitch attitude.

50 While approaching the outer marker, the tower informs you of a microburst. You will expect to encounter:

- **A** Convection motion of air mass
- **B** Supercooled water
- **C** Windshear (vertical and horizontal)

Answer: C. A microburst produces both vertical and horizontal wind shear.

51 One of the main characteristics of windshear is that it:

- **A** Occurs only at low altitude (2000 ft) and never in the horizontal plane
- **B** Can occur at any altitude in both the vertical and horizontal planes
- **C** Can occur at any altitude and only in the horizontal plane

Answer: B. Wind shear is not altitude- or plane-restricted.

52 Which magnitude will be the first to change its value when penetrating a wind shear?

- **A** Indicated airspeed
- **B** Groundspeed
- **C** Vertical speed

Answer: A. A sudden wind change is felt first as an IAS change.

53 An aircraft with a 40 kt headwind heading towards the centre of a microburst may expect, on crossing it, a windshear of:

- **A** 40 kt
- **B** 80 kt
- **C** 20 kt

Answer: B. A 40 kt headwind reverses into a 40 kt tailwind — an 80 kt shear.

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54 When a commercial transport passenger airplane is equipped with a door in the flight-crew compartment area, this door must include:

- **A** A locking system to prevent any unauthorized access
- **B** A sealing system allowing maintenance of cockpit pressure during a depressurization
- **C** Distinctive red or yellow coloured markings indicating the access area

Answer: A. The flight-deck door must lock out unauthorised access.

55 The State in which an aircraft has landed after an act of unlawful interference must immediately notify the:

- **A** State of Registry of the aircraft, the State of the operator and ICAO
- **B** State of Registry of the aircraft and the State of the operator only
- **C** State of Registry of the aircraft and the DGCA

Answer: A. Registry State, operator State and ICAO must all be informed.

56 In case of a hi-jack, the squawk code is:

- **A** 2000
- **B** 7500
- **C** 7700

Answer: B. 7500 = unlawful interference; 7700 = general emergency; 7600 = radio failure.

57 The flight deck door should be capable of being:

- **A** Remotely locked from either inside or outside the compartment
- **B** Remotely locked by cabin crew operation from outside the compartment
- **C** Locked from within the compartment

Answer: C. Control of the lock stays with the flight crew, from inside.

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58 When flight crew members are at their duty stations they must:

- **A** Wear a communications head set
- **B** Keep the seat belts fastened
- **C** Fully raise the seat

Answer: B. Seat belts must stay fastened at the duty station.

59 The State where an aircraft has landed after an act of unlawful interference must immediately notify the:

- **A** State of Registry of the aircraft, the State of the operator and ICAO
- **B** State of Registry of the aircraft and the State of the operator only
- **C** State of the operator, the J.A.A. and ICAO

Answer: A. Registry State, operator State and ICAO.

60 Mist in the cabin, pressure and temperature drop characterize:

- **A** A fast depressurization
- **B** A plastic fire
- **C** A slow depressurization

Answer: A. Sudden fog plus temperature drop are fast-decompression signatures.

61 Products are considered dangerous goods if defined as such by:

- **A** The IATA document "Regulations governing the transportation of dangerous goods by air"
- **B** The directive of the Community Union
- **C** The ICAO document "Technical Instructions for the safe transportation of dangerous goods by air"

Answer: C. The ICAO Technical Instructions are the defining reference.

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62 ICAO Appendix 18 is a document dealing with:

- **A** The air transport of live animals
- **B** The safety of the air transport of hazardous materials
- **C** The noise pollution of aircraft

Answer: B. Annex/Appendix 18 governs the safe transport of dangerous goods.

63 A runway is considered damp when:

- **A** Its surface is not dry, and surface moisture does not give a shiny appearance
- **B** It is covered with a film of water of less than 3 mm
- **C** Surface moisture gives it a shiny appearance

Answer: A. Damp = not dry but no shine.

64 A runway is considered wet when: (1) covered with water/loose-slushy snow ≤ 3 mm equivalent; (2) moisture modifies colour but not shiny; (3) moisture makes it reflective but no large stagnant sheets; (4) bears stagnant sheets of water. Correct combination:

- **A** 1, 2
- **B** 1, 2, 3
- **C** 1, 3

Answer: C. Wet = up to 3 mm or reflective — stagnant sheets mean flooded/contaminated.

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65 A runway covered with 4 mm thick water is said to be:

- **A** Contaminated
- **B** Flooded
- **C** Damp

Answer: A. More than 3 mm of water makes the runway contaminated.

66 A runway is considered damp when:

- **A** It is covered with a film of water of less than 1 mm
- **B** Its surface is not dry, and surface moisture does not give it a shiny appearance
- **C** It is covered with a film of water of less than 3 mm

Answer: B. Damp is defined by the absence of shine, not a water depth.

67 The effect whereby a tyre is lifted from the runway due to aeroplane speed along the runway is known as:

- **A** Surface tension
- **B** Hydroplaning
- **C** Aqua-skimming

Answer: B. Hydroplaning lifts the tyre onto a film of water.

68 Viscous hydroplaning is caused by:

- **A** A smooth and dirty runway surface
- **B** Bald tyres
- **C** A rough runway surface

Answer: A. A thin film on a smooth, dirty surface causes viscous hydroplaning.

69 What is the shortest distance in a sequence for landing between a "Heavy" aircraft preceding a "Light" aircraft?

- **A** 3 NM
- **B** 6 NM
- **C** 2 km

Answer: B. Distance-based minimum for Heavy followed by Light is 6 NM.

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70 A separation minimum applies between a LIGHT/MEDIUM and a HEAVY (and between LIGHT and MEDIUM) when the heavier aircraft makes a low/missed approach and the lighter lands opposite-direction on the same or a parallel runway separated by:

- **A** Less than 730 m
- **B** Less than 760 m
- **C** 730 m

Answer: B. The trigger spacing is parallel runways less than 760 m apart.

71 Same scenario, but the lighter aircraft is utilising an opposite-direction runway for take-off. This minimum is:

- **A** 2 minutes
- **B** 3 minutes
- **C** 1 minute

Answer: A. A 2-minute minimum applies for the opposite-direction take-off case.

72 To meet wake turbulence criteria, what minimum separation applies when a MEDIUM aircraft takes off behind a HEAVY aircraft, both using the same runway?

- **A** 1 minute
- **B** 3 minutes
- **C** 2 minutes

Answer: C. Heavy leading, Medium following, same runway = 2 minutes.

73 For arriving aircraft using timed approaches, what minima apply to an aircraft landing behind a heavy or medium aircraft?

- **A** Medium behind other medium — 2 minutes
- **B** Medium behind heavy — 3 minutes
- **C** Medium behind heavy — 2 minutes

Answer: C. Arriving Heavy followed by Medium is 2 minutes (time-based table).

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74 What is the minimum wake turbulence separation when a LIGHT aircraft takes off behind a MEDIUM aircraft, both using the same runway?

- **A** 1 minute
- **B** 3 minutes
- **C** 2 minutes

Answer: C. Medium leading, Light following, same runway = 2 minutes.

75 The letter "L" is written in the wake-turbulence box of a flight-plan form when the MCTOW of an aircraft is less than or equal to:

- **A** 14000 kg
- **B** 20000 kg
- **C** 7000 kg

Answer: C. Light category = MCTOW of 7,000 kg or less.

Memory Hook

"15 – 5 – Dry." Crosswind 15 kt, tailwind 5 kt, runway not dry — any one of these and noise abatement loses its vote on runway selection.

Memory Hook

"Heavy, Clean, Slow" — the three words that mean maximum wake. The most dangerous wind while landing or taking off behind a large aircraft is a **light quartering tailwind**.

Amber / dashed boxes — Memory aids & exam traps.

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Chapter 18 — Aeronautical Communications

Memory Aid — Bearings & Tracks

- **QDM** — *"D for Direction to"* — Magnetic HEADING **TO** the station (no wind).
- **QDR** — *"R for Radial / from"* — Magnetic BEARING **FROM** the station.
- **QTE** — *"True"* — TRUE bearing FROM the station.
- **QUJ** — *"U-Track True"* — TRUE TRACK TO the station.
- **QFE** — sub-scale for **HEIGHT** above aerodrome.
- **QNH** — sub-scale for **ELEVATION** (i.e. altitude AMSL).
- **QNE** — sub-scale set to **1013.2 mb** / **29.92 in** = Standard / Flight Level.

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Chapter 19 — Aircraft Accident and Incident Investigation

Practice Questions

1 The objective of the investigation of an accident or incident shall be the:

- **A** Prevention of accidents or incidents and to help the judges
- **B** Prevention of accidents or incidents and to help the manufacturers in design
- **C** Prevention of accidents or incidents

Answer: C. . The sole objective of an investigation is the prevention of accidents and incidents — not to apportion blame or liability (Annex 13). Helping judges or helping manufacturers is not the objective. Refer to Section 1 above for the formal statement.

2 Which of the following, according to ICAO Annex 13, shall be entitled to appoint an accredited representative to participate in the investigation?

- **A** State conducting the investigation and State of design and manufacturing
- **B** All ICAO members States
- **C** Any State which, on request, provides information, facilities or experts to the State conducting the investigation

Answer: C. . Per Annex 13 — entitlement to appoint an accredited representative is tied to the State that contributes information, facilities, or experts on request.

3 Who is responsible, under Annex 13 of the Chicago Convention, for the initiation of an accident investigation?

- **A** The law enforcement authorities of the state in which the aircraft is registered
- **B** The aircraft manufacturer
- **C** The government of the state in which the accident took place

Answer: C. . Under Annex 13, initiation of investigation lies with the State of Occurrence — i.e. the government of the state where the accident physically happened.

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4 Just before arriving on the apron, taxiing inadvertently on the grass, a wheel falls into a hole, which seriously damages the aircraft and obliges the crew to delay departure.

- **A** This is an accident and the crew must follow the procedure relevant to this case
- **B** This is an irregularity in the operation, the crew must inform the operator of the delay caused by necessary repair
- **C** Since no physical injury has been noticed and the flight is over, the actions to be taken are related only to insurance, to the repair man, the operator and the persons in charge of the runways and taxiways

Answer: A. . The aircraft sustained damage that adversely affects structural strength / performance while persons were still on board with the intent of flight (definition of "accident" extends from boarding to disembarking). The damage is not among the listed exceptions (tires/brakes/wheels alone). Treat as an accident and follow the relevant procedure.

5 Where an accident or incident occurs to an aircraft in India, then the Pilot-in-Command of the aircraft shall send notice thereof to:

- **A** AAIB, DGCA, DM and O-i-C of nearest police station — as soon as is reasonably practicable but in any case not later than 24 hours
- **B** AAIB, DGCA, DM and O-i-C of nearest police station — as soon as is reasonably practicable but in any case not later than 48 hours
- **C** AAIB, DGCA, DM and AAI — as soon as is reasonably practicable but in any case not later than 24 hours

Answer: A. . The PIC must notify AAIB + DGCA + District Magistrate + O-i-C of the nearest Police Station , within 24 hours of becoming aware. The agency is the police station — not AAI.

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6 In the case of an accident, the aircraft or any parts or contents thereof may be removed without the authority of DG/AAIB:

- **A** To look for evidence
- **B** If so directed by the P-i-C
- **C** To extricate the body of a dead dog

Answer: C. . Under Clause (2) Proviso (a), aircraft/parts/contents may be moved by SAR-authorised personnel to extricate persons or animals — dead or alive . The other options (looking for evidence; PIC direction alone) are not permitted exceptions.

7 "An incident involving circumstances indicating that an accident or a serious incident could have occurred, if the risk had not been managed within safety margins" is known as?

- **A** Reportable Incident
- **B** Significant Incident
- **C** Flight Safety Incident

Answer: B.

8 CVR and airborne image recording data:

- **A** Can be used for disciplinary, civil, administrative and criminal proceeding against the employees, operational personnel or organizations
- **B** Can NOT be used for disciplinary, civil, administrative and criminal proceeding against the employees, operational personnel or organizations
- **C** Can be used for disciplinary, civil, administrative and criminal proceeding against the employees, operational personnel or organizations with the permission of DGCA

Answer: B. . The use of CVR & airborne image data is strictly protected — it shall not be used for any disciplinary, civil, administrative or criminal proceedings against employees, operational personnel or organizations. This protection preserves the safety-investigation purpose and protects "just culture".

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Memory Aid
Think of it as a **"near-near-miss"**: it would have escalated into an accident or serious incident, but timely risk management kept it within safe limits.

Memory Aid – Severity Ladder

Incident (occurrence)	→ Significant Incident (could have escalated) → Serious Incident (high prob. of accident) → ACCIDENT

Memory Aid – "24-Hour Rule"

Remember: TWO independent 24-hour clocks

- **Notification** of accident/incident to AAIB, DGCA, DM & police — **24 hrs** from PIC becoming aware.
- **Air Traffic Incident Report Form** submission — **24 hrs** after landing.

Try each question before you read the answer. Use the explanation to understand the rule or principle.

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Chapter 20 — Facilitation

Practice Questions

1 The ICAO Annex which deals with entry and departure of persons and their baggage, cargo and other articles on international flights is:

- **A** Annex 9
- **B** Annex 15
- **C** Annex 8

Answer: A. A — Annex 9 ("Facilitation"). Annex 15 = Aeronautical Information Services. Annex 8 = Airworthiness.

Memory Hook

- **Lading vs Unlading** — Lading is "loading on", Unlading is "removing off". Easy mnemonic: *"L = Load, U = Unload"*.
- **Embarkation vs Disembarkation** — both **exclude** through-passengers and through-crew. Only new boarders / new exiters count.
- **Cargo** — exam trap: cargo does **NOT** include mail, stores, or baggage.

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Chapter 21 — Security and Unlawful Interference

Practice Questions

1 A State shall take adequate measures for the safety of passengers and crew of an aircraft which is subjected to an act of unlawful interference,

- **A** Until their journey can be continued
- **B** During a period of investigation
- **C** And arrange for them to return to their country of origin

Answer: A. The Annex uses the exact wording " until their journey can be continued ". The duty of care extends until passengers and crew can resume their travel — not merely until an investigation finishes, and not until they are flown home.

2 When an aircraft subject to unlawful interference has landed, it shall notify by the most expeditious means of the State of registry of the aircraft and the State of the operator of the landing and shall similarly transmit all other relevant information to the:

- **A** Two aforementioned States, each State whose citizens suffered fatalities or injuries, each State whose citizens were detained as hostages, each State whose citizens are known to be on board the aircraft and the ICAO
- **B** Two aforementioned States, each State whose citizens suffered fatalities or injuries on board the aircraft and the ICAO
- **C** Two aforementioned States, each State whose citizens suffered fatalities or injuries, each State whose citizens are known to be on board the aircraft and the ICAO

Answer: A. Option A is the only one that includes all five categories: aforementioned States + fatalities/injuries + hostages + on-board citizens' States + ICAO. Options B and C drop the "hostages" line, which is the deliberate examiner trap.

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3 Referring to the operational aspects in the unlawful seizure acts, it can be said:

- **A** The contracting States will not assist with navigation aids, air transit services, etc, to an aircraft affected by an unlawful seizure act
- **B** The contracting State will make provisions to ensure that an aircraft affected by an unlawful seizure act, which has landed in their territory, would be retained in all cases
- **C** The contracting State will make provisions to ensure that an aircraft affected by an unlawful seizure act, which has landed in their territory, would be retained, unless its departure is justified to protect lives

Answer: C. Option A contradicts the duty to provide NAP assistance. Option B is too absolute — it omits the life-saving exception. Option C reproduces the exact rule: retain the aircraft, unless departure is justified to protect lives .

Memory Aid — The "5 Pillars" of Annex 17 Scope

Remember **"CC-CAA"**: **C**abin baggage, **C**hecked baggage, **C**argo & other goods, **A**ccess control, **A**irport design.

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Memory Aids & Quick-Recall Mnemonics

Try each question before you read the answer. Use the explanation to understand the rule or principle.

Mnemonic — Annex 17 Scope: "CC-CAA"

Cabin baggage • **C**hecked baggage • **C**argo & other goods • **A**ccess control • **A**irport design

Mnemonic — Five Notifications After Hijacked Aircraft Lands: "R-O-FIH-OB-I"

Registry State • **O**perator State • States of **F**atalities / **I**njuries / **H**ostages • **O**n-**B**oard citizens' States • **I**CAO

Mnemonic — Assistance to a Hijacked Aircraft: "NAP"

Navigation aids • **A**ir traffic services • **P**ermission to land

Mnemonic — Commissioner's Five Verbs: "E-D-I-M-R"

Establish • **D**evelop • **I**mplement • **M**aintain • **R**eview the national civil aviation security programme

Mnemonic — Indian JVC Airports: "DMC-HB"

DIAL=Delhi • **M**IAL=Mumbai • **C**IAL=Cochin • **H**IAL=Hyderabad • **B**IAL=Bangalore

Mnemonic — Two Conditions for Foreign Weapons: "Prior & PIC"

Prior State-to-State notification **+** Notification of the **PIC** (Pilot-in-Command)

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Chapter 22 — Human Factors: The Basics

Practice Questions

1 As a cause of accidents, the human factor:

- **A** Is cited in approximately 73% of aviation accidents.
- **B** Has increased accidents exponentially since 1970.
- **C** Has been generally due to high workload in the cockpit.

Answer: A.

2 The rate of accidents in commercial aviation:

- **A** Is approximately 52 accidents per million airport movements.
- **B** Has worsened considerably since 1970.
- **C** Is approximately 1 accident per million airport movements.

Answer: C.

3 Most aircraft accidents are linked to:

- **A** Deficiencies in human performance.
- **B** Technical failures.
- **C** Adverse weather conditions.

Answer: A.

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Memory aid — "SAFE PILOT"
Situation awareness · **A**ircraft handling · **F**lexibility · **E**ffective communication · **P**hysical fitness · **I**nformed decision making · **L**eadership (teamwork/CRM) · **O**rganisation (task management) · **T**hreat & error management.

Memory aid: *"Pilots Eat Pizza, So Always"* → **P**sychology, **E**ngineering, **P**hysiology, **S**ociology, **A**nthropometry.

Chapter 23 — Crew Resource Management, Threat and Error Management, LOFT

Practice Questions

1 Human errors:

- **A** Are now considered as being inherent to the cognitive function of human and can be totally eliminated by CRM training.
- **B** Are now considered as being inherent to the cognitive function of human and are generally inescapable.
- **C** Are likely to be eliminated in the near future by higher degree of automation.

Answer: B. The foundational premise of modern Human Factors and TEM is that errors are a normal and inescapable by-product of human cognition. CRM does not try to eliminate errors — it equips crews to manage them (avoid, trap, mitigate).

2 The error in checklist is related to an interface mismatch between:

- **A** Liveware – Liveware.
- **B** Liveware – Software.
- **C** Liveware – Environment.

Answer: B. Checklists are explicitly listed under the Liveware–Software interface as part of "Written Information" (along with manuals, NOTAMs, SOPs, charts). Therefore an error in a checklist is a human–software mismatch.

3 A congested cockpit restricting movement is an interface mismatch between:

- **A** Liveware – Software.
- **B** Liveware – Environment.
- **C** Liveware – Hardware.

Answer: C.

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Website YouTube Instagram
Telegram channel Telegram community Subject study group

4 High noise level in a cockpit is a mismatch between:

- **A** Liveware – Liveware.
- **B** Liveware – Environment.
- **C** Liveware – Hardware.

Answer: B. Noise is part of the environment. In the SHEL model the Environment is the conditions inside and outside the cockpit, so a high noise level is a Liveware-Environment mismatch.

5 The main consideration in modern airline operations should be:

- **A** Efficient utilisation of resources.
- **B** Maximum utilisation of resources.
- **C** Safety.

Answer: C. Safety is the cardinal principle of aviation. Efficiency and resource utilisation matter, but never above safety. The entire CRM/TEM/LOFT framework exists to contribute to incident and accident prevention (see Section 1.2). Safety is the overarching objective.

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6 The elements of the SHEL model are:

- **A** Software, hardware, electronics, and livewire.
- **B** Shareware, hardware, education, and limitations.
- **C** Software, hardware, environment, and livewire.

Answer: C.

7 Which is true regarding the interaction between various elements of the SHEL model?

- **A** All components of the SHEL model are equally important, as conflict and error can arise from the various mismatches.
- **B** The hardware is the hub for interactions between the different components of the system, hence other components should be matched to design most suitable hardware.
- **C** The liveware is the hub of the SHEL model; therefore, the nonhuman components should be adapted and matched to this central component.

Answer: C.

8 Pilots have now become responsible for the management of hardware and software interfaces, through which they are required to direct the operation of the aircraft. These technological advances have:

- **A** Significantly eliminated human errors.
- **B** Given rise to new forms of error.
- **C** Are aimed to eliminate pilots from the cockpit.

Answer: B.

9 About what proportion of civil air accidents result from inadequate human performance?

- **A** 89%
- **B** 73%
- **C** 57%

Answer: B.

10 High Arousal Overload results in:

- **A** A real danger of attention becoming narrowed.
- **B** An attention becoming sharpest.
- **C** An optimum arousal for more difficult or intellectually (cognitive) tasks.

Answer: A.

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11 What airplane equipment marked a substantial decrease in hull loss rates in the eighties?

- **A** GPWS
- **B** TCAS
- **C** SSR

Answer: A.

Memory Aid — SHELL (with second L)
Think **"S–H–E–L–L"**: **S**oftware → **H**ardware → **E**nvironment → **L**iveware (you) → **L**iveware (others).

Memory Aid — "PPPP" of Liveware Factors
Physical → **P**hysiological → **P**sychological → **P**sychosocial.
Remember: **"Physiological and Psychosocial effects are MORE pronounced."**

Memory Aid — "EACOC + Cabin" (6 threats)
Environmental · **A**TC · A**i**rcraft · **C**rew Support · **O**perational · **C**abin.

Memory Aid — "I PCPO" (5 errors)
Intentional non-compliance · **P**rocedural · **C**ommunication · **P**roficiency · **O**perational decision.

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Chapter 24 — Aviation Psychology

Exam Tip — SA Interference Mnemonic
"SBF-EPI" → *Stress, Boredom, Fatigue, Emotional disturbance, Poor communication, Interruptions*.

Master Reference Tables & Mnemonics

Purpose
Every numerical value and key rule from the chapter consolidated for quick exam-night revision.

M-2. Mnemonics & Memory Aids

Concept	Mnemonic
Aviation Psychology aims	**D-P-U-I** → Describe, Predict, Understand, Influence
Information-processing stages	**D-P-D-A-F** → Detection, Perception, Decision, Action, Feedback
Workload drivers	**DPST** → Difficulty, Parallel, Series, Time
Anderson model phases	**C-A-A** → Cognitive, Associative, Automatic
STM expansion tools	**"Chunk & Chain"** → Chunking + Association
Long-term memory types	**S-E-P** → Semantic, Episodic, Procedural
SA-killers	**SBF-EPI** → Stress, Boredom, Fatigue, Emotional, Poor comm., Interruptions
Decision-making seven steps	**R-C-A-D-S-E-F** → Recognize, Consider, Analyze, Develop, Select, Execute, Follow-up
Communication barriers	**4-A + 2-I-R** → Aggressiveness, Arrogance, Anti-authoritarian, (im)Pulsiveness, Invulnerability, Resignation

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Human-centred automation qualities	**"A-SPA-CFDIE²"** → Accountable, Subordinate, Predictable, Adaptable, Comprehensible, Flexible, Dependable, Informative, Error-resistant, Error-tolerant
Control design principles	**S-F-S-I-V-S-S-W** → Standardization, Frequency, Sequence, Importance, Visual-tactile, Symbolism, Simultaneous, Warnings
Stress observables	**"P-FISh-DP-FB"** → Perspiration, Flushed Skin, Dilated Pupils, Fast Breathing
Sleep stages	**1-2-3-4-R** → NREM 1 (transition) → 2 (light, 50%) → 3 (onset deep) → 4 (deep) → REM (dreams)
Jet-lag rule	**"East is Least, West is Best"** → East 50% slower; 1.5 d/zone east vs 1 d/zone west
Mental health priority targets	**D-A-S** → Depression, Anxiety, Substance misuse

Exam Tip — Workload Drivers
Mnemonic **"DPST"** — *Difficulty, Parallel tasks, Series switching, Time available*.

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Chapter 25 — Aviation Physiology

Practice Questions

1 Linear deceleration:

- **A** Causes sensory illusions on the pitch axis.
- **B** Can create the illusion of being in a nose up attitude.
- **C** Application of full power during a missed instrument approach can lead to this illusion.

Answer: A. Linear DECELERATION (e.g. lowering flaps, air-brakes, throttle-back) deflects the otoliths in such a way that the pilot perceives a pitch-down attitude — i.e. a sensory illusion on the pitch axis. (B) is the wrong direction — linear acceleration (not deceleration) creates the nose-up illusion. (C) describes application of FULL POWER → linear acceleration , again the wrong direction (this is the Head-Up illusion / take-off somatogravic). The distractors are flipped on purpose; deceleration affects the pitch axis but produces a "nose-down" illusion.

2 High blood pressure or hypertension is:

- **A** A major cause of unfitness in pilots.
- **B** Responsible for reduced tolerance to G forces.
- **C** May cause chokes during flight.

Answer: A.

3 Unusual fatigue and a loss of consciousness are symptoms of:

- **A** Leans
- **B** Presbyopia
- **C** Hypoxia

Answer: C.

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4 At 8,000 ft, the standard barometric pressure is 77 kPa (574 mmHg). This means that there is:

- **A** 57% of the oxygen available at sea level.
- **B** 65% of the oxygen available at sea level.
- **C** 76% of the oxygen available at sea level.

Answer: C.

5 The atmospheric gas pressure:

- **A** Decreases slower at lower altitudes compared with higher levels and equivalent altitude changes.
- **B** Rises with altitude.
- **C** Drops faster at lower altitudes in comparison to the same altitude changes at higher altitudes.

Answer: C. Atmospheric pressure falls exponentially with altitude — most of the air is in the lowest few kilometres. From SL to 18,000 ft pressure halves (~50 % drop in just 18,000 ft); from 18,000 to 36,000 ft it halves again (a further 25 %); from 36,000 to 54,000 ft another halving. So the absolute pressure drop per unit altitude is larger near the surface than at altitude. (B) is wrong — pressure decreases, doesn't rise.

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6 The chemical composition of the earth's atmosphere approximately is:

- **A** 78% nitrogen, 21% oxygen, 0.9% argon, 0.03% carbon dioxide.
- **B** 70% nitrogen, 28% oxygen, 1.9% argon, 0.03% carbon dioxide.
- **C** 78% nitrogen, 20% oxygen, 0.9% carbon dioxide, 1.03% argon.

Answer: A. Standard atmospheric composition (by volume): N_2 ≈ 78 %, O_2 ≈ 21 %, Ar ≈ 0.93 %, CO_2 ≈ 0.03–0.04 % . Option A matches exactly. (B) has too much O_2. (C) swaps Ar and CO_2 values — note that CO_2 is the tiny trace, not Ar.

7 The volume percentage of oxygen in the atmosphere is 21% which:

- **A** Is constant for all altitudes in troposphere.
- **B** Decreases with increasing altitude.
- **C** Increases with increasing altitude.

Answer: A. The percentage composition of atmospheric gases is essentially constant throughout the troposphere (sea level to ~ 36,090 ft / tropopause) due to atmospheric mixing. What changes with altitude is the partial pressure of O_2 (and total pressure), not the volume percentage. This is the classic exam trap — students confuse "less oxygen" (true in terms of partial pressure) with "lower percentage" (false). Hypoxia at altitude is caused by reduced pO_2, not reduced O_2 percentage.

8 Pain in the middle ear during descent may be eased by:

- **A** Increasing the rate of descent.
- **B** Sneezing deliberately with the nostrils wide open.
- **C** Leveling off and possibly climbing.

Answer: C. " If the problem is not resolved, the rate of descent should be decreased or stopped at intervals... This is also known as STEP DESCENT. Pilot may resort to even CLIMB if pressure persists. " Levelling off allows the Eustachian tube time to equalise; if that fails, a brief climb reduces external pressure and the differential. (A) increasing descent makes it worse. (B) deliberate sneezing is not the DGCA-listed technique — Valsalva (blow with nostrils pinched closed) is.

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9 Gases of physiological importance to man are:

- **A** Oxygen and carbon dioxide.
- **B** Ozone and carbon dioxide.
- **C** Oxygen and Ozone.

Answer: A. The two gases the body actively exchanges are oxygen (taken in for cellular respiration) and carbon dioxide (expelled as metabolic waste). Ozone (O_3) is toxic to humans and exists at high concentrations in the stratosphere; it is a hazard to high-altitude pilots, but not a "physiologically important" gas in normal metabolism.

10 Oxygen, combined with hemoglobin in blood is transported by:

- **A** Platelets.
- **B** Blood plasma.
- **C** Red blood cells.

Answer: C. Haemoglobin is contained within Red Blood Cells (RBCs/erythrocytes) — approximately 280 million Hb molecules per RBC. Each Hb binds up to 4 O_2 molecules. (A) Platelets are for clotting. (B) Plasma carries dissolved gases too but in vastly smaller quantities than Hb in RBCs — over 98 % of O_2 in blood is carried by Hb in RBCs.

11 According to the ICAO standard atmosphere, the temperature lapse rate of the troposphere is approximately:

- **A** -2°C every 1,000 feet.
- **B** -10°C every 10,000 feet.
- **C** -1°C every 190 meters.

Answer: A. The ISA lapse rate is 1.98 °C per 1,000 ft — which rounds to roughly 2 °C/1,000 ft . So option A is the closest match. (B) –10°C per 10,000 ft = 1°C per 1,000 ft — too low. (C) –1°C per 190 m → ≈ 1°C/623 ft → ≈ 1.6°C per 1,000 ft — close but not the standard expression. The DGCA-preferred answer in this exam is (A) –2°C/1,000 ft.

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12 The barometric pressure drops to 1/2 of the pressure at sea level at:

- **A** 30,000 feet.
- **B** 18,000 feet.
- **C** 10,000 feet.

Answer: B. " By 18,000 ft ASL (5,486 m) atmospheric pressure is HALVED . " This is also the threshold above which DCS symptoms may begin to develop. (A) 30,000 ft is roughly quarter-SL pressure (a second halving from 18,000 ft to ~36,000 ft). (C) 10,000 ft pressure is ~70 % of SL — not halved.

13 The total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases which compose the mixture corresponds to:

- **A** Charle's law.
- **B** Dalton's law.
- **C** Henry's law

Answer: B. Dalton's Law of Partial Pressures : the total pressure of a gas mixture equals the sum of the partial pressures of each component gas. This is the basis of pO_2 calculations at altitude. (A) Charles's Law: $V \propto T$ at constant P. (C) Henry's Law: gas dissolved in a fluid is proportional to its partial pressure — explains DCS.

14 Boyle's law is directly applicable in case of:

- **A** The transfer of gas between the alveoli and the blood.
- **B** The expansion of trapped gases in the human body with increasing altitude.
- **C** Pressure of a mixture of gases at high altitude.

Answer: B.

15 Which data compose the ICAO standard atmosphere?

- **A** Pressure, Temperature, Humidity and positive lapse rate.
- **B** Density, Pressure, Temperature and lapse rate.
- **C** Density, Pressure, dryness and negative lapse rate.

Answer: B. ICAO Standard Atmosphere defines: SL pressure 1013.25 hPa, SL temperature 15°C, SL density 1.225 kg/m³, and lapse rate 1.98°C/1,000 ft . Humidity is NOT part of ISA (ISA assumes dry air). The lapse rate is decreasing (temperature falls with altitude) — but the standard refers to it as "the lapse rate," not "negative" or "positive." Option (B) gives the four ISA parameters correctly.

16 Dalton's law explains the occurrence of:

- **A** Ischemic hypoxia.
- **B** Histotoxic hypoxia.
- **C** Hypoxic hypoxia.

Answer: C.

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17 Henry's law explains the occurrence of:

- **A** Hyperventilation.
- **B** Hypoxia.
- **C** Decompression sickness.

Answer: C. " Henry's Law explains the occurrence of decompression sickness. " Henry's Law: gas dissolved in a fluid is proportional to its partial pressure at the surface. When ambient pressure drops rapidly, dissolved N_2 comes out of solution in the blood as bubbles → bends, creeps, chokes, staggers.

18 In aviation any mismatch between what we sense and what we expect is:

- **A** An effect of motion sickness.
- **B** An illusion.
- **C** A symptom of Hypoxia.

Answer: B. " In aviation any mismatch between what we sense and what we expect is an illusion. "

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19 On ascent the gases in the digestive tract will:

- **A** Shrink.
- **B** Escape to lungs.
- **C** Expand.

Answer: C.

20 The most serious hypoxia:

- **A** Causes headache and tingling in hands and feet.
- **B** Causes fornication, flushed cheeks and cherry red lips.
- **C** Interferes with reasoning, gives rise to unusual fatigue and, finally, results in a loss of consciousness/death.

Answer: C.

21 One of the symptoms of decompression sickness are:

- **A** The bends.
- **B** Convulsions.
- **C** Impaired night vision and slow reaction time.

Answer: A.

22 The tendency to develop the bends increases with:

- **A** Low rates of climb, age, obesity, physical activity and high temperatures.
- **B** High rates of climb, age, obesity, physical activity and low temperatures.
- **C** High rates of descend age, obesity, physical activity and high temperatures.

Answer: B. FIVE factors: " High rates of climb, age, obesity, physical activity, low temperatures. " Option B matches exactly. (A) flips climb rate and temperature. (C) refers to "descend" instead of "climb" — and high temperature is the wrong direction.

23 The cabin pressure in airline operation is:

- **A** Normally not exceeding 4,000 to 5,000 feet.
- **B** Normally not exceeding 2,000 to 3,000 feet.
- **C** Normally not exceeding 6,000 to 8,000 feet.

Answer: C. typical airliner cabin pressure ≈ 6,000 ft , maximum 8,000 ft . So "not exceeding 6,000 to 8,000 ft" is correct. Modern aircraft (787, A350) maintain lower cabin altitudes (~6,000 ft cruise) for passenger comfort and reduced fatigue — but the DGCA-listed regulatory ceiling remains 8,000 ft.

24 The circulation system transports:

- **A** Oxygen, carbon monoxide and adrenaline.
- **B** Oxygen and carbon dioxide only.
- **C** Chemical messengers, oxygen and carbon dioxide.

Answer: C. DGCA-listed five functions of blood include: respiratory transport (O_2, CO_2), endocrine transport (hormones / "chemical messengers") , defence, temperature regulation, waste removal. (A) Carbon monoxide is a toxin — not normally transported. (B) is incomplete — circulation transports far more than just gases.

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25 Physiological problems at high altitude are caused by:

- **A** Increased atmospheric pressure.
- **B** Reduced oxygen percentage in atmosphere.
- **C** Decreased atmospheric pressure.

Answer: C. The root cause of altitude-related physiological problems is the decrease in atmospheric pressure , which lowers the partial pressure of oxygen (Dalton), reduces gas dissolution (Henry), expands trapped body gases (Boyle), and degrades cognitive performance. (B) is the classic trap — the percentage of O_2 is constant; it's the partial pressure that falls. (A) is the wrong direction.

26 Which gas will diffuse from the blood to the alveoli:

- **A** Nitrogen.
- **B** Carbon Dioxide.
- **C** Oxygen.

Answer: B. Gas exchange in alveoli: O_2 moves FROM alveoli INTO blood (high pO_2 in alveoli → lower pO_2 in venous blood); CO_2 moves FROM blood INTO alveoli (high pCO_2 in venous blood → lower pCO_2 in alveoli). Both exchanges happen by simple diffusion down partial-pressure gradients. (A) Nitrogen is inert and roughly in equilibrium across the membrane — no net diffusion either way in normal breathing.

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27 The symptoms caused by gas bubbles under the skin following a decompression are called:

- **A** Leans.
- **B** Creeps.
- **C** Chokes.

Answer: B. DGCA's four DCS syndromes: Bends (joints), Creeps (skin), Chokes (lungs), Staggers (brain) . Skin bubbles → Creeps . (A) Leans is a vestibular illusion in IMC — not DCS. (C) Chokes is bubbles in pulmonary capillaries.

28 The Oxygen-Carbon Dioxide exchange takes place through:

- **A** The walls of the capillaries.
- **B** Red blood cells.
- **C** Platelets.

Answer: A. Gas exchange happens across the alveolar-capillary membrane — the thin walls separating alveolar air from blood in pulmonary capillaries. RBCs are the carriers of O_2 once it has crossed the membrane, but the exchange site itself is the capillary wall.

29 The normal arterial blood-pressure of a healthy adult is:

- **A** 140/100 mm HG.
- **B** 120/60 mm HG.
- **C** 120/80 mm HG.

Answer: C. normal BP: 120/80 mm Hg (systolic/diastolic). (A) 140/100 is hypertension. (B) 120/60 — diastolic is too low.

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30 Decompression symptoms are:

- **A** Bends, chokes, skin manifestations, neurological symptoms and circulatory shock.
- **B** Bends, chokes, Personality change, Impaired memory and circulatory shock.
- **C** Bends, chokes, Personality change, Impaired memory and Impaired vision.

Answer: A. " Symptoms of decompression sickness are: Bends, Chokes, Skin manifestations, Neurological symptoms, Circulatory shock. " Options B and C substitute hypoxia-style symptoms (personality change, memory, vision) — wrong category.

31 Baroreceptor reflex is triggered by:

- **A** Body's need to adjust blood pressure.
- **B** Body's need to adjust to extreme temperatures.
- **C** Body's need to adjust to extreme cold.

Answer: A. " Any change in your body's demand for blood can trigger your baroreceptor reflex. " The receptors are pressure sensors in the carotid arteries and aortic arch — they respond to BP changes, not temperature. Temperature regulation is a separate autonomic loop.

32 While climbing from sea level to 40,000 ft the difference in barometric pressure is greatest between:

- **A** Sea level and 10,000 ft.
- **B** 10,000 and 20,000 ft.
- **C** 30,000 and 40,000 ft.

Answer: A. Atmospheric pressure decreases exponentially with altitude — so the absolute drop per 10,000 ft is greatest at the bottom . From SL (1013 hPa) to ~10,000 ft (697 hPa) = drop of ~316 hPa. From 30,000 to 40,000 ft (300→187 hPa) = drop of ~113 hPa. The biggest absolute change is the SL → 10,000 ft band. This is why hypoxic risk rises steeply in the first few thousand feet.

33 The blood-pressure which is measured during flight medical checks is the:

- **A** Pressure in the artery of the left upper arm (representing the pressure at left arm).
- **B** Pressure in the artery of the upper arm (representing the pressure at heart level).
- **C** Pressure in the artery of the upper arm (representing the cardiac pressure).

Answer: B. Standard clinical BP measurement is at the brachial artery (upper arm) , with the arm positioned at heart level — so the reading represents systemic arterial pressure at heart level. (A) "left arm" is artificial — either arm is used, both should agree (significant difference indicates pathology). (C) "cardiac pressure" is wrong terminology — cardiac pressures are measured inside the heart, not in an arm artery.

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34 What is the average Time of Useful Consciousness after a rapid decompression at 40,000 ft?

- **A** About 12 seconds.
- **B** More than 1 minute.
- **C** About 40 seconds.

Answer: A.

35 What is the Time of Useful Consciousness?

- **A** The amount of time an individual is able to function effectively (e.g. perform flying duties) in an environment of inadequate oxygen supply.
- **B** The time period during which there is an inability of the cells of the body to use the oxygen available.
- **C** The period of time when there is inadequate blood flow to body tissues.

Answer: A. (B) describes histotoxic hypoxia. (C) describes ischemic / stagnant hypoxia. The hallmark of TUC is the phrase "function effectively in an environment of inadequate oxygen supply " — the test is functional usefulness, not consciousness alone.

36 Which is the procedure to be followed when symptoms of decompression sickness occur?

- **A** 100% Oxygen, immediate descent, land as soon as possible, seek medical assistance immediately on landing.
- **B** 100% Oxygen, immediate descent, land in case not comfortable, seek medical assistance if required.
- **C** Descend below 10,000 feet, symptoms will disappear after few minutes.

Answer: A. If symptoms of decompression sickness appear, descend at once, land as soon as possible, give 100% oxygen and get urgent medical help on landing, even if the person seems to have recovered.

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37 What is decompression sickness?

- **A** A rapid reduction in ambient pressure, may cause the nitrogen in our blood to come out of solution as small bubbles.
- **B** A rapid increase in ambient pressure, may cause the nitrogen in our blood to come out of solution as small bubbles.
- **C** A rapid reduction in ambient pressure, may cause the oxygen in our blood to come out of solution as small bubbles.

Answer: A. Henry's Law application: " A rapid REDUCTION in ambient pressure may cause the NITROGEN in our blood to come out of solution as small bubbles. " Not oxygen (which is consumed metabolically and present at low dissolved levels); not an increase in pressure (which would dissolve more gas, not release it).

38 The part of blood without cell is called:

- **A** Serum.
- **B** Lymph.
- **C** Plasma.

Answer: C. Blood = formed elements (RBCs, WBCs, platelets — >90 % RBCs) + plasma (the cell-free liquid component containing water, proteins, hormones, salts, dissolved gases). (A) Serum is plasma minus the clotting factors — what's left after blood clots. (B) Lymph is a different fluid in the lymphatic system, not part of blood.

39 At rest, with a heart rate of 72 beats per minute and a stroke volume of 70 ml the cardiac output is about:

- **A** 7 liters/min.
- **B** 5 liters/min.
- **C** 6 liters/min.

Answer: B. Cardiac Output = Heart Rate × Stroke Volume = 72 × 70 ml = 5,040 ml/min ≈ 5 L/min . This is the resting cardiac output for a typical adult. At exercise it can rise to 20+ L/min.

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40 Hemoglobin is:

- **A** In the white blood cells.
- **B** Dissolved in the plasma.
- **C** In the red blood cells.

Answer: C. Haemoglobin (Hb) lives inside red blood cells — ~280 million Hb molecules per RBC. WBCs (leukocytes) are immune cells, not O_2 carriers. Plasma carries only a small fraction of total blood O_2 as dissolved gas.

41 The heart muscle is supplied with blood from:

- **A** The pulmonary veins.
- **B** The coronary arteries.
- **C** Coronary capillaries.

Answer: B. The heart muscle (myocardium) is supplied with oxygenated blood by the coronary arteries — branches off the aorta just above the aortic valve. (A) Pulmonary veins return oxygenated blood from lungs to the left atrium. (C) Coronary capillaries are the downstream microvessels — the supply is from the arteries.

42 Bubbles in the joints cause rheumatic-like pain, called the:

- **A** Staggers.
- **B** Chokes.
- **C** Bends.

Answer: C. " Joints: Bubbles in the joints cause rheumatic-like pain, called the BENDS. " The mnemonic: Joints Bend, Skin Creeps, Lungs Choke, Brain Staggers.

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43 Which symptom does not belong to the Decompression Sickness:

- **A** Bends.
- **B** Creeps.
- **C** Convulsions.

Answer: C.

44 In a severe case of hypoxia:

- **A** Only brain and heart functions are maintained.
- **B** Only blood pressure and heart functions are maintained.
- **C** Only blood pressure and brain functions are maintained.

Answer: B. In severe (terminal) hypoxia, higher cortical functions fail first → loss of consciousness. Reflex BP and cardiac function are maintained by brainstem mechanisms even as the cortex shuts down. The brain (cortex) is in fact the most sensitive tissue to hypoxia and fails before the heart. Hence "blood pressure and heart functions" (autonomic/brainstem) are retained while higher brain function is lost.

45 Generally, active pilots should not donate blood, because:

- **A** The loss of blood causes permanent disturbances to the circulation.
- **B** The fluid reduction caused by donating one unit of blood is generally replaced within several weeks.
- **C** The effects at ground level are minimal, flying during this period may entail a risk.

Answer: C. " While the effects at ground level are minimal, flying during this period may entail a risk. " (A) is too strong — "permanent" disturbances would prevent donation entirely. (B) is factually wrong on the time scale — fluid volume is replaced in hours , while red cell mass takes weeks.

46 In mild cases, hypoxia causes only:

- **A** Inattentiveness, poor judgment, and brain stem reflexes.
- **B** Inattentiveness, poor judgment, and uncoordinated movement.
- **C** State of complete unawareness and unresponsiveness.

Answer: B. Mild hypoxia attacks cognitive and motor functions: inattention, impaired judgement, slow reactions, and uncoordinated movement (clumsy fine motor). (A) — brainstem reflexes are retained in mild hypoxia, not lost. (C) — complete unawareness is severe/terminal hypoxia.

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47 The symptoms of CO poisoning are:

- **A** Memory Impairment, Sensory Loss, Tunnel Vision, Cyanosis (a bluing of the body extremities).
- **B** Headache, Dizziness, Nausea, Impaired vision.
- **C** Impaired Judgment, Headache, Tingling in hands & feet, Hyperventilation.

Answer: B. Classic CO poisoning: headache, dizziness, nausea, impaired vision , progressing to confusion and unconsciousness. (A) Cyanosis is the wrong colour — CO poisoning gives cherry-RED lips/skin (carboxyhaemoglobin is bright red), not blue. (C) Tingling and hyperventilation cluster matches hyperventilation/hypoxia signs, not CO specifically.

48 In severe hypothermia, core body temperature can drop below:

- **A** 95 degrees.
- **B** 90 degrees.
- **C** 82 degrees.

Answer: C. normal 98.6 °F · hypothermia < 95 °F · severe hypothermia: core ≤ 82 °F (or lower) . Option A (95 °F) is the threshold for "mild" hypothermia entry, not severe.

49 A pilot, climbing in a non-pressurized aircraft and without using supplemental oxygen will pass the "critical threshold" at approximately:

- **A** 22,000 feet.
- **B** 18,000 feet.
- **C** 10,000 feet.

Answer: A.

50 While flying solo, suddenly your mood changes to intense feelings of well-being and happiness. Your actions should be:

- **A** Plan a date with your girlfriend.
- **B** Enjoy the flight singing favorite songs.
- **C** Suspect onset of hypoxia. Take oxygen and descend below 10,000 feet.

Answer: C.

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51 A pressurized cabin is necessary to protect against:

- **A** Decompression sickness, expansion of gases in the intestines and hypoxia.
- **B** Decompression sickness, UV rays and hypoxia.
- **C** Expansion of gases in the intestines, heart attack and hypoxia.

Answer: A.

52 What is the "Time of Useful Consciousness" for a progressive decompression at 20,000 ft?

- **A** Between 15 and 20 minutes depending on the physical activities being performed.
- **B** Between 45 seconds and 2 minutes depending on the physical activities being performed.
- **C** Between 5 and 10 minutes depending on the physical activities being performed.

Answer: C. DGCA TUC table: FL220 (22,000 ft) — normal ascent TUC = 10 minutes ; rapid decompression at FL220 = 5 minutes. So a "progressive" (slow) decompression at 20,000 ft falls between FL180 (20-30 min) and FL220 (10 min) — approximately 5-10 minutes . Option (C) is the closest band.

53 Hypobaric hypoxia is a result of:

- **A** Exposure to high temperatures.
- **B** High blood alcohol levels in blood.
- **C** Exposure to high altitude.

Answer: C. Hypobaric hypoxia = "low-pressure" hypoxia = the hypoxia caused by reduced atmospheric pressure at altitude. It's another name for hypoxic hypoxia . (A) heat → heat stress / dehydration, not hypoxia per se. (B) alcohol → histotoxic hypoxia (cells can't use O_2), a different sub-type.

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54 The release of nitrogen bubbles due to decompression can adversely affect:

- **A** Joints, skin, respiratory system and brain.
- **B** Joints, skin, inner ear and brain.
- **C** Vision, liver, respiratory system and brain.

Answer: A. The four DCS syndromes target: Joints (Bends) · Skin (Creeps) · Respiratory system (Chokes) · Brain (Staggers) . Option A matches all four. (B) inner ear is not in the canonical list. (C) liver / vision are not classical DCS targets.

55 Myocardial infarction refers to:

- **A** A brain attack when high blood flow causes damage to brain nerves.
- **B** A heart attack when low blood flow causes the heart to starve for oxygen.
- **C** Expansion of trapped gases in intestine at high altitude causing severe pain in the chest.

Answer: B. "Myocardial infarction" = "death of (infarction) heart muscle (myocardium)" — i.e. a heart attack , caused when a coronary artery is blocked (typically by a plaque + clot) and the muscle distal to the blockage is starved of oxygen. (A) describes a stroke (brain attack), not MI. (C) describes gas-expansion chest pain — wrong mechanism.

56 Dehydration due to low relative humidity at high altitude can be prevented with:

- **A** Excessive fluid intake.
- **B** Adequate fluid intake.
- **C** Diuretics like coffee or tea to be taken frequently.

Answer: B. " Over-all dehydration can be prevented with ADEQUATE fluid intake. Diuretics like coffee or tea should be AVOIDED. " (A) "excessive" is wrong — over-hydration is also problematic (water intoxication, frequent bathroom breaks). (C) is the exact opposite of what DGCA prescribes — coffee and tea are diuretics.

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57 Oxygen in blood is carried by, which is found in:

- **A** Hemoglobin, white blood cells.
- **B** Hemoglobin, red blood cells.
- **C** Red blood cells, platelets.

Answer: B. Oxygen binds to haemoglobin , which is contained inside red blood cells . (A) Hb is NOT in WBCs — WBCs are immune cells. (C) Garbled — platelets are for clotting, not gas transport.

58 Smoking:

- **A** Has decreased susceptibility to Carbon Monoxide poisoning.
- **B** Results in a 'masked hangover', rendering pilot unfit for flying.
- **C** Reduces capacity to carry oxygen resulting in hypoxia at lower altitudes.

Answer: C.

59 What does Visual Acuity stand for?

- **A** Is the capacity of the eye to determine small detail, undistorted, at a given distance.
- **B** Time taken for night vision adaptation.
- **C** Distance at which one can see objects undistorted under dim and bright Illumination.

Answer: A. " Visual acuity is a measure of the capacity of the eye to determine SMALL DETAIL, UNDISTORTED, at a given distance. " (B) describes dark adaptation. (C) is a vague distractor.

64 Symptoms caused by gas bubbles in the lungs, following a decompression are called:

- **A** Chokes
- **B** Bbends
- **C** Staggers

Answer: A. Bubbles in lungs (respiratory system) → CHOKES (shortness of breath, burning gnawing piercing chest pain). Mnemonic: Joints Bend · Skin Creeps · Lungs CHOKE · Brain Staggers.

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65 The chance of a black-out during positive G Maneuvers can be reduced by:

- **A** Engaging autopilot.
- **B** A tilted back seat.
- **C** Applying negative G.

Answer: B. " A tilted back seat can reduce the chance of a black-out during positive G-manoeuvres. " The reclined seat shortens the hydrostatic blood column from heart to brain, preserving cerebral perfusion at higher G. Negative G (C) would cause red-out — not a useful counter to blackout!

66 During sustained positive G-forces the order of symptoms is:

- **A** Diminished awareness, Tiredness, Lack of coordination, Black out.
- **B** Internal organs displaced, Onset of tunnel vision, Grey out, Possible black out.
- **C** Memory lapses, Frequent mistakes in speech and actions, Rapid changes of mood, Grey out.

Answer: B. order: 1. Increase in body weight · 2. Mobility impaired · 3. Internal organs displaced · 4. Onset of tunnel vision · 5. Grey out · 6. Possible blackout. Option B preserves the correct 3→4→5→6 sequence. (A) and (C) describe hypoxia-style mental symptoms, not the G cascade.

67 Hemoglobin, the oxygen-carrying chemical in the blood, picks up:

- **A** Carbon monoxide over 200 times more readily than it picks up oxygen.
- **B** Carbon dioxide over 10 times more readily than it picks up oxygen.
- **C** Oxygen over 5 times more readily than it picks up nitrogen.

Answer: A. haemoglobin binds CO ~200 times more readily than O_2 — which is why even small CO concentrations are catastrophic. (B) the CO_2-Hb relationship is via carbamino bonds and is reversible — not the right framing. (C) Hb doesn't significantly bind nitrogen.

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68 Carbon monoxide is a:

- **A** Light blue, odorless, tasteless gas.
- **B** Colorless, odorless, tasteless gas.
- **C** Colorless, foul smelling, tasteless gas.

Answer: B. CO = colourless, odourless, tasteless → the "silent killer". This is precisely why CO detectors are required in modern light aircraft and homes — humans cannot detect CO with their senses.

69 Why most pilots can learn to tolerate moderate increases in positive "G", but many find even the smallest exposure to negative "G" to be unpleasant?

- **A** Negative "G" manoeuvres decrease the flow of blood to the head, the face becomes white, and the vision becomes blurred.
- **B** Negative "G" manoeuvres increase the flow of blood to the head, the face becomes very flushed, and the eyes bulge.
- **C** Black out occurs in quick succession after a red out, allowing hardly any reaction time.

Answer: B. Negative G pushes blood UP into the head → face flushed, eyes bulge, "red out". The human body has no compensatory reflex for this direction (the baroreceptor reflex is built to maintain cerebral pressure under positive G, not under negative G).

70 Barotrauma is caused by:

- **A** The trapped gases inside the aircraft which create pain as ambient pressure increases.
- **B** The trapped gases inside the body which create pain as ambient pressure decreases.
- **C** The trapped gases inside the outer ear, which cause damage to the ossicles or the ear drum.

Answer: B. Barotrauma is pain or damage from gas trapped inside the body that changes volume as the surrounding pressure changes. Trapped gas expands as ambient pressure falls in the climb.

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71 The retina has about 100 million rods on its peripheral zone. Rods can:

- **A** Only detect black and white but are much more sensitive at lower light levels.
- **B** All colors but are much more sensitive black and white.
- **C** Only detect black and white but are much less sensitive at lower light levels.

Answer: A. Rods = peripheral retina, ~100M, monochrome (B&W only), but extremely sensitive in dim light — they are the basis of night vision. (B) is wrong — rods do NOT detect colour. (C) is reversed — rods are MORE sensitive at low light, not less.

72 Depth perception when objects are close is achieved through:

- **A** Central Vision.
- **B** Visual Acuity.
- **C** Binocular Vision.

Answer: C. " Depth perception when objects are close is achieved through binocular vision. " Each eye sees the same object from slightly different angles; the brain fuses the two images → stereo depth. For distant objects, monocular cues (perspective, motion parallax, atmospheric perspective) dominate.

73 Which scanning technique should be used when flying at night?

- **A** Increase intensity of cabin lights.
- **B** Look to the side of the object.
- **C** Look directly at the object.

Answer: B.

74 A B-787 pilot feels nausea and dizziness while flying through low clouds. This could be due to:

- **A** Somatogravic Effect, take 100% oxygen and climb to a higher altitude.
- **B** The leans, get on to the instruments and correct the aircraft attitude.
- **C** Stroboscopic Effect, switch off anti-collision lights.

Answer: C. " The bounce back from rotating beacons on aircraft which have penetrated clouds [can cause flicker vertigo]. If the beacon is bothersome, shut it off during these periods. " Anti-collision strobes reflecting off cloud particles create exactly the 4-20 Hz flicker condition → flicker vertigo (nausea, dizziness). Switching off strobes in cloud is the prescribed cockpit action.

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75 The primary and most reliable sense of spatial orientation is ____. The balance sensors situated in the ____ provide us with a secondary system.

- **A** Ear, Eyesight.
- **B** Eyesight, ear.
- **C** Vestibular apparatus, Somatosensory system.

Answer: B. " The primary and most reliable sense of spatial orientation is EYESIGHT. The balance sensors situated in the EAR provide us with a secondary system. " Option B matches the DGCA wording exactly. Option A is reversed.

76 The Vestibular apparatus (Otoliths + Semi-circular canals) helps maintain spatial orientation. The Otoliths detect ____. The Semi-Circular Canals detect ____.

- **A** Angular acceleration, linear acceleration.
- **B** Equilibrium, orientation
- **C** Linear acceleration, angular acceleration.

Answer: C. Otoliths = LINEAR acceleration (takeoff, braking, gravity). Semi-Circular Canals = ANGULAR acceleration (turning, pitching, rolling). (A) is reversed. Mnemonic: Otoliths Line · Canals Curve .

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77 A pilot approaching a runway which is narrower than normal may feel that he is ____ and make an ____.

- **A** High, under shooting approach.
- **B** Low, over shooting approach.
- **C** Disoriented, instrument approach (2D).

Answer: A. a narrower-than-usual runway appears to be farther/higher → pilot feels HIGH → reduces descent rate → flies a shallow approach → ends up LOW on final / undershoot / lands short. Wide runway = opposite (feels LOW, steep, lands long). The mnemonic: Narrow → Feels High → Undershoots .

78 Human beings maintain spatial orientation using a combination of three factors. Out of these, most reliable is ____, insufficiently reliable is ____ and not at all reliable especially in IMC is ____.

- **A** The Somatosensory system ("seat of the pants" feeling)/G-force, The sense of vision, The Vestibular Apparatus.
- **B** The Vestibular Apparatus, The Somatosensory system ("seat of the pants" feeling)/G-force, The sense of vision.
- **C** The sense of vision, The Vestibular Apparatus, The Somatosensory system ("seat of the pants" feeling)/G-force.

Answer: C. reliability ranking: Vision (most) → Vestibular (partial) → Somatosensory (not reliable at all) . Option C presents them in this exact order. Mnemonic: V·V·S — Vision · Vestibular · Somatosensory.

79 Coriolis illusion, causing spatial disorientation is the result of:

- **A** Simultaneous head movements during aircraft manoeuvres can give the feeling to pilots that they are pitching, yawing, and rolling simultaneously.
- **B** The absence of visual cues during IMC manoeuvres.
- **C** Because of the lack of stable visual references and the erroneous mental models, the pilot is at a disadvantage.

Answer: A. Coriolis (also called somatogyral cross-coupling) occurs when head movement during a sustained turn excites multiple semi-circular canals simultaneously → brain receives signals as if the aircraft is pitching, yawing, and rolling all at once. The cure: keep head stationary in IMC turns, lock eyes on the attitude indicator.

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80 The Eustachian tube:

- **A** Equalizes pressure between middle ear and external atmosphere.
- **B** Provides us with our sense of balance.
- **C** Helps maintain spatial orientation.

Answer: A. " The Eustachian tube allows pressure in the middle ear to equalize across the ear drum with outside or ambient pressure when climbing or descending. " (B) Balance is the function of the vestibular apparatus, NOT the Eustachian tube. (C) Spatial orientation comes from vestibular + vision + somatosensory.

81 Night vision can be affected by:

- **A** Lack of exercise, High blood pressure, Lack of exercise, Smoking.
- **B** Obesity, Too much fat in your diet, Alcohol, Smoking.
- **C** Age, hypoxia, altitudes above 8,000 ft, smoking and alcohol.

Answer: C.

82 Dark adaptation is impaired by:

- **A** Altitudes above 8,000 feet, inhaling carbon monoxide, myopia and exposure to bright sunlight.
- **B** Altitudes above 5,000 feet, vibrations, deficiency of Vitamin C and exposure to bright sunlight.
- **C** Altitudes above 5,000 feet, inhaling carbon monoxide, deficiency of Vitamin A and exposure to bright sunlight.

Answer: C. four dark-adaptation impairers: cabin altitude > 5,000 ft · CO (smoking, exhaust) · Vitamin A deficiency · prolonged exposure to bright sunlight . Option C matches all four. (A) "8,000 ft" is wrong threshold; myopia is unrelated. (B) "Vitamin C" is wrong vitamin — Vitamin A is essential for rod function.

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83 Myopia is more commonly known as ____ and Hypermetropia as ____.

- **A** Long-sightedness, Short-sightedness.
- **B** Short-sightedness, Long-sightedness.
- **C** Defective vision, color-defective vision.

Answer: B.

84 Hypothermia is potentially:

- **A** A life-threatening condition caused by core body temperature dropping below 95 deg F. It needs emergency medical attention.
- **B** Not a life-threatening condition, shivering tends to ease discomfort and is followed by full recovery with 100% oxygen.
- **C** A shivering condition caused due to low cabin temperature. However, it doesn't have any adverse effect on physical and mental abilities.

Answer: A. hypothermia begins when core temp drops below 95 °F, is life-threatening, requires emergency medical attention. (B) is dangerously wrong — 100% O_2 does nothing for hypothermia. (C) is wrong — hypothermia DOES affect physical AND mental abilities (directly).

85 The absence of anything to focus on in a featureless sky (empty field myopia) affects the pilot's capacity to scan for traffic. It can be overcome by:

- **A** Periodically and deliberately focusing on objects at a distance.
- **B** Periodically and deliberately focusing on objects, both close and at a distance.
- **C** Periodically and deliberately focusing on objects at the horizon.

Answer: B. " Pilots should minimize the risks associated with empty visual field by periodically and deliberately focusing on objects, BOTH CLOSE AND AT A DISTANCE. " The full range exercises the ciliary muscles and breaks the empty-field resting focus (which sits at 1-2 m, inside the cockpit). (A) and (C) only exercise distance vision.

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88 The illusions are:

- **A** A natural phenomena, organized formal training is the best protective measure against illusions.
- **B** Abnormal, require psychiatric treatment.
- **C** Visual cues with no effective counter measures available to overcome them.

Answer: A. " The illusions are NATURAL PHENOMENA. Organized formal training is the BEST PROTECTIVE MEASURE against illusions. " (B) — illusions affect every pilot regardless of mental health. (C) is fatalistic and wrong — many counter-measures exist (instrument flying, training, briefings, etc.)

89 Bright runway / app. lights give illusion of being:

- **A** High, resulting in low approaches.
- **B** Low, resulting in high approaches.
- **C** On correct approach, resulting in an inadvertent speed loss.

Answer: B. " Down-sloping runway or terrain, wider than usual runway and bright runway/app. lights give illusion of being LOW, resulting in HIGH approaches. " Bright lights make the runway look closer → pilot feels low → increases descent rate → steep approach → land long. Mnemonic: Bright = feels low = flies high .

90 A dark scene spread with ground lights and stars and obscured sky can create illusions of:

- **A** Not being aligned correctly with the actual horizon.
- **B** Being aligned correctly with the actual horizon.
- **C** Being aligned with the artificial horizon.

Answer: A.

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6 Self-Check & Memory Aids

9.6 Self-Check and Memory Aids

Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — 4 Types of Hypoxia **"HAIH"** → **H**ypoxic · **A**nemic · **I**schemic (Stagnant) · **H**istotoxic. Or memorably — "**How Aviators Inhale Hypoxia**".

Mnemonic — Hypoxia symptoms **"He Hits The Hammer Making Many Strange Tunnel Cuts Forming"** → **H**eadache · **H**yperventilation · **T**ingling · **H**ot/cold sensation · **M**uscular impairment · **M**emory impairment · **S**ensory loss · **T**unnel vision · **C**yanosis · **F**ormication.

Mnemonic — Hypoxia actions **"O down, A down"** → (1) **OXYGEN** on, (2) **ALTITUDE** down — below 10,000 ft or MSA if higher.

Mnemonic — CO Poisoning actions **"H V O L M N"** = **H**eat off · **V**ents open · **O**xygen on · **L**and ASAP · **M**edical aid · **N**o flying till cleared.

Mnemonic — Hyperventilation low-altitude causes **"I FAMSVHG"** (I Fight Aero-Med Symptoms Via Heated G-loads) = **I**ntense concentration · **F**ear · **A**nxiety · **M**otion sickness · **S**hock · **V**ibration · **H**eat · **G**-forces.

Similar symptoms: do not diagnose by altitude alone

Hypoxia (too little oxygen reaching the tissues) and hyperventilation (breathing more than the body needs) can cause similar early symptoms. If the cause is uncertain, treat possible hypoxia first: use the oxygen system as prescribed and check the supply and equipment. If symptoms continue, consciously control the rate and depth of breathing. Follow the aircraft's approved procedures and obtain help as needed.

Mnemonic — CO's 200× rule "**Two hundred times faster than air**" — Hb grabs CO 200× more readily than O_2.

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Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Smoking effects on pilot **"LBCRID"** = **L**ung cancer · **B**reathing problems · **C**irculatory problems · **R**educed G-tolerance · **I**ncreased heart-attack risk · **D**egraded night vision. *"Lighting Burns Cells; Risks Increase Daily"*.

Mnemonic — the smoker's penalty **"Smoker = minus 3,000."** A smoker hits hypoxia 3,000 ft lower than a non-smoker (7,000 vs 10,000 ft).

Mnemonic — bottle-to-throttle **"24 hours, full stop."** Forget the 8-hour minima — DGCA wants 24.

Mnemonic — alcohol metabolism **"15 / 100 / 60."** 0.015 g per 100 ml per hour (≈ 1 drink/hour in men). The body works on a fixed clock — no coffee shortcut.

Mnemonic — alcohol units **Men 5/21 · Women 3/14.** Daily / weekly. The "21" and "14" are equal to (5×4)+1 and roughly (3×4)+2 — but just memorise.

Mnemonic — "1 = 2" **"Six grand — drinks double."** At 6,000 ft, one drink = two at sea level (because hypoxic + histotoxic hypoxia stack).

Mnemonic — BP normal **"One twenty over eighty."** Higher number = systolic (with each beat). Lower number = diastolic (between beats / "permanent" pressure).

Mnemonic — Hypertension causes **"SS PO LAN"** = **S**tress · **S**moking · **P**oor diet · **O**besity · **L**ack of exercise · **A**ge · **N**arrowing of arteries.

Mnemonic — Diastolic vs Systolic **"D = Default. S = Spike."** Diastolic = default/baseline/resting pressure. Systolic = spike with each heartbeat. *Or:* "D for Down number; S for Spike number."

Mnemonic — BP determinants **"COVE"** = **C**ardiac output · **O**bstruction (peripheral resistance) · **V**olume + viscosity · **E**lasticity of arteries.

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Why this section may be the most heavily tested block for jet/transport aspirants
The **Time of Useful Consciousness (TUC) table**, **cabin pressurization**, **decompression sickness**, and **flying-after-diving 24-hour rule** sit at the heart of every operational HPL stem question. Get the TUC numbers wrong by a few seconds and you've lost the question — and in a real rapid decompression, you've lost the aircraft.

This section also opens with two short topics easily missed by students: **the baroreceptor reflex** (how your body keeps BP steady when you stand up, get scared, or accelerate) and **blood donation** (with its specific 48-hour rule for active pilots).

Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Baroreceptor triggers **"Stand-Scare-Sprint"** = the three DGCA-listed

triggers. Stand up, see something scary, switch from walking to running.

Mnemonic — Wait-before-flying rules **"12 - 24 - 24 - 48"** = 12 hours after rapid decompression · 24 hours after the last drink · 24 hours after diving · 48 hours after blood donation. *Each step gets longer — easier to remember.*

Mnemonic — The 4 DCS syndromes (B-C-C-S) **"Bends · Creeps · Chokes · Staggers"** in head-to-toe order from joints to brain. Or memorably: *"Joints Bend, Skin Creeps, Lungs Choke, Brain Staggers."*

Mnemonic — DCS treatment **"WODL-M"** = **W**arm + 100% **O**xygen · **D**escend immediately · **L**and ASAP · **M**edical aid on landing.

Mnemonic — Bends predisposing factors **"CAOP-T"** = **C**limb rate · **A**ge · **O**besity · **P**hysical activity · **T**emperature (low). *Or: "Climb Aged-Obese Physical Cold"*.

Mnemonic — TUC at FL350 jet cruise **"Half a minute, half a minute, half a minute onward!"** TUC at FL350 ≈ 30 s – 1 min normal; halved to 15–30 s on rapid decompression. *The mask must be on your face inside 10 seconds.*

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Mnemonic — Soda-bottle physics **"Cap-off, bubbles-on."** Open a soda → bubbles appear because pressure dropped. Same thing in your blood when cabin altitude rises past 18,000 ft. Henry's Law.

Mnemonic — Cabin altitudes **"6 normal · 8 max"** = 6,000 ft typical cabin altitude · 8,000 ft maximum cabin altitude in commercial pressurised flight.

Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Humidity rule **"40–60 normal, <20 dry."** Modern humidifiers raise 3–8 % to **22 %**. *Avoid coffee & tea — they're diuretics.*

Mnemonic — Temperature reference **"20 fine · 30 hot · >15 deviation cold."** Optimal 20 °C; heat stress at 30 °C; cold loss of fine motor at >15 °C deviation from comfort.

Mnemonic — Vibration body parts **"S(kull) 1-4 · CA(chest/abdo) 4-10 · B(ack) 8-12 · H(ead-pain) 10-12 · V(ision) 30-40"**. *Or: low frequency = low body, high frequency = high body.*

Mnemonic — Natural resonance **"Skull skips at 1–4. Eyes jiggle at 30–40."**

Mnemonic — Obvious incapacitation attributes **"SUUPC-F"** = **S**udden · **U**npredictable · **U**navoidable · **P**rolonged · **C**omplete · **F**requent (in stress).

Mnemonic — Obvious incapacitation causes **"CMICE"** = **C**ardiac arrest · **M**yocardial infarction · **I**ntracranial haemorrhage · **C**erebral apoplexy (stroke) · **E**pilepsy.

Mnemonic — Subtle > Obvious **"What you can't see can kill you."** Subtle incapacitation is more dangerous than obvious because the other crew may not detect it.

Mnemonic — Epilepsy & flying **"Any fit → no flight."** Any fit (major or minor) is associated with unpredictable LoC and is an **absolute bar** to a flying licence.

Mnemonic — Faint causes **"Stand-Shock-Spill-Starve-Stress"** = Stand up too quickly · Sudden Shock · Spill (loss of blood) · Starve (lack of food/fluid) · other physiological Stress.

Mnemonic — Syncope warning trio **"Light-Weak-Dizzy"** before fainting. *Light-headedness · Muscle weakness · Dizziness*.

Mnemonic — Faint vs Fit, the licence rule **"Faint = Fly. Fit = Forbidden."** A faint (with a clear cause) is OK; a fit is not.

Why Vision deserves its own section

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The eye is — by DGCA's own description — "the most sensitive of our sensory organs" and "delivers to the brain information about the outside world at a much faster rate than any other sensory organ". In an aircraft **~80 % of the pilot's situational-awareness input is visual**: outside reference, instruments, charts, fellow traffic. Every visual fact and number in this section is HPL-exam material.

Memorise the **100 million rods / 6 million cones** ratio. Memorise **7 minutes vs 30 minutes** for cone vs rod dark adaptation. Memorise the **five visual defects** (myopia, hypermetropia, presbyopia, astigmatism, glaucoma). Memorise the **"sunglasses six"** characteristics. The DGCA examiner pulls one of these every paper.

Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Rods vs Cones **"Rods are Plenty & Peripheral — see at Night."** 100 million Rods, peripheral retina, monochrome, dim-light.
"Cones are Clustered & Central — see in Colour." 6 million Cones, central retina (fovea), colour, daylight.

Mnemonic — Cornea contribution **"70 to 80 cornea-focus."** The cornea — not the lens — does most of the eye's focusing.

Mnemonic — Dark adaptation times **"10 seconds bright, 7 minutes cones, 30 minutes rods."** The three iconic numbers — the DGCA examiner mixes them in MCQs every year.

Mnemonic — Dark-adaptation killers **"ACCB"** = **A**ltitude above 5,000 ft cabin · **C**arbon monoxide (smoking/exhaust) · **C**arrots-low (Vitamin A deficiency) · **B**right sun exposure.

Mnemonic — Off-centre viewing **"Look 15 – 20 to the side at night."** Avoid the cone-only fovea; use the rod-rich periphery.

Mnemonic — Empty field focus **"Empty sky, eye at one to two metres."** Pump the eyes by re-focusing on wingtip/horizon/ground feature periodically.

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Mnemonic — Sunglasses 85/10–15 **"Eighty-five absorb, ten to fifteen pass."** ≥85 % light absorption, 10–15 % luminescence transmittance.

Mnemonic — Sunglasses characteristics **"I-T-P-O-L-F"** = **I**mpact resistant · **T**hin frames · **P**olycarbonate coating · **O**ptical quality good · **L**uminescence 10–15 % · **F**iltration appropriate.

Mnemonic — Refractive defects vs lens **"Myopia → Mini (concave) · Hyperopia → Hyper (convex) · Presbyopia → Plus (convex) · Astigmatism → Axis (cylindrical)."**

Mnemonic — Presbyopia age **"40-to-50, lens stops being fifty-fifty."** Accommodation fails between 40 and 50 years of age.

Mnemonic — Glaucoma trio **"Narrow-field · Sneaky-onset · High-pressure."** Progressive narrowing of visual field · insidious/concealed onset · raised intra-ocular pressure.

Mnemonic — Pilot's "spare-pair" rule **"If you fly with glasses, fly with spares."** Mandatory.

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Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Pilot uses of colour (7) **"NRGCMEL"** = **N**av lights · **R**unways & airfields · **G**round obstructions · **C**ockpit displays · **M**aps · **E**mergency flares · **L**ight signals.

Mnemonic — Ossicles head-to-tail **"MIS"** = **M**alleus → **I**ncus → **S**tapes (Hammer → Anvil → Stirrup) — same order from eardrum to oval window.

Mnemonic — Ear clearing (3 steps) **"Swallow · Yawn · Valsalva"** in that order. If still blocked: **step-descend or even re-climb**.

Mnemonic — Audible range & sound qualities **"16 to 20K — Pitch, Loud, Quality."** Audible 16–20,000 Hz. Three qualities of sound: pitch (frequency), loudness (amplitude), quality (harmonious/noisy).

Mnemonic — Noise thresholds **"80 maybe, 90 measurable."** >80 dB = task *may* be impaired. >90 dB = *measurable* impairment.

Mnemonic — Presbycusis **"Old ears lose the highs first."** Age-related hearing loss = presbycusis; high tones cut off first.

Mnemonic — Vestibular components **"Otoliths Line · Canals Curve."** Otoliths detect linear acceleration. Semi-circular canals detect angular (curve) acceleration.

Mnemonic — Three orientation factors **"V·V·S"** = **V**ision (best) · **V**estibular (partial) · **S**omatosensory (worst).

Mnemonic — Why instruments matter in steady turn **"No flow = no clue."** In a steady-speed turn, vestibular fluid stops flowing → vestibular thinks "no turn". **Only the eyes & instruments will save you.**

Mnemonic — Definition of illusion **"Sense ≠ Expect = Illusion."**

Mnemonic — Gestalt six **"PCSSSC"** = **P**roximity · **C**ontinuity · **S**imilarity · **S**ymmetry · **S**implicity · **C**losure.

Why this section is the most operationally lethal block of this chapter
Spatial disorientation kills pilots. In every aviation jurisdiction (legal power over a person, place or event)'s accident database, a significant share of fatal accidents — particularly at night, in IMC, or in mountainous terrain — trace back to **a pilot acting on what their senses said while ignoring what the instruments said**. This part covers every illusion the DGCA syllabus names — by far the longest illusion catalogue in any aviation textbook — plus motion sickness and the start of G-force physiology.

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The unifying principle, repeated almost as a mantra, is: **"If you suspect disorientation, concentrate on and believe in aircraft instruments in IMC. If in VMC, look out at the horizon."** Every other line in this section is detail underneath that one sentence.

Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Master rule **"IMC = Instruments. VMC = horizon."** Two words, two states, two actions. Every other illusion-defence rule reduces to these.

Mnemonic — 6 outside-reference traps **"TWHGSG"** = **T**ake-off (immediately after) · **W**ater · **H**illy terrain · **G**ently sloping terrain · **S**loping cloud · **G**round sloping on approach.

Mnemonic — Shallow approach triggers (5) **"URFRH"** = **U**p-slope · **R**unway narrow · **F**eatureless terrain · **R**ain on screen · **H**aze. All cause the illusion of "too high".

Mnemonic — Steep approach triggers (3) **"DWB"** = **D**own-slope · **W**ide runway · **B**right lights. All cause the illusion of "too low".

Mnemonic — Ground proximity cues (4) **"SSWT"** = **S**peed of ground objects · **S**ize of lights · **W**idth of runway · **T**exture of ground — all increase / change as you descend.

Mnemonic — Motion sickness symptoms (7) **"NH-V-PCHD"** = **N**ausea + fear · **H**yperventilation · **V**omiting · **P**allor · **C**old sweating · **H**eadache · **D**epression.

Mnemonic — Linear-acceleration illusions **"Accel-Up, Decel-Down"** Forward acceleration = pilot feels nose UP → pushes forward = dive. Deceleration = pilot feels nose DOWN → pulls back = STALL on approach.

Mnemonic — Elevator illusion **"Updraft → push DOWN trap. Downdraft → pull UP trap."** The pilot's reflex is always wrong — the illusion is opposite to reality.

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Mnemonic — Black hole **"Black hole = short."** No visual cues between aircraft and runway → pilot thinks "too high" → flies shallow → lands short. Killed many pilots over unlit water at night.

Mnemonic — Flicker vertigo band **"4 to 20 — flicker hurts."** 4–20 Hz flickering light can cause nausea, dizziness, LoC, even epilepsy-like fits. Defence: **turn away · shade · close eyes**.

Mnemonic — Load factor & bank **"60 → 2G, 70 → 3G"** A 60° level turn = 2 G; a 70° level turn = 3 G; aerobatic light aircraft = +6 G limit.

Mnemonic — G-relief technique **"Tense + Ease"** = Tense thigh & stomach + Ease backward pressure on stick. The "AGSM" + stick-relax combination.

Mnemonic — Vection three **"CLR — Circular · Linear · Roll"** The three forms of self-motion illusion induced by moving surrounds.

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Self-Check, Cheat-Sheet and Mnemonics

Mnemonics — burn these into long-term memory

Mnemonic — Six G effects in order **"WMO-TGB"** = **W**eight ↑ · **M**obility ↓ · **O**rgans displaced · **T**unnel vision · **G**rey-out · **B**lackout. *(Or remember the chain: heavy → can't move → guts sag → vision narrows → grey → black.)*

Mnemonic — Bank-angle load factors **"30→1.15 · 45→1.41 · 60→2.0 · 70→3.0"** — the four numbers every DGCA candidate must know. **45° / 1.41 G** is the DGCA "normal flying" reference.

Mnemonic — G-tolerance killers (6) **"ASF-HOS"** = **A**lcohol · **S**moking · **F**atigue · **H**eat · **O**besity · **S**ickness. *(All six are also independent flight-fitness risks — they degrade G-tolerance on top of their other effects.)*

Mnemonic — Black-out vs Red-out **"+G drops blood DOWN (black-out). −G shoves blood UP (red-out)."** Positive G blanks the screen black; negative G tints it red — face flushed, eyes bulging.

Mnemonic — BMI **"Men 20-25, Women 19-24."** Body Mass Index = weight (kg) ÷ height (m)². Men one notch higher than women.

Mnemonic — Exercise prescription **"2 × HR for 20 minutes, 3 times a week."** Double your resting heart rate, ≥20 min, 3 days/wk.

Mnemonic — Three major minerals **"Ca·P·Fe — Bones · Brain-chem · Blood."** Calcium = bones & teeth; Phosphorous = chemical reactions; Iron = haemoglobin.

Mnemonic — Carbohydrate sources **"GVNF"** = **G**rains · **V**egetables · **N**uts · **F**ruit.

Mnemonic — Brain-glucose rule **"Brain can store nothing. Feed it continuously."** Never fly on an empty stomach — hypoglycemia is a DGCA-listed subtle-incapacitation cause.

Mnemonic — Food contamination (4) **"UUUS"** = **U**nclean prep · **U**ndercooked meats · **U**nwashed salads · **S**eafood / local ice cream / mayonnaise.

Mnemonic — Gastroenteritis symptoms (5) **"NVAD-L"** = **N**ausea · **V**omiting · **A**bdominal pain · **D**iarrhoea · **L**oss of appetite. *Pilot grounded even on meds — gas expands at altitude.*

Mnemonic — Obesity multiplier **"Obesity = every risk + 1."** The 13-item list in §67.3 hits heart, BP, hypoxia (lower), DCS, G-tolerance, joints, varicose veins, infections, sweating, diabetes & gout. It is the single biggest *compounding* health factor for pilots.

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Mnemonic — Sinus & ear: same problem, same phase **"Descent destroys."** Both ear (§38) and sinus block (§71) happen worst on descent. Both: slow/stop descent — or better, don't fly congested.

Mnemonic — Six dental causes **"D-D-D-P-C-I"** = **D**ental disease · **D**ental caries · **D**efective restoration · **P**ulpitis · **C**ysts · **I**mpacted teeth.

Mnemonic — Wait-before-flying master list **"12 - 12 - 24 - 24 - 48 - 48 - 48-72"** hr after:

- **12 hr** — Rapid Decompression
- **12 hr** — Local anaesthetic
- **24 hr** — Alcohol (bottle-to-throttle)
- **24 hr** — Diving
- **48 hr** — Blood donation
- **48 hr** — General anaesthetic
- **48-72 hr** — Major dental work

Mnemonic — Four reasons medicines disqualify **"DSDL"** = **D**isease itself disqualifies · **S**ide effects · **D**rug-flight interaction · **L**ingering effect after stopping treatment.

Mnemonic — Caffeine limit **"250-300 mg, 2-3 cups."** Most widely used drug. Addictive. Sources: **Coffee · Tea · Cocoa · Chocolate · Cola**.

Mnemonic — Hypothermia thresholds (°F) **"98.6 normal · <95 hypo · ≤82 severe."** Late-stage sign: **shivering CEASES + apathy**. Hypothermia is life-threatening — emergency medical aid required.

Mnemonic — Six toxic-hazard sources **"F-A-F-A-F-B"** = **F**urnishings & baggage · **A**cetone & turpentine · **F**uels/lubricants/propellants · **A**nti-icing fluid · **F**ire extinguishing agents · **B**attery fumes.

Mnemonic — Common-ailments rule **"In doubt? On ground."** If any doubt about fitness to fly, stay on the ground.

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Mnemonic**"Never Ought An Cool Naval Helmsman Make His Nasty Owl"** — Nitrogen, Oxygen, Argon, Carbon dioxide, Neon, Helium, Methane, Hydrogen, Nitrous oxide, Ozone — in descending order of abundance. (Optional aid — feel free to invent your own.)

Mnemonic — 5 Gas Laws**"Bad Hens Cluck, Geese Dance"** → **B**oyle · **H**enry · **C**harles · **G**raham · **D**alton.

Mnemonic — Zones (low → high)**P – P-D – PS – TS** → Physiological · Physiological-Deficient · Partial Space-equivalent · Total Space-equivalent. *(Or: "Pilots Probably Prefer Tea" — P, PD, PS, TS.)*

Mnemonic — Halving rule**"Half by Eighteen, Quarter by Thirty-Six"** → Half of sea-level O_2 at **18,000 ft**; one-quarter at **36,000 ft**.

Mnemonic — 3 NS divisions **"Cops Patrol Automatically"** → **C**NS (Central — Brain + Cord, the HQ) · **P**NS (Peripheral — all peripheral nerves, the patrol) · **A**NS (Autonomic — runs without orders).

Mnemonic — Cardiac Output **CO = HR × SV** → Cardiac Output = Heart Rate × Stroke Volume. **72 × 70 ≈ 5040 ml ≈ 5 L/min.**

Mnemonic — Right vs Left heart **"Right is BLUE, Left is RED"** → Right side handles de-oxygenated blood (to lungs); Left side handles oxygenated blood (to body).

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6.2 Common DGCA-style probe questions

Try answering these without looking back

1. State the percentage composition of nitrogen and oxygen in the atmosphere. *Does it change with altitude?*
2. Give the ISA sea-level values in **five** different units.
3. What is the ISA lapse rate, and up to what altitude does it apply?
4. Why does hypoxia occur with altitude, when the % of O_2 is unchanged? *(Hint: Dalton's Law.)*
5. At what altitude does a pilot normally start using supplementary oxygen?

6. State the four physiological zones with their altitude boundaries.
7. At what altitude does an unpressurised pilot, NOT using supplemental O_2, hit the critical threshold?
8. Why is a pressure suit mandatory above 50,000 ft? Quote Armstrong's Line value.
9. Explain Boyle's Law and give two flight-medical situations where it applies.
10. How does Henry's Law explain both hypoxia *and* decompression sickness?

DGCA-style probe questions

Try these without looking back

1. Trace the path of air from the nose to the alveoli — name **every** structure.
2. What is the difference between External and Internal Respiration?
3. What is the normal adult respiration rate, and what is the primary regulator of breathing?
4. Define cardiac output. Compute it for HR 72 bpm, SV 70 ml.
5. What is the difference between Systole and Diastole? Describe the two-stage contraction.
6. State the two main components of blood and the three types of formed elements.
7. How many haemoglobin molecules are inside each RBC? How many hemes per Hb?
8. List the five principal functions of blood.
9. What are the two principal failure modes of the circulatory system? Name the medical term for a heart attack.
10. List at least eight risk factors for angina/heart attack relevant to a DGCA medical.
11. What is plaque, and why does it matter to coronary arteries?

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DGCA-style probe questions

Try these without looking back

1. Define *cerebral hypoxia*. How does it usually present (mild vs severe symptom set)?
2. List the FOUR types of hypoxia, with one cause for each.
3. A pilot at FL150 in an unpressurised aircraft reports euphoria and an inability to do mental arithmetic. What type of hypoxia, and what are the 2 immediate actions in correct order?
4. State the altitude at which **night vision** begins to be affected, and explain why.
5. By what factor does haemoglobin prefer CO over O_2? What is the maximum CO percentage in piston-engine exhaust?
6. List the six immediate actions if CO poisoning is suspected (in order). State the DGCA mandatory prevention rule about cabin heating.

7. Why does CO poisoning cause cherry-red lips? Contrast this with the skin sign of hypoxic hypoxia.
8. Define hyperventilation. Give five low-altitude causes.
9. At 8,000 ft a passenger develops tingling fingers, light-headedness and pale clammy skin. What is the most likely diagnosis, and how do you treat it?
10. Why does the same pilot reach his "hypoxia ceiling" earlier if he smokes? Give the approximate equivalent altitude figure.
11. Define *formication* and *cyanosis*. In which condition would you expect each?
12. State the difference between "hypoxic hypoxia" and "anemic hypoxia". Which one is caused by CO poisoning?

DGCA-style probe questions

Try these without looking back

1. State the effective hypoxia altitude for a smoker vs a non-smoker. What is the size of the penalty in feet?
2. By what percentage does a one-pack-per-day smoker reduce his oxygen-carrying capacity?
3. List the SIX medical effects of smoking on a pilot, as quoted in DGCA syllabus.
4. State the recommended bottle-to-throttle separation.
5. What is the body's metabolism rate for alcohol? Can coffee or medication speed it up?
6. At what altitude does one drink have the effect of two at sea level? Why?
7. Define "one unit" of alcohol. State the daily and weekly maxima for men and women.
8. At what blood alcohol concentration do pilot errors spike significantly? List four effects at this level.
9. What is a "masked hangover" and why is it especially dangerous?
10. State the four factors that determine blood pressure.
11. Define *systolic* and *diastolic* pressure. Which is higher? Which represents the "permanent pressure on the arterial wall"?
12. State the normal BP for a healthy young adult. Where is BP measured at the medical?
13. List the seven causes of hypertension. Which are modifiable?
14. Why is hypertension described as "a major cause of unfitness in pilots"?
15. How does smoking link directly to the angina/heart-attack risk-factor list from §9.5?
16. Why does a smoker have a lowered night-vision capability?
17. Explain why the "1 = 2" rule for alcohol at altitude makes physiological sense — name the two types of hypoxia involved.

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DGCA-style probe questions

Try these without looking back

1. What is the baroreceptor reflex? Name three triggers.
2. State the DGCA recommendation for active pilots regarding blood donation. What is the minimum wait before flying?
3. Define Time of Useful Consciousness (TUC). What is its other name (acronym EPT)?
4. State three factors that determine TUC following a rapid decompression.
5. At FL350 in a transport jet, what is the typical TUC range during normal cruise?
6. By what factor does rapid decompression typically reduce TUC?
7. Reproduce the TUC table for FL220, FL280, FL350, and FL430 — both columns.
8. What is the maximum altitude without supplemental oxygen at which flying efficiency is not impaired?
9. At what altitude is atmospheric pressure halved? What is the inert gas making up 80 % of our atmosphere?
10. At what unpressurised flight altitude does the risk of the bends become significant?
11. State the typical cabin altitude maintained by a commercial airliner flying at FL300. What is the maximum cabin altitude?
12. Give four signs of a fast decompression. Give two causes of a slow decompression.
13. List the FIVE factors that increase the tendency to develop the bends.
14. State Henry's Law verbatim. How does it explain decompression sickness?
15. Name the FOUR DCS syndromes by tissue affected and their colloquial names.
16. State the four steps in the treatment of DCS.
17. How long after rapid decompression must a pilot wait before flying — even if feeling fit?
18. At what minimum diving depth does compressed-air diving create a fly-risk? How long must a diver wait before flying?
19. State the symptom-onset window for DCS after surfacing from a dive.
20. Compare and contrast the four "wait-before-flying" rules: alcohol, blood donation, diving, rapid decompression.

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DGCA-style probe questions

Try these without looking back

1. State the normal range of relative humidity. At what RH does minor discomfort begin? List the four areas of dryness affected.
2. Why should diuretics like coffee and tea be avoided in flight?
3. By how much do modern aircraft humidifiers (A350, 787, future 777X) raise cabin RH?
4. List three potential problems of installing humidifiers on aircraft.
5. At what temperature is the average pilot most comfortable? At what temperature do HR, BP and sweating increase?

6. State the natural resonance frequency of (a) the skull and (b) the eyeball.
7. Reproduce the DGCA vibration-frequency effect table — five rows.
8. At what vibration frequency is vision interfered with? Why?
9. What two effects does UV radiation from sunlight have on the pilot's eyes?
10. State the six attributes that make in-flight incapacitation a serious safety threat.
11. What measure minimizes the risk of incapacitation? How does its frequency change with age?
12. Define obvious incapacitation. List five causes.
13. Define subtle / insidious incapacitation. State the key warning about the appearance of the affected crew member.
14. List eight causes of subtle incapacitation.
15. Why is subtle incapacitation often considered *more* dangerous than obvious incapacitation?
16. Define a fit or seizure. What does the term "epilepsy" refer to?
17. Compare Grand Mal and Petit Mal epilepsy in three respects.
18. What is the DGCA medical implication of any fit, major or minor?
19. Define a faint and list five common causes.
20. Define vasovagal syncope. What three warning symptoms typically precede the fainting?
21. What is the flying-licence implication of a faint, provided the cause is clearly understood?

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DGCA-style probe questions

Try these without looking back

1. State the percentage contribution of the cornea to the total focusing ability of the eye.
2. How many rods and how many cones does the retina contain? Where is each type located?
3. Define the fovea. Why does it provide the best day vision but no night vision?
4. Define binocular vision and state two advantages of having two eyes.
5. Define visual acuity. Why is normal acuity (with or without spectacles) essential for pilots?
6. What is accommodation? Why do middle-aged pilots wear bi-focals?
7. List six factors that affect the sharpness of central vision.
8. How long does it take for (a) full bright adaptation, (b) cone dark adaptation, (c) rod dark adaptation?
9. Why is red cockpit lighting useful for preserving dark adaptation? What is its main drawback?
10. List four factors that impair dark adaptation.
11. At what angle off-axis should a pilot look to use rod (peripheral) vision at night?
12. Explain what the "blind spot" is and how binocular vision compensates for it.

13. Define empty visual field. At what distance does the eye naturally focus in an empty field? How should the pilot mitigate this?
14. State four protective measures against flash blindness in a thunderstorm.
15. State three required properties of good aviation sunglasses (light absorption, UV/IR, equal-colour absorption). What is the luminescence transmittance range?
16. List all six DGCA-listed characteristics of pilot sunglasses.
17. Are photochromatic lenses allowed in flight?
18. Define myopia and hypermetropia. Name the corrective lens type for each.
19. Define presbyopia. At what age does it typically arise? What lens corrects it?
20. Define astigmatism. What causes it?
21. State the DGCA rule on a pilot who wears corrective lenses regarding the carriage of spectacles.
22. Define glaucoma. State three of its characteristics. Why is it of particular aero-medical concern?

DGCA-style probe questions

Try these without looking back

1. List seven reasons pilots need good colour vision.
2. What causes colour blindness, anatomically?
3. State the two distinct functions of the ear.
4. Name the three ossicles. What do they do?
5. What does the Eustachian tube do? When can a pilot NOT fly because of it?
6. During which phase of flight is ear-clearing most difficult? Why?
7. Define Otis Barotrauma. State the three deliberate clearing techniques.
8. What is meant by "step descent"? When else may the pilot resort to a climb?
9. Name the three qualities of sound and what each depends on.
10. State the audible frequency range of a fit young person.
11. List the four DGCA-listed causes of hearing impairment.
12. Define conductive deafness. Define presbycusis. Which tones disappear first in presbycusis?
13. State the dB threshold above which task performance may be impaired, and the dB threshold for measurable impairment.
14. Explain the counter-intuitive "high-noise may improve vigilance" effect.
15. Define orientation. Which sense is primary, and which is secondary?
16. Define proprioception.
17. What two components make up the vestibular apparatus? What does each detect?
18. Rank the three orientation factors by reliability.
19. What is the somatosensory system? How reliable is it?
20. What happens (in fluid-flow terms) when the body is in a steady-speed turn? What can detect motion in this condition?

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21. Define an illusion in aviation context.
22. How can spatial disorientation from illusions be prevented?
23. Explain the "atmospheric perspective" illusion. What direction of error does flying in clear air cause?
24. List the six Gestalt principles of perceptual organisation.

DGCA-style probe questions

Try these without looking back

1. What illusion does blowing snow during a taxi-into-headwind create?
2. State the Somatogravic illusion on take-off. What is the dangerous reflex response?
3. List the SIX scenarios where outside references give a false impression.
4. Define auto-genesis. What may the pilot try to do that is dangerous?
5. State the three stages of the approach-and-landing visual task.
6. Define "Visual Angle".
7. Explain the narrow-runway illusion and the wide-runway illusion — both perception and outcome.
8. Define the Black Hole Effect. Give the typical environments. What is the result?
9. List FIVE triggers of a shallow-approach illusion.
10. List THREE triggers of a steep-approach illusion.
11. Explain "texture flow" and the inadvertent-speed-loss trap on final approach.
12. List FOUR ground-proximity height-assessment cues.
13. List the SIX DGCA-listed protective measures against illusions.
14. State the DGCA "master rule" for handling suspected disorientation — both IMC and VMC variants.
15. State the vibration band that contributes to airsickness. List SEVEN symptoms of motion sickness.
16. Define Head-Up Illusion. When does it typically occur? Why is the reflex response dangerous?
17. Define Head-Down Illusion. Why is the reflex response especially dangerous on final approach?
18. Define the Elevator Illusion for updraft and downdraft separately. What is each reflex response?
19. State the FOUR DGCA-listed triggers of False Horizon.
20. Name the three forms of vection illusion with one-line definitions.
21. Define Geometric Perspective Illusion.
22. State the flicker-vertigo frequency band and list 4 possible reactions. Give 3 in-flight causes and 3 preventative actions.
23. List the SIX statements in the Disorientation Summary section.
24. State the numerical value of 1 G in both ft/s² and m/s². State the load factor at 60° and 70° banks. State the positive G limit of a light aerobatic aircraft.

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25. State the two counter-measures against adverse G effects.

DGCA-style probe questions

Try these without looking back

1. List the SIX effects, in order, that a pilot experiences during increased "G" manoeuvres.
2. What single physical/mechanical action in the cockpit can reduce the chance of blackout during positive-G manoeuvres?
3. State the load factor experienced in a level turn at 45° angle of bank. Why is this number the upper limit of "normal flying"?
4. State the SIX DGCA-listed factors that reduce G-tolerance.
5. What in-flight scenario typically produces an exposure to negative G? What is the dangerous reflex that causes it?
6. Define "Red-out". List the three physical symptoms. State the corrective action.
7. State the BMI formula. Give the normal range for men and women.
8. What is the most efficient way to lose weight ?
9. State the DGCA exercise prescription — target heart rate, duration, frequency.
10. Are pilots allowed to take appetite suppressants?
11. List at least eight of the thirteen conditions that being badly overweight predisposes a pilot to.
12. State the four DGCA-listed sources of carbohydrates.
13. Define vitamins. What do they help the body do?
14. Name the three major minerals and the role of each (one-liner).
15. What is the brain's fuel? Why is hypoglycemia particularly bad for cognitive functioning?
16. List the four major causes of food contamination.
17. Why is a pilot with gastroenteritis (even if on medication) considered unfit to fly?
18. List the five DGCA-listed symptoms of gastroenteritis.

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DGCA-style probe questions

Try these without looking back

1. During which phase of flight does sinus block most frequently occur? What is the best in-flight action? What is the best preventative?
2. Define barodontalgia. List the six dental conditions that can cause it.
3. State the DGCA no-fly window after major dental work. What in-flight action gives relief?

4. What effect does dehydration have on a pilot's performance?
5. State the DGCA "stay-on-the-ground" rule regarding common ailments.
6. State the DGCA absolute rule on drugs/medication. Who must clear a pilot for flight while on medication?
7. State the FOUR DGCA-listed reasons medicines have flight-qualification consequences.
8. State the recommended maximum daily caffeine intake (both mg and cups of coffee). List the five sources of caffeine in the DGCA list.
9. State the no-fly periods after LOCAL and GENERAL anaesthetics. What does the DGCA say about strong analgesics?
10. State (i) normal body temperature, (ii) hypothermia threshold, (iii) severe hypothermia threshold — all in °F.
11. Describe what happens to shivering and consciousness as hypothermia progresses to severe.
12. List the SIX DGCA-listed toxic-hazard sources in aviation.
13. State the DGCA "even mild toxic effects…" warning and the recommended action after toxic exposure.
14. Define Dangerous Cargo. Why is it the chapter-closing topic?
15. State the four "wait-before-flying" rules in this chapter — diving / RD / blood donation / alcohol / general anaesthetic / dental — match each to its hours.

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